



Experimental results of motions and structural loads on the 372 catamaran model in head and oblique waves.

Ir. Riaan van 't Veer

Report 1130

December 1998

*In Co-operation with MARIN, P.O. Box 28,
Wageningen, The Netherlands*

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Abstract

This report presents the measurement results of a series of experiments carried out in the seakeeping basin at MARIN, Wageningen, with a catamaran model. The measurements were carried out as part of a PhD project.

The purpose of the test was:

- to determine the motion RAO's in the centre of gravity,
- to determine the connecting forces on the centreline of the vessel.

The comparison of the measurements with numerical results is presented in the PhD thesis Van 't Veer (1998a). Another series of experiments with the catamaran 372 in head waves is presented in the measurement report Van 't Veer (1998b). The latter experiments were carried out at Delft University.

A preliminary data report, Report No. 257608-1-ZT, has been published by MARIN (June 1997) concerning the same experiments. Several pictures or pages from this report are included (photocopied) in this report as well. From the test runs 8703 to 8789 a video has been made, of which a copy is available at Marin and Delft University.

This report presents:

- the model,
- the test set-up,
- the first-harmonic analysis of the measurement signals,
- tables and figures of the motion RAO's,
- the force RAO's in the COG.

List of Symbols

a	-	subscript for amplitude
Δ	[kg]	displacement
ϵ_3	[deg]	heave phase angle
ϵ_5	[deg]	pitch phase angle
η_3	[mm]	heave motion
η_5	[rad]	pitch motion
λ	[m]	wave length
ρ	[kg/m ³]	water density (1000 kg/m ³)
θ	[rad]	pitch angle
ω	[rad/sec]	wave frequency
ω_e	[rad/sec]	wave encounter frequency
ζ	[mm]	wave height
A_{ij}	[]	added mass value in the i -direction due to a force in the j -direction
B_{ij}	[]	fluid damping value in the i -direction due to a force in the j -direction
B	[m]	overall vessel beam
C_{ij}	[]	restoring coefficient in the i -direction due to a force in the j -direction
C_B	[-]	block coefficient
F_3	[N]	Vertical force (in the z -direction)
F_5	[Nm]	Moment around the y -axis
F_n	[-]	Froude number; $F_n = U/\sqrt{Lg}$
H	[m]	distance between the centre lines of the two hulls
I_{55}	[kgm ²]	moment of inertia, around y -axis
KG	[m]	distance of centre of gravity above keel
LCG	[m]	longitudinal centre of gravity, measured from AP
L_{0A}	[m]	overall ship length
L	[m]	ship length between perpendiculars
M	[kg]	mass

R	[N]	overall ship resistance
T	[m]	ship draught
T_{AP}	[m]	draught at aft perpendicular
T_{FP}	[m]	draught at fore perpendicular
U	[m/sec]	steady forward speed of the vessel
b	[m]	breadth of the demihull
g	[m/sec ²]	gravitational acceleration; 9.81 m/sec ²
k	[rad/m]	wave number; $k = \omega^2/g$
k_{yy}	[m]	pitch radius of gyration
x	[m]	x-coordinate
y	[m]	y-coordinate
y_w	[m]	local beam of vessel, centre line to waterline
z	[m]	z-coordinate
z	[mm]	heave motion

1 Description of the experiments

1.1 Introduction

To obtain insight in the behaviour of fast catamaran vessels in waves, a series of experiments have been carried out. The test program consisted of:

- Resistance measurements, including trim and sinkage registration,
 $F_n = 0.18$ to $F_n 0.75$
- Motion Test in regular waves, wave heading 180, 195 and 225 degrees
 $F_n = 0.30, 0.60$ and 0.75

The catamaran used in the experiments was designed at Delft University. The ratio of several overall dimensions were chosen from realistic values found in the literature of recently build catamarans.

The transom was given a zero depth. It is understood that for practical reasons the transom stern of a catamaran has a certain depth, since this is necessary to install the waterjet propulsion system (on full scale). However, to illuminate the highly nonlinear behaviour of such a transom stern, a stern with zero depth was chosen.

The range of forward speed in which the catamaran model was tested, was selected from $F_n = 0.30$ up to $F_n = 0.75$, which is a realistic operating range for existing catamaran vessels.

The model itself was constructed from GRP in the model shop laboratory at Delft University.

1.2 Experimental set-up

Model experiments with the 372 catamaran model were carried out in the Seakeeping basin at MARIN. The dimensions of the towing tank are $L \times B \times D = 100 \times 24 \times 2.5$ [m]. During all experiments the catamaran was free-floating and only connected with the towing tank carriage via electronic wires. The model was equipped with a waterjet propulsion system. This propulsion system was available at Marin and the necessary equipment was build into the model at Marin as well.

The main particulars of the catamaran vessel are given in Table 1.1 and a lines plan of the catamaran is presented in Figure 1.1 and on page 9 (larger scale). A table of offsets is included as well. During the experiments at Delft University with the same model, the mass inertia terms were different than during the experiments at Marin which was due to the experimental constraints of the waterjet system.

Main Particulars	Symbol	Value
Length over all	L_{OA}	3.11 m
Length between perpendiculars	L	3.00 m
Beam overall	B	0.94 m
Beam demihull	b	0.24 m
Distance between center of hulls	H	0.70 m
Draught	T	0.15 m
Displacement	Δ	87.07 kg
Draught, AP	T_{AP}	0.15 m
Draught, FP	T_{FP}	0.15 m
Vertical center of gravity	KG	0.278 m
Longitudinal center of gravity	LCG	1.41 m
Roll radius of gyration	k_{xx}	0.389 m
Pitch radius of gyration	k_{yy}	0.810 m
Yaw radius of gyration	k_{zz}	0.930 m
Length over beam ratio	L/b	12.5
Length over draught ration	L/T	20.0
Block coefficient	C_B	0.403

Table 1.1: *Main characteristics of the Catamaran 372 model.*

Several pages are included from the MARIN report which present among others the definitions of the different signals and the position where the signals were measured.

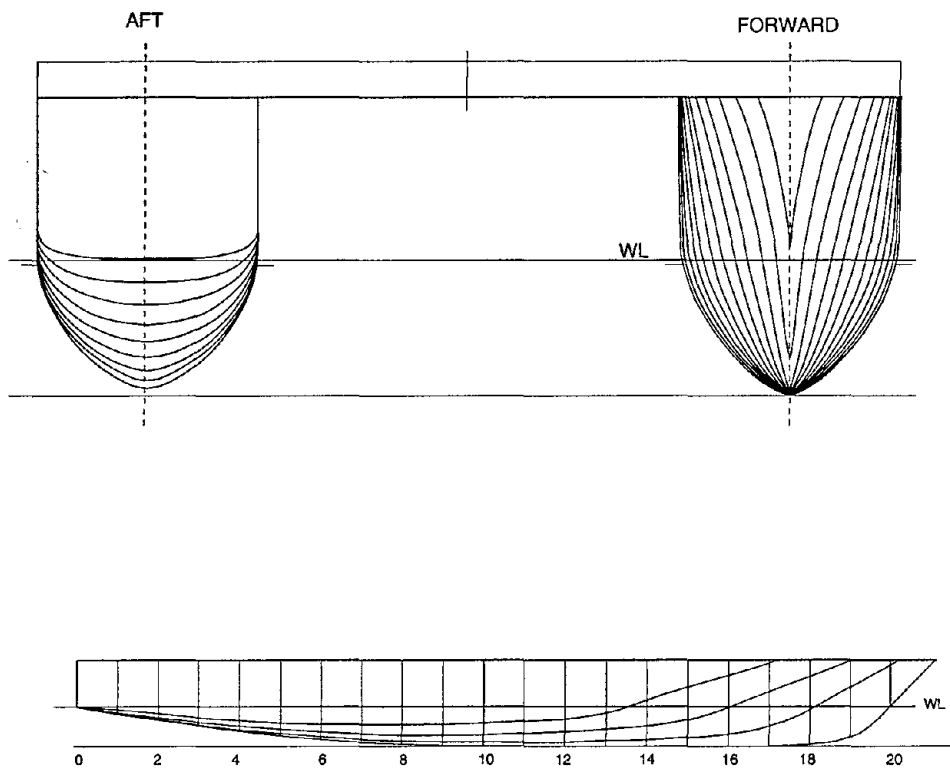


Figure 1.1: *Lines plane Catamaran 372.*

TABLE OF OFFSETS

x=-15.00		x=-14.25		x=-13.50		x=-12.75		x=-12.00		x=-11.25	
z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20
1.90	1.20	1.83	1.20	1.80	1.198	1.80	1.20	1.80	1.199	1.78	1.20
1.80	1.193	1.661	1.15	1.60	1.141	1.578	1.15	1.60	1.170	1.509	1.15
1.70	1.141	1.551	1.05	1.50	1.065	1.425	1.05	1.50	1.132	1.321	1.05
1.57	1.00	1.469	0.90	1.40	0.936	1.301	0.90	1.40	1.072	1.160	0.90
1.545	0.80	1.427	0.75	1.339	0.80	1.228	0.75	1.227	0.90	1.059	0.75
1.492	0.60	1.405	0.60	1.293	0.60	1.187	0.60	1.088	0.60	0.995	0.60
1.475	0.30	1.381	0.30	1.260	0.30	1.144	0.30	1.033	0.30	0.928	0.30
1.470	0.00	1.372	0.00	1.247	0.00	1.127	0.00	1.011	0.00	0.901	0.00

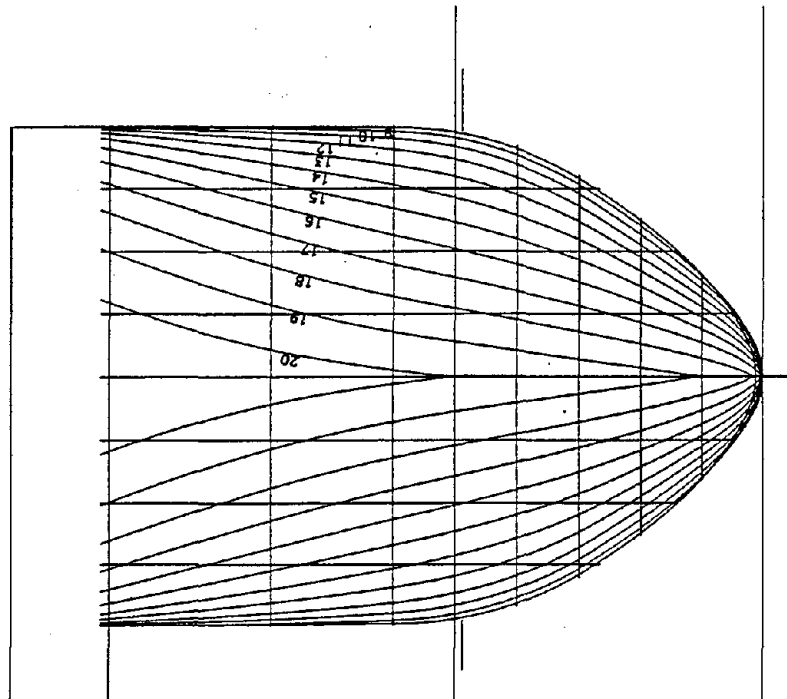
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z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20	2.00	1.20
1.80	1.199	1.80	1.20	1.80	1.20	1.70	1.20	1.70	1.20	1.70	1.20
1.60	1.182	1.70	1.198	1.60	1.194	1.50	1.186	1.50	1.187	1.50	1.187
1.50	1.157	1.50	1.170	1.50	1.180	1.30	1.138	1.30	1.143	1.30	1.138
1.30	1.066	1.30	1.104	1.40	1.157	1.10	1.057	1.10	1.067	1.10	1.067
1.098	0.90	1.10	0.987	1.20	1.084	0.90	0.938	0.90	0.960	0.90	0.963
0.909	0.60	0.90	0.808	1.00	0.969	0.70	0.780	0.70	0.817	0.70	0.830
0.827	0.30	0.757	0.60	0.80	0.805	0.50	0.553	0.50	0.638	0.50	0.670
0.795	0.00	0.645	0.30	0.632	0.60	0.361	0.30	0.40	0.521	0.30	0.459
		0.60	0.00	0.489	0.30	0.30	0.124	0.260	0.30	0.189	0.30
				0.429	0.00	0.284	0.00	0.20	0.163	0.136	0.20
								0.169	0.00	0.099	0.10
										0.085	0.00

x=-1.50		x=0.00		x=1.50		x=3.00		x=4.50		x=6.00	
z	y	z	y	z	y	z	y	z	y	z	y
2.00	1.194	2.00	1.179	2.00	1.151	2.00	1.106	2.00	1.040	2.00	0.953
1.80	1.191	1.80	1.172	1.80	1.138	1.80	1.086	1.80	1.012	1.80	0.917
1.50	1.168	1.50	1.142	1.50	1.100	1.50	1.039	1.50	0.957	1.50	0.855
1.20	1.091	1.20	1.061	1.20	1.016	1.20	0.953	1.20	0.870	1.20	0.771
0.90	0.949	0.90	0.919	0.90	0.874	0.90	0.816	0.90	0.742	0.90	0.655
0.60	0.747	0.60	0.720	0.60	0.679	0.60	0.628	0.60	0.569	0.60	0.502
0.30	0.485	0.30	0.479	0.30	0.453	0.30	0.416	0.30	0.375	0.30	0.329
0.148	0.30	0.133	0.30	0.139	0.30	0.161	0.30	0.090	0.20	0.119	0.20
0.089	0.20	0.066	0.20	0.063	0.20	0.072	0.20	0.025	0.10	0.035	0.10
0.048	0.10	0.023	0.10	0.015	0.10	0.018	0.10	0.00	0.00	0.00	0.00
0.034	0.00	0.009	0.00	0.00	0.00	0.00	0.00				

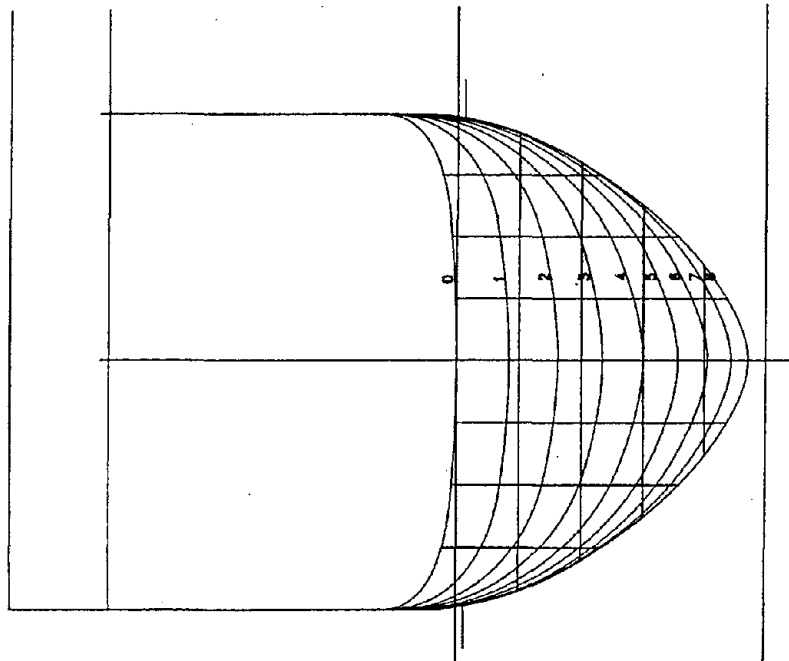
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z	y	z	y	z	y	z	y	z	y	x	z
2.00	0.848	2.00	0.724	2.00	0.584	2.00	0.426	2.00	0.250	15.435	2.00
1.80	0.805	1.80	0.678	1.80	0.537	1.80	0.382	1.80	0.213	15.00	1.525
1.50	0.739	1.50	0.610	1.50	0.472	1.50	0.324	1.50	0.166	14.50	1.018
1.20	0.658	1.20	0.536	1.20	0.407	1.20	0.269	1.20	0.123	14.00	0.600
0.90	0.557	0.90	0.450	0.90	0.336	1.00	0.232	0.90	0.082		
0.60	0.428	0.60	0.347	0.60	0.256	0.80	0.195	0.60	0.041		
0.30	0.280	0.30	0.224	0.30	0.161	0.60	0.155	0.317	0.00		
0.10	0.151	0.10	0.116	0.146	0.10	0.40	0.110				
0.00	0.00	0.00	0.00	0.002	0.00	0.20	0.059				
						0.049	0.00				

LINES PLANE DUT CATAMARAN MODEL 372

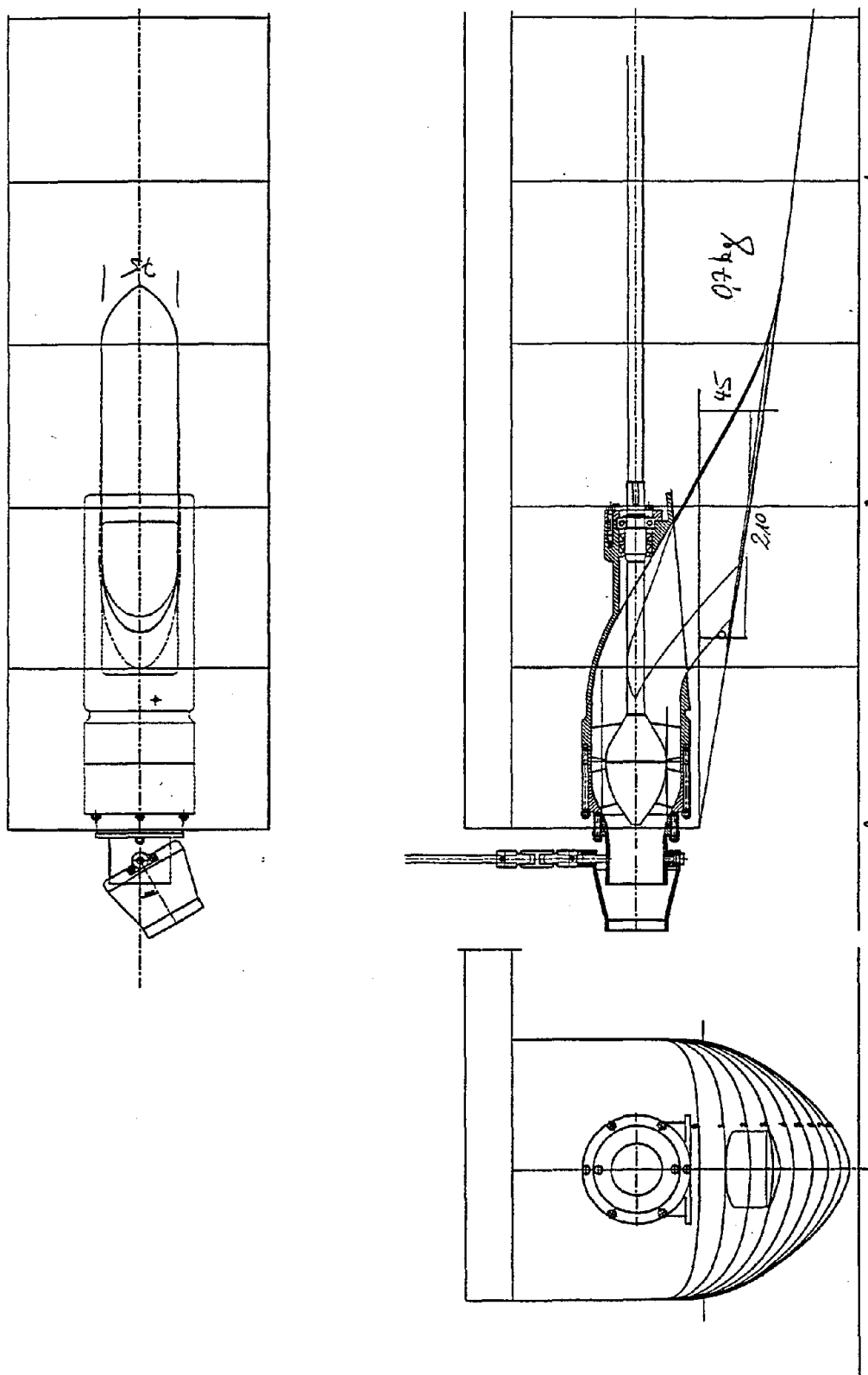
FORWARD

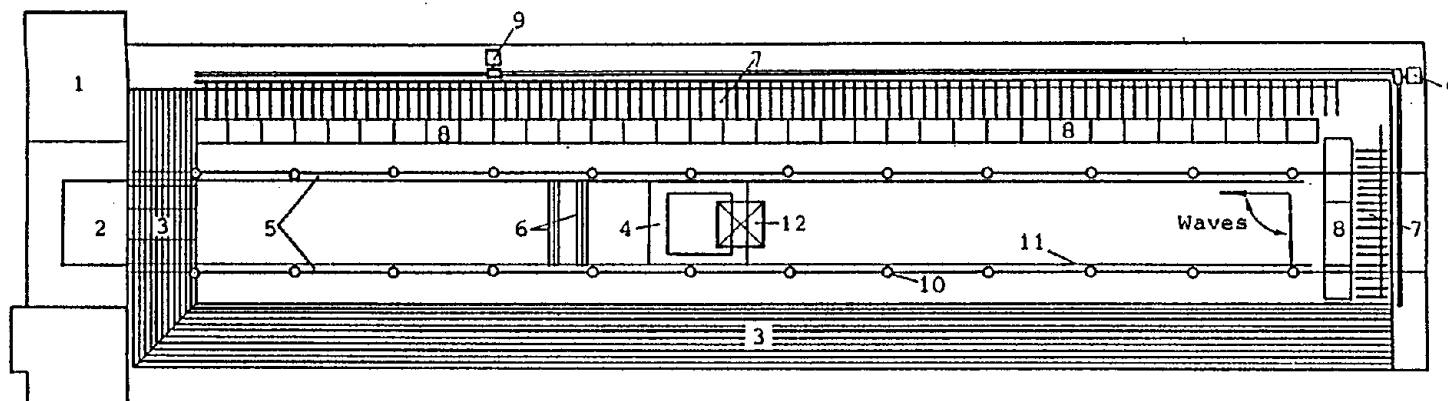


AFT



EXPERIMENTAL SET-UP, MODEL DETAILS



SEAKEEPING LABORATORY

- | | | | |
|---|---------------------|----|------------------------------------|
| 1 | Workshop | 7 | Wave generator |
| 2 | Working pond | 8 | Wire mesh packages |
| 3 | Beach | 9 | Wave generator motor |
| 4 | Carriage | 10 | Support piles |
| 5 | Wave guiding boards | 11 | Rails |
| 6 | Auxiliary carriage | 12 | Coverable pit
(water depth 6 m) |

SEAKEEPING BASIN

Length overall	100	m
Breadth overall	24	m
Breadth between the wave guiding boards	6.7	m
Water depth	2.5	m

In case of generating head waves guiding boards can be positioned vertically underneath the rails.

1.3 Test conditions

1.3.1 Test run overview

All experiments were carried out in regular waves. The tests were performed at three different forward speed which were Froude number 0.30, 0.60 and 0.75. The wave conditions at which the experiments are performed are presented in the test run overview, tables 1.2 to 1.4.

Marin test no.	Snelheid U [m/s]	Frequency ω_0 [rad/s]	Period T [s]	λ/L	Wave A [cm]
871501	1.602	3.589	1.750	1.60	1.73
871401	1.718	3.835	1.638	1.40	1.24
871302	1.623	4.137	1.518	1.20	1.45
871201	1.612	4.519	1.390	1.01	2.54
871102	1.706	4.763	1.319	0.91	2.61
871002	1.645	5.061	1.241	0.80	2.47
870602	1.596	5.890	1.067	0.59	1.49
872201	3.196	3.258	1.929	1.94	1.38
872101	3.354	3.434	1.830	1.74	1.41
872001	3.347	3.583	1.754	1.60	1.79
871901	3.287	3.849	1.632	1.39	1.95
871801	3.229	4.115	1.527	1.21	2.35
871701	3.240	4.497	1.397	1.02	2.56
871602	3.242	4.738	1.326	0.92	2.61
872901	4.090	3.278	1.917	1.91	1.38
872801	4.101	3.375	1.862	1.80	1.70
872701	4.095	3.544	1.773	1.64	1.45
872601	4.102	3.583	1.754	1.60	1.79
872501	4.037	3.818	1.646	1.41	2.10
872401	4.033	4.169	1.507	1.18	2.07
872301	4.002	4.538	1.385	1.00	2.41

Table 1.2: *Test run overview of experiments in 180 degrees*

Marin test no.	Snelheid U [m/s]	Frequency ω_0 [rad/s]	Period T [s]	λ/L	Wave A [cm]
874702	1.590	3.210	1.957	1.99	2.78
874401	1.615	3.470	1.811	1.71	2.81
874302	1.579	3.826	1.642	1.40	2.43
873701	1.520	4.155	1.512	1.19	2.54
873801	1.668	4.132	1.521	1.20	2.51
873601	1.556	4.473	1.405	1.03	2.33
873102	1.618	5.063	1.241	0.80	2.19
873001	1.598	5.827	1.078	0.61	2.19
874801	3.327	3.209	1.958	2.00	2.72
874501	3.185	3.462	1.815	1.71	2.80
874201	3.246	3.823	1.644	1.41	2.39
873901	3.251	4.193	1.498	1.17	2.03
873501	3.295	4.569	1.375	0.98	1.70
873201	3.327	5.033	1.248	0.81	2.23
874902	4.081	3.187	1.972	2.02	2.60
874601	4.031	3.467	1.812	1.71	2.75
874102	4.129	3.834	1.639	1.40	2.33
874001	4.010	4.131	1.521	1.20	2.41
873401	4.030	4.506	1.394	1.01	2.29
873301	4.022	5.064	1.241	0.80	2.10

Table 1.3: *Test run overview of experiments in 195 degrees*

Marin test no.	Snelheid U [m/s]	Frequency ω_0 [rad/s]	Period T [s]	λ/L	Wave A [cm]
875002	1.710	3.267	1.923	1.92	0.89
875102	1.612	3.193	1.968	2.02	1.53
875402	1.631	3.543	1.773	1.64	1.14
877002	1.710	3.652	1.720	1.54	2.41
875902	1.643	3.881	1.619	1.36	2.02
878401	1.620	4.055	1.549	1.25	2.32
876001	1.611	4.114	1.527	1.21	2.10
878901	1.623	4.204	1.495	1.16	2.20
876501	1.608	4.589	1.369	0.97	1.99
876602	1.661	5.076	1.238	0.80	2.72
876901	1.627	5.927	1.060	0.58	2.15
875202	3.371	3.177	1.977	2.04	1.52
875501	3.320	3.478	1.807	1.70	1.67
877101	3.243	3.650	1.721	1.54	2.35
875801	3.246	3.843	1.635	1.39	2.56
878501	3.259	4.099	1.533	1.22	2.10
876101	3.227	4.199	1.496	1.17	1.72
878801	3.237	4.280	1.468	1.12	1.79
876401	3.168	4.522	1.389	1.00	2.73
876702	3.393	5.173	1.483	0.77	1.98
875301	4.211	3.356	2.286	1.82	0.85
875601	3.934	3.470	2.211	1.71	1.71
877201	4.024	3.729	2.057	1.48	2.01
875702	4.184	3.822	2.007	1.41	2.51
878601	4.028	4.056	1.891	1.25	2.26
876201	4.061	4.117	1.864	1.21	2.04
878701	4.133	4.188	1.832	1.17	2.10
876301	3.975	4.509	1.702	1.01	2.70
876801	3.986	5.064	1.515	0.80	2.69

Table 1.4: Test run overview of experiments in 225 degrees

1.3.2 Instrumentation

The following instrumentation was used:

Vessel motions:

Roll and Pitch were determined using a Gyroscope, Surge, Sway, Heave and Yaw were measured using the optical tracking system (infrared). Seven different accelerations were measured, one (1) longitudinal, three (3) transverse, and three (3) in vertical direction.

Forces on the hull:

The forces between the two demi-hulls were measured on the centreline of the catamaran. Six force transducers were used in a measuring frame, see the experimental set-up for more detail (page 17). A seventh transducer was used between the two waterjet nozzles (one nozzle was connected with the auto-pilot).

Waves:

The wave profiles in the towing tank were measured using a wave prob and a resistance type wave height meter. The signal of Wave 1 (the wave prob) was measured in front of the vessel, see the experimental set-up for more detail (page 18).

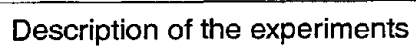
Measurements, Definitions and Notations:

Definitions and sign conventions are given on page 16.

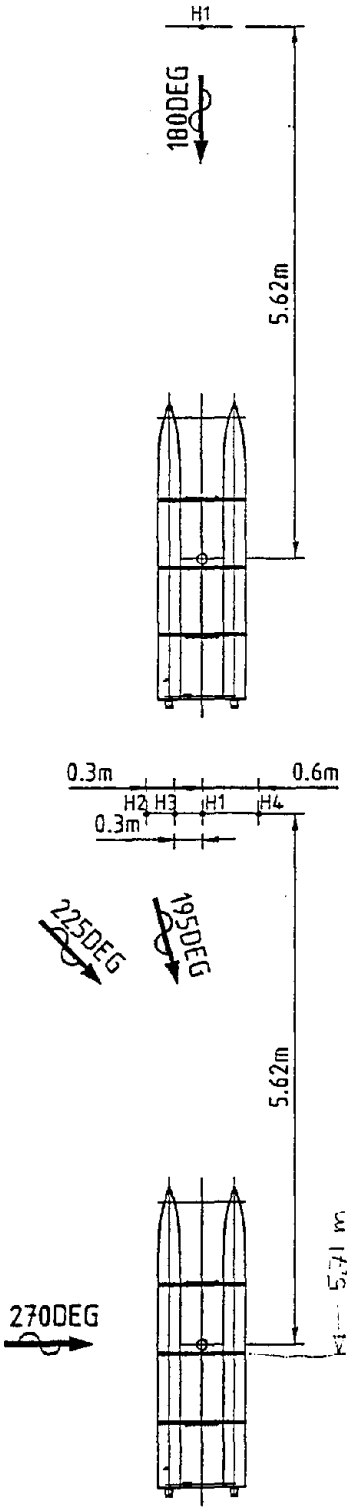
MEASUREMENTS, DEFINITIONS and NOTATIONS

QUANTITY	NOTATION	POSITIVE WHEN:	MEASURED BY:
VESSEL MOTIONS			
Roll	ROLL	Stb. down	Gyroscope
Pitch	PITCH	Bow down	Gyroscope
Yaw	YAW	Bow to port	Infrared system
Surge	SURGE	Vessel forward	Optical
Sway	SWAY	Vessel to port	tracking system
Heave	HEAVE	Vessel upwards	"
Longitudinal acceleration	AX COG	Vessel accel. forwards	Accelerometer
Transverse acceleration	AY COG	Vessel accel. to port	Accelerometer
Vertical acceleration	AZ COG	Vessel accel. upwards	Accelerometer
Transverse acceleration	AY PS FORE	Vessel accel. to port	Accelerometer
Vertical acceleration	AZ PS FORE	Vessel accel. upwards	Accelerometer
Transverse acceleration	AY PS AFT	Vessel accel. to port	Accelerometer
Vertical acceleration	AZ SB FORE	Vessel accel. upwards	Accelerometer
FORCES ON HULL			
Longitudinal force	FX	SB Hull fixed	
Transverse force fore	FY 1	PS Hull fore	Strain gauge
Transverse force mid.	FY 2	PS Hull to port	transducers
Transverse force aft	FY 3	"	
Vertical force fore	FZ 1	PS Hull up	
Vertical force aft	FZ 2	"	
Transverse force rod	FY ROD	PS hull to SB	Strain gauge tr.
NOZZLES			
P Nozzle starboard	PNOZZLE SB	Sailing ahead	Strain gauge
P Nozzle port side	PNOZZLE PS	Sailing ahead	transducers
RPM	RPM	Sailing ahead	Tachometer
Nozzle angle	NOZZLE ANG	Nozzle to port	
Velocity	SPEED	Forward	Tachometer
Wave 1	WAVE 1	Elevation upwards	Resistance type
Wave 2	WAVE 2		wave probe
Wave 3	WAVE 3		
Wave 4	WAVE 4		

Note : For the exact locations see Figures



EXPERIMENTAL SET-UP, DETAIL OF WAVE METERS



1.4 Calculations

The first harmonic components of all measured motions, accelerations and forces are given in the detailed list of results (page 47 to 116).

The forces in the centre of gravity are calculated as follows:

$$F_x = f_x$$

$$F_y = f_{y1} + f_{y2} + f_{y3} + f_{yrod}$$

$$F_z = f_{z1} + f_{z2}$$

$$M_x = -0.30 f_{y3} - 0.14 f_{yrod}$$

$$M_y = 0.722 f_{z2} - 0.722 f_{z1}$$

$$M_z = 0.722 f_{y1} - 0.722 f_{y3} - 1.39 f_{yrod}$$

The summation is carried out before the harmonic analysis is carried out to obtain the amplitude and phase of the force signals in the centre of gravity. The arm of the different force components can be found on page 17.

2 Results

2.1 Overview of presented results

An overview of all the measured data is included in this chapter. Since the experiments are not related to any full-scale vessel, only model data are presented.

The following results are presented:

Page	item
21	Table, motion results 180 degrees, $F_n = 0.30, 0.60, 0.75$
22	Table, motion results 195 degrees, $F_n = 0.30, 0.60, 0.75$
23	Table, motion results 225 degrees, $F_n = 0.30$
24	Table, motion results 225 degrees, $F_n = 0.60, 0.75$
25	Table, forces in cog, 180 degrees, $F_n = 0.30, 0.60, 0.75$
26	Table, forces in cog, 195 degrees, $F_n = 0.30, 0.60, 0.75$
27	Table, forces in cog, 225 degrees, $F_n = 0.30$
28	Table, forces in cog, 225 degrees, $F_n = 0.60, 0.75$
29-34	Figures, motion results 180 degrees
35-40	Figures, motion results 195 degrees
41-46	Figures, motion results 225 degrees
47-67	Harmonic analysis, experiments 180 degrees
68-87	Harmonic analysis, experiments 195 degrees
88-116	Harmonic analysis, experiments 225 degrees

MOTION RESULTS, 180 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	Surge [cm/cm] [deg]		Sway [cm/cm] [deg]		Heave [cm/cm] [deg]		Roll [deg/cm] [deg]		Pitch [deg/cm] [deg]		Yaw [deg/cm] [deg]	
Fn 0.30															
8715	3.589	5.692	1.595	0.216	63	0.185	29	0.865	10	0.048	41	0.964	240	0.097	235
8714	3.835	6.412	1.397	0.185	337	0.326	324	1.009	20	0.043	336	1.222	225	0.198	219
8713	4.137	6.970	1.200	0.111	124	0.240	102	1.330	21	0.030	305	1.362	203	0.185	200
8712	4.519	7.876	1.006	0.238	173	0.078	179	1.647	344	0.021	262	0.848	170	0.101	167
8711	4.763	8.710	0.906	0.176	180	0.116	226	1.444	297	0.012	251	0.714	163	0.111	141
8710	5.061	9.358	0.802	0.138	110	0.125	95	0.657	254	0.019	200	0.552	138	0.076	137
8706	5.890	11.537	0.592	0.016	128	0.181	307	0.153	36	0.018	47	0.094	62	0.008	61
Fn 0.60															
8722	3.258	6.714	1.935	0.202	164	0.106	340	1.820	15	0.025	96	1.028	221	0.122	210
8721	3.434	7.465	1.742	0.052	243	0.203	156	2.670	358	0.034	124	1.083	197	0.170	190
8720	3.583	7.964	1.601	0.281	153	0.027	87	2.802	312	0.077	40	0.865	160	0.136	150
8719	3.849	8.815	1.387	0.279	115	0.105	303	2.310	285	0.078	340	0.688	156	0.088	145
8718	4.115	9.689	1.214	0.202	59	0.033	82	1.254	241	0.041	266	0.567	132	0.064	127
8717	4.497	11.177	1.016	0.157	32	0.014	105	0.387	214	0.011	213	0.326	109	0.028	105
8716	4.738	12.159	0.915	0.092	145	0.031	11	0.154	209	0.002	188	0.211	103	0.019	107
Fn 0.75															
8729	3.278	7.758	1.912	0.464	167	0.031	201	2.729	343	0.085	95	0.781	185	0.104	168
8728	3.375	8.135	1.804	0.571	117	0.059	226	2.691	302	0.103	25	0.645	162	0.083	148
8727	3.544	8.789	1.636	0.696	69	0.071	71	2.776	311	0.110	14	0.674	184	0.100	167
8726	3.583	8.949	1.600	0.838	112	0.004	100	2.466	272	0.093	326	0.678	156	0.088	140
8725	3.818	9.819	1.409	0.191	121	0.029	93	1.380	236	0.045	262	0.542	134	0.054	125
8724	4.169	11.315	1.182	0.118	168	0.034	210	0.673	236	0.017	234	0.392	137	0.045	127
8723	4.538	12.938	0.998	0.045	100	0.038	3	0.218	219	0.007	191	0.207	116	0.019	108

MOTION RESULTS, 195 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	Surge [cm/cm] [deg]		Sway [cm/cm] [deg]		Heave [cm/cm] [deg]		Roll [deg/cm] [deg]		Pitch [deg/cm] [deg]		Yaw [deg/cm] [deg]	
Fn 0.30															
8747	3.210	4.825	1.994	0.337	111	0.081	58	0.966	4	0.124	53	0.735	253	0.074	201
8744	3.470	5.386	1.706	0.151	78	0.103	120	0.900	9	0.164	41	0.886	246	0.072	210
8743	3.826	6.104	1.404	0.110	90	0.087	140	0.897	18	0.157	11	1.118	235	0.076	216
8737	4.155	6.740	1.190	0.189	150	0.106	132	1.230	34	0.154	338	1.550	227	0.135	222
8738	4.132	6.938	1.203	0.200	178	0.099	120	1.374	19	0.086	329	1.355	203	0.136	202
8736	4.473	7.541	1.027	0.262	127	0.130	317	1.565	344	0.043	242	0.990	160	0.174	166
8731	5.063	9.149	0.802	0.095	128	0.044	4	0.933	266	0.075	157	0.665	148	0.104	168
8730	5.827	11.172	0.605	0.005	81	0.020	133	0.114	20	0.110	72	0.103	68	0.043	162
Fn 0.60															
8748	3.209	6.587	1.995	0.109	157	0.086	83	2.115	3	0.101	23	1.112	207	0.140	197
8745	3.462	7.223	1.714	0.299	157	0.079	97	2.725	335	0.119	11	1.115	171	0.137	160
8742	3.823	8.492	1.406	0.364	105	0.049	136	2.419	284	0.140	348	0.690	150	0.068	129
8739	4.193	9.819	1.169	0.164	47	0.068	110	1.307	266	0.097	330	0.561	157	0.051	165
8735	4.569	11.348	0.984	0.063	110	0.124	78	0.475	241	0.045	282	0.358	140	0.052	157
8732	5.033	13.339	0.811	0.040	347	0.101	335	0.045	228	0.006	136	0.119	94	0.032	138
Fn 0.75															
8749	3.187	7.273	2.023	0.568	119	0.090	93	2.999	325	0.145	24	0.822	164	0.091	138
8746	3.467	8.242	1.710	0.412	126	0.094	100	2.735	291	0.196	341	0.619	161	0.053	128
8741	3.834	9.819	1.397	0.845	105	0.077	96	1.534	247	0.116	289	0.586	143	0.035	136
8740	4.131	10.872	1.204	0.104	174	0.041	109	0.748	229	0.052	286	0.414	130	0.020	136
8734	4.506	12.568	1.012	0.185	207	0.067	46	0.246	218	0.031	225	0.227	112	0.038	143
8733	5.064	15.226	0.801	0.049	314	0.036	343	0.033	252	0.021	138	0.080	106	0.026	151

MOTION RESULTS, 225 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	Surge [cm/cm] [deg]		Sway [cm/cm] [deg]		Heave [cm/cm] [deg]		Roll [deg/cm][deg]		Pitch [deg/cm][deg]		Yaw [deg/cm][deg]	
Fn 0.30															
8750	3.267	4.580	1.925	1.145	12	0.432	148	0.882	22	0.505	93	0.456	281	0.115	299
8751	3.193	4.379	2.016	0.206	109	0.368	99	0.929	6	0.531	82	0.466	266	0.177	170
8754	3.543	5.022	1.637	0.170	138	0.278	110	0.985	6	0.775	54	0.651	257	0.211	171
8770	3.652	5.301	1.540	0.184	72	0.233	125	0.941	5	0.565	42	0.712	251	0.126	162
8759	3.881	5.667	1.364	0.324	115	0.115	177	0.867	17	0.482	65	0.767	256	0.095	189
8784	4.055	5.979	1.249	0.194	71	0.363	177	0.926	3	0.782	34	0.949	229	0.151	177
8760	4.114	6.083	1.214	0.092	142	0.446	152	0.843	12	0.901	26	0.889	234	0.102	159
8789	4.204	6.278	1.162	0.169	90	0.374	153	0.926	15	0.825	9	0.989	230	0.095	179
8765	4.589	7.034	0.976	0.036	160	0.271	135	1.157	21	0.282	349	1.125	212	0.131	202
8766	5.076	8.167	0.797	0.133	137	0.229	37	1.412	307	0.110	347	0.572	149	0.132	152
8769	5.927	10.057	0.585	0.013	105	0.020	31	0.172	231	0.157	94	0.267	113	0.076	173

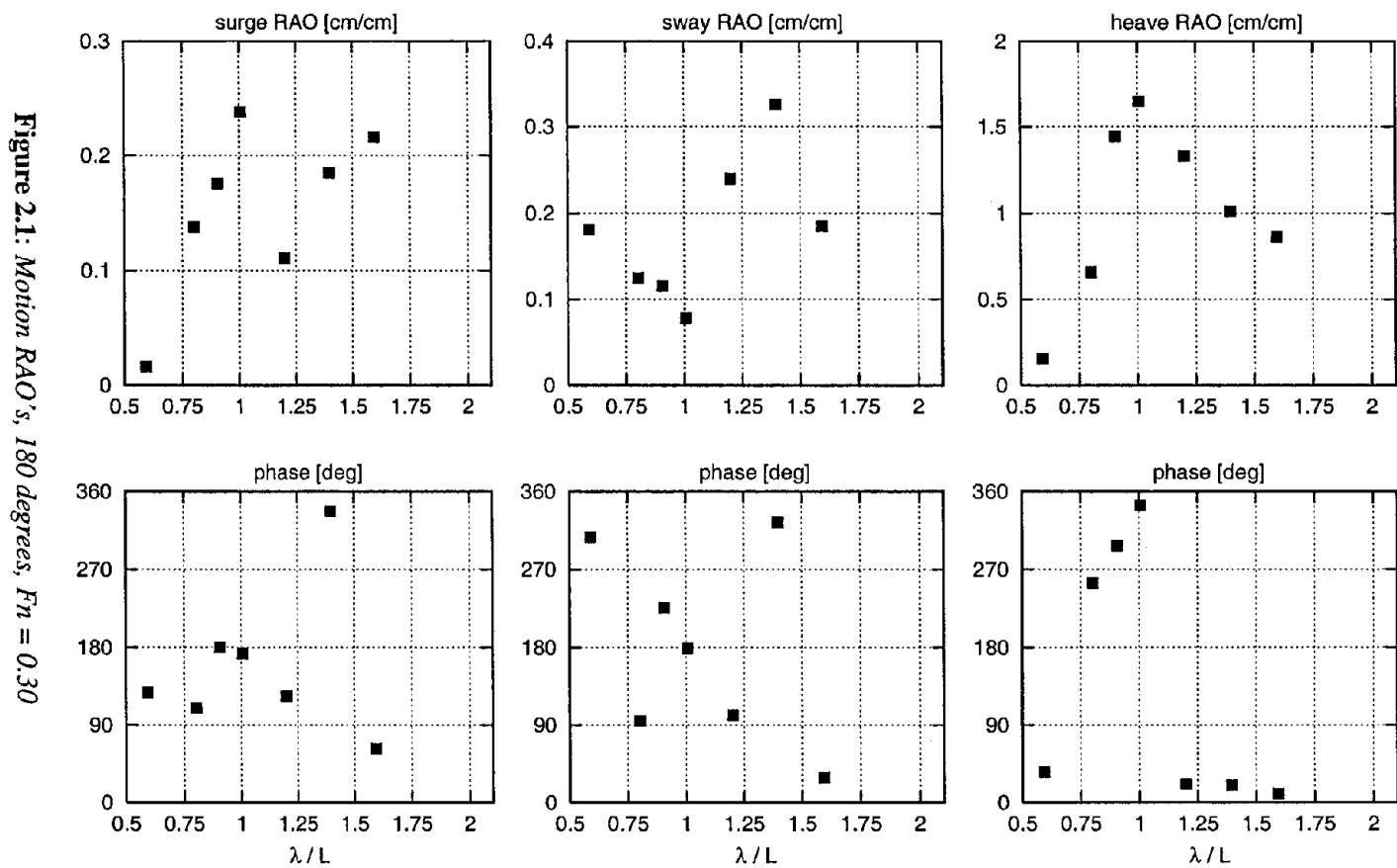
MOTION RESULTS, 225 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	Surge [cm/cm] [deg]		Sway [cm/cm] [deg]		Heave [cm/cm] [deg]		Roll [deg/cm][deg]		Pitch [deg/cm][deg]		Yaw [deg/cm][deg]	
Fn 0.60															
8752	3.177	5.636	2.035	0.359	253	0.560	85	1.268	3	0.549	57	0.648	231	0.150	186
8755	3.478	6.380	1.699	0.193	228	0.100	221	1.577	357	0.635	24	0.819	208	0.160	170
8771	3.650	6.772	1.542	0.402	192	0.217	108	1.993	360	0.505	18	0.957	204	0.108	167
8758	3.843	7.307	1.391	0.461	169	0.174	104	2.216	343	0.403	24	0.862	181	0.155	166
8785	4.099	8.057	1.223	0.309	119	0.448	165	2.654	309	0.681	16	0.774	155	0.149	136
8761	4.199	8.308	1.165	0.457	172	0.415	138	2.551	331	0.745	23	0.712	181	0.047	140
8788	4.280	8.563	1.122	0.348	153	0.477	159	2.170	302	0.687	354	0.576	161	0.078	150
8764	4.522	9.196	1.005	0.406	78	0.151	104	1.656	258	0.335	326	0.561	145	0.017	206
8767	5.173	11.730	0.768	0.040	79	0.229	51	0.326	234	0.097	287	0.240	129	0.074	170
8753	3.356	6.781	1.825	1.015	216	0.226	165	1.974	55	0.745	92	0.869	263	0.133	202
Fn 0.75															
8756	3.470	6.891	1.707	0.254	155	0.027	163	2.212	344	0.704	11	0.810	183	0.144	133
8772	3.729	7.772	1.477	0.753	152	0.254	112	2.738	340	0.558	6	0.673	184	0.072	124
8757	3.822	8.233	1.407	0.237	143	0.183	109	2.491	301	0.500	356	0.533	170	0.055	141
8786	4.056	8.837	1.249	0.361	118	0.364	81	2.190	273	0.600	338	0.565	160	0.025	288
8762	4.117	9.089	1.212	0.281	92	0.366	101	1.961	271	0.650	327	0.567	162	0.048	204
8787	4.188	9.426	1.171	0.764	90	0.208	60	1.508	246	0.483	296	0.489	140	0.101	160
8763	4.509	10.344	1.011	0.078	319	0.104	74	0.883	233	0.210	299	0.403	135	0.037	178
8768	5.064	12.444	0.801	0.058	72	0.290	66	0.228	222	0.061	255	0.198	119	0.052	162

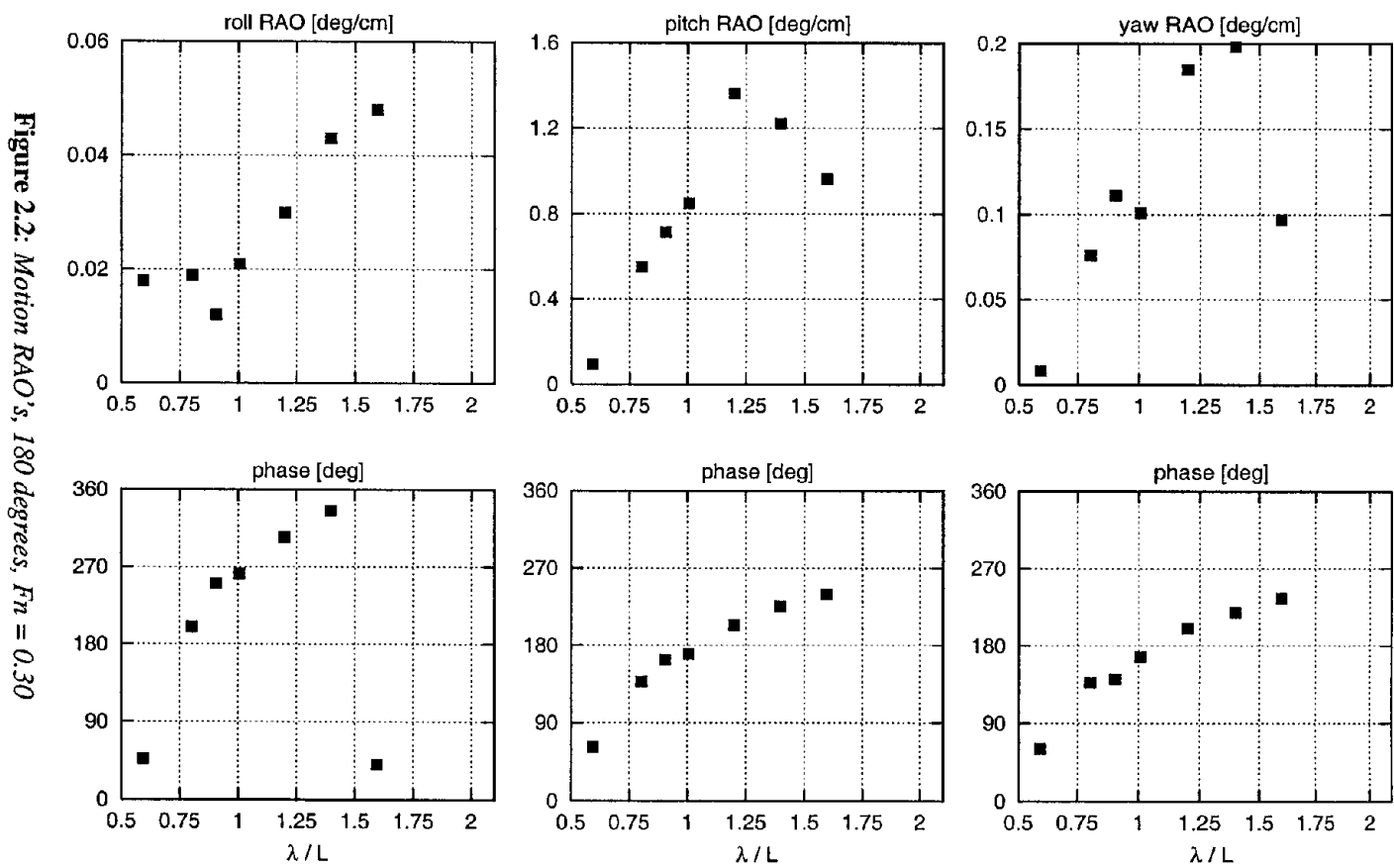
FORCES IN COG, 180 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	FX [N]	[deg]	FY [N]	[deg]	FZ [N]	[deg]	MX [Ncm]	[deg]	MY [Ncm]	[deg]	MZ [Ncm]	[deg]
Fn 0.30															
8715	3.590	5.691	1.594	0.318	41	3.295	248	0.236	204	0.788	226	0.367	256	1.555	145
8714	3.831	6.398	1.400	0.491	21	6.526	209	0.334	126	1.465	197	1.224	244	2.441	72
8713	4.134	6.949	1.202	0.564	2	10.008	168	0.345	99	2.163	166	1.743	208	3.853	27
8712	4.520	7.861	1.006	0.425	323	11.980	85	0.338	63	2.193	97	1.228	165	5.947	270
8711	4.760	8.689	0.907	0.241	295	10.390	11	0.189	17	1.607	38	1.141	153	6.957	185
8710	5.068	9.356	0.800	0.137	352	7.798	318	0.133	2	1.063	338	0.955	121	5.345	134
8706	5.872	11.468	0.596	0.188	3	2.832	239	0.123	216	0.587	231	0.183	111	1.811	41
Fn 0.60															
8722	3.206	6.545	1.999	0.950	14	3.412	191	0.199	72	1.040	184	1.256	196	0.328	9
8721	3.378	7.260	1.801	1.308	359	3.610	166	0.454	6	1.581	165	1.469	177	0.896	239
8720	3.590	7.963	1.594	1.256	329	3.620	135	0.356	271	1.957	135	1.721	151	1.982	187
8719	3.831	8.727	1.400	0.982	300	2.829	110	0.313	174	1.903	105	1.786	133	2.624	142
8718	4.134	9.701	1.202	0.650	283	1.773	95	0.200	84	1.264	76	1.363	111	1.761	102
8717	4.520	11.196	1.006	0.331	275	1.098	74	0.173	83	0.567	59	0.937	68	0.255	100
8716	4.760	12.135	0.907	0.277	289	1.075	74	0.125	72	0.330	59	0.515	66	0.396	170
Fn 0.75															
8729	3.206	7.455	1.999	1.011	344	3.648	163	0.603	288	1.788	153	1.563	156	0.968	179
8728	3.378	8.055	1.801	1.088	333	4.117	143	0.554	223	2.093	132	1.638	142	1.502	154
8727	3.471	8.394	1.705	1.124	329	3.953	134	0.583	193	2.090	120	1.788	129	1.116	144
8726	3.590	8.944	1.594	1.250	314	4.569	119	0.483	152	2.324	102	2.061	119	1.291	123
8725	3.831	9.818	1.400	0.887	283	3.698	90	0.202	121	1.699	72	1.348	95	0.793	113
8724	4.134	11.121	1.202	0.696	263	2.634	70	0.099	115	1.046	55	0.948	95	0.666	155
8723	4.520	12.723	1.006	0.419	267	1.822	51	0.132	48	0.538	45	0.575	85	0.904	167

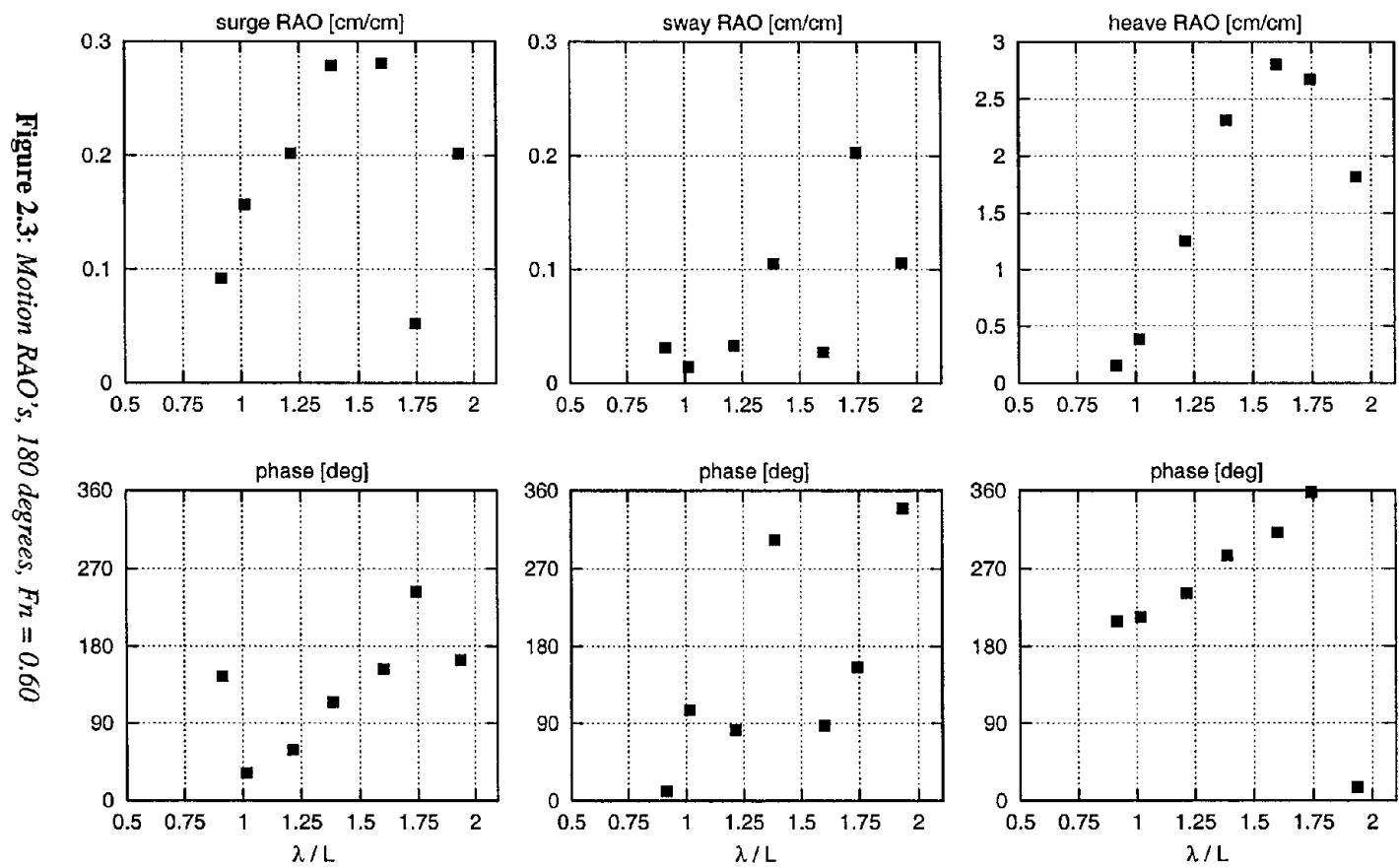
FORCES IN COG, 195 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	FX [N]	[deg]	FY [N]	[deg]	FZ [N]	[deg]	MX [Ncm]	[deg]	MY [Ncm]	[deg]	MZ [Ncm]	[deg]
Fn 0.30															
8747	3.206	4.807	1.999	0.528	360	2.095	289	0.328	193	0.400	258	1.181	332	1.080	190
8744	3.471	5.380	1.705	0.717	3	2.934	268	0.676	187	0.625	243	1.558	338	1.396	162
8743	3.831	6.085	1.400	0.772	9	4.377	236	1.022	168	1.032	218	1.697	335	2.236	122
8737	4.134	6.641	1.202	1.047	16	7.294	201	1.433	138	1.676	193	1.766	337	3.187	79
8738	4.134	6.935	1.202	1.133	10	9.446	169	1.298	131	2.021	167	0.999	333	3.569	15
8736	4.520	7.543	1.006	1.579	355	11.641	111	1.478	89	2.262	118	2.300	329	4.601	309
8731	5.068	9.095	0.800	1.545	21	9.591	335	0.303	28	1.373	357	2.721	9	7.174	146
8730	5.872	11.220	0.596	1.239	55	3.164	236	0.814	263	0.600	232	0.148	134	1.858	73
Fn 0.60															
8748	3.206	6.545	1.999	1.264	0	2.921	195	0.605	107	1.112	185	0.989	257	0.584	330
8745	3.471	7.213	1.705	1.479	340	4.059	156	0.593	109	1.727	158	0.740	185	1.543	231
8742	3.831	8.473	1.400	1.092	325	2.757	106	0.815	158	1.852	109	1.309	89	3.040	148
8739	4.134	9.582	1.202	0.991	344	1.647	81	1.219	137	1.334	75	2.552	59	2.471	106
8735	4.520	11.010	1.006	1.774	345	1.019	46	1.265	102	0.716	48	4.167	14	1.158	97
8732	5.068	13.318	0.800	1.858	1	0.950	354	0.643	101	0.182	15	3.783	10	0.860	111
Fn 0.75															
8749	3.206	7.306	1.999	1.212	326	2.326	160	0.158	103	1.640	151	0.592	165	0.164	309
8746	3.471	8.213	1.705	1.129	312	3.246	121	0.814	118	2.106	115	1.503	100	0.769	104
8741	3.831	9.666	1.400	1.426	325	2.817	82	0.989	109	1.630	70	2.250	66	1.115	65
8740	4.134	10.792	1.202	1.180	341	1.998	56	1.078	128	1.069	52	2.935	54	0.894	76
8734	4.520	12.516	1.006	1.685	339	1.551	56	1.230	86	0.631	51	4.432	11	1.113	96
8733	5.068	15.091	0.800	1.754	358	0.631	23	0.375	81	0.203	54	3.988	1	1.479	123

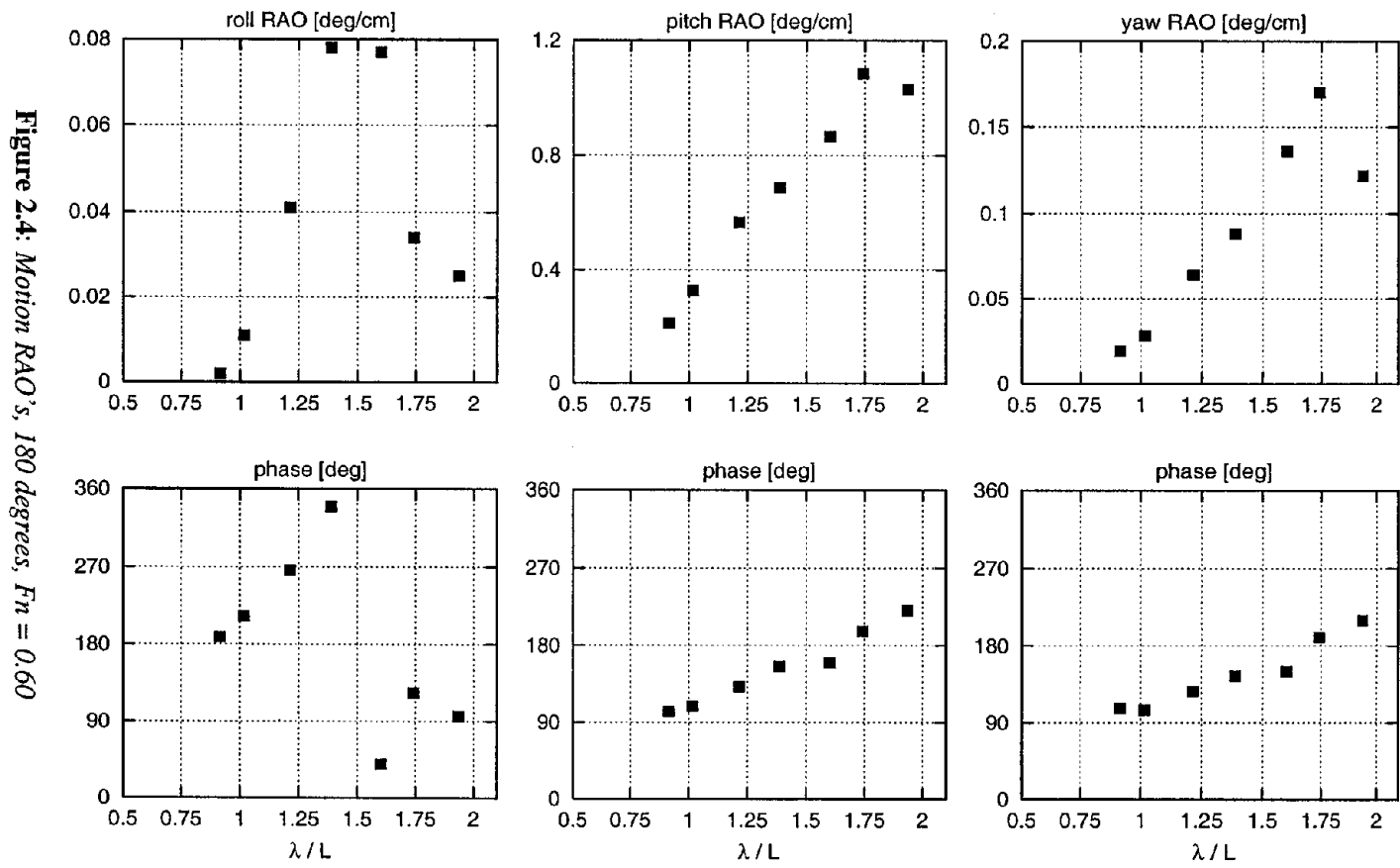
FORCES IN COG, 225 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	FX [N]	[deg]	FY [N]	[deg]	FZ [N]	[deg]	MX [Ncm]	[deg]	MY [Ncm]	[deg]	MZ [Ncm]	[deg]
Fn 0.30															
8750	3.206	4.442	1.999	1.457	16	3.639	346	1.007	220	0.543	338	3.984	13	0.163	216
8751	3.206	4.363	1.999	1.488	11	3.523	344	0.915	225	0.554	336	3.884	10	0.441	239
8754	3.471	4.888	1.705	2.357	345	4.545	320	2.006	187	0.703	311	5.944	343	0.435	122
8770	3.653	5.277	1.540	1.934	351	5.236	319	2.055	190	0.796	307	5.053	350	1.184	108
8759	3.831	5.549	1.400	1.746	14	5.205	318	2.180	209	0.784	305	4.560	15	1.812	112
8784	4.054	5.969	1.250	3.281	8	5.802	289	4.695	187	0.925	270	9.103	10	2.991	86
8760	4.134	6.097	1.202	3.853	9	4.983	296	5.854	185	0.776	273	11.631	11	3.369	96
8789	4.217	6.277	1.155	3.841	358	4.673	287	6.018	165	0.736	259	12.047	355	4.260	89
8765	4.520	6.879	1.006	2.549	353	3.619	189	3.596	128	0.956	170	7.276	340	6.376	46
8766	5.068	8.166	0.800	2.939	334	10.356	17	2.378	86	1.810	35	7.433	325	2.227	142
8769	5.872	9.973	0.596	3.431	21	4.248	275	1.007	196	0.642	323	7.470	359	10.108	57
Fn 0.60															
8752	3.206	5.653	1.999	2.037	6	2.149	316	1.486	186	0.285	271	4.024	351	1.058	24
8755	3.471	6.374	1.705	3.123	341	2.268	275	2.730	147	0.553	221	5.666	321	0.924	18
8771	3.653	6.764	1.540	2.498	347	1.742	293	2.532	149	0.634	208	3.633	317	1.091	2
8758	3.831	7.272	1.400	2.512	350	0.643	327	2.168	152	0.761	165	2.248	332	0.548	305
8785	4.054	7.920	1.250	3.461	345	1.675	358	4.312	151	1.201	110	3.920	19	1.505	121
8761	4.134	8.055	1.202	4.050	0	2.619	34	5.946	152	1.374	120	5.680	21	2.577	140
8788	4.217	8.352	1.155	4.267	337	2.967	9	5.930	126	1.387	90	5.643	5	2.968	108
8764	4.520	9.191	1.006	2.843	1	4.420	356	3.571	122	1.609	51	6.134	27	5.568	89
8767	5.068	11.365	0.800	3.828	328	4.929	325	3.052	77	1.209	352	9.562	351	5.372	56

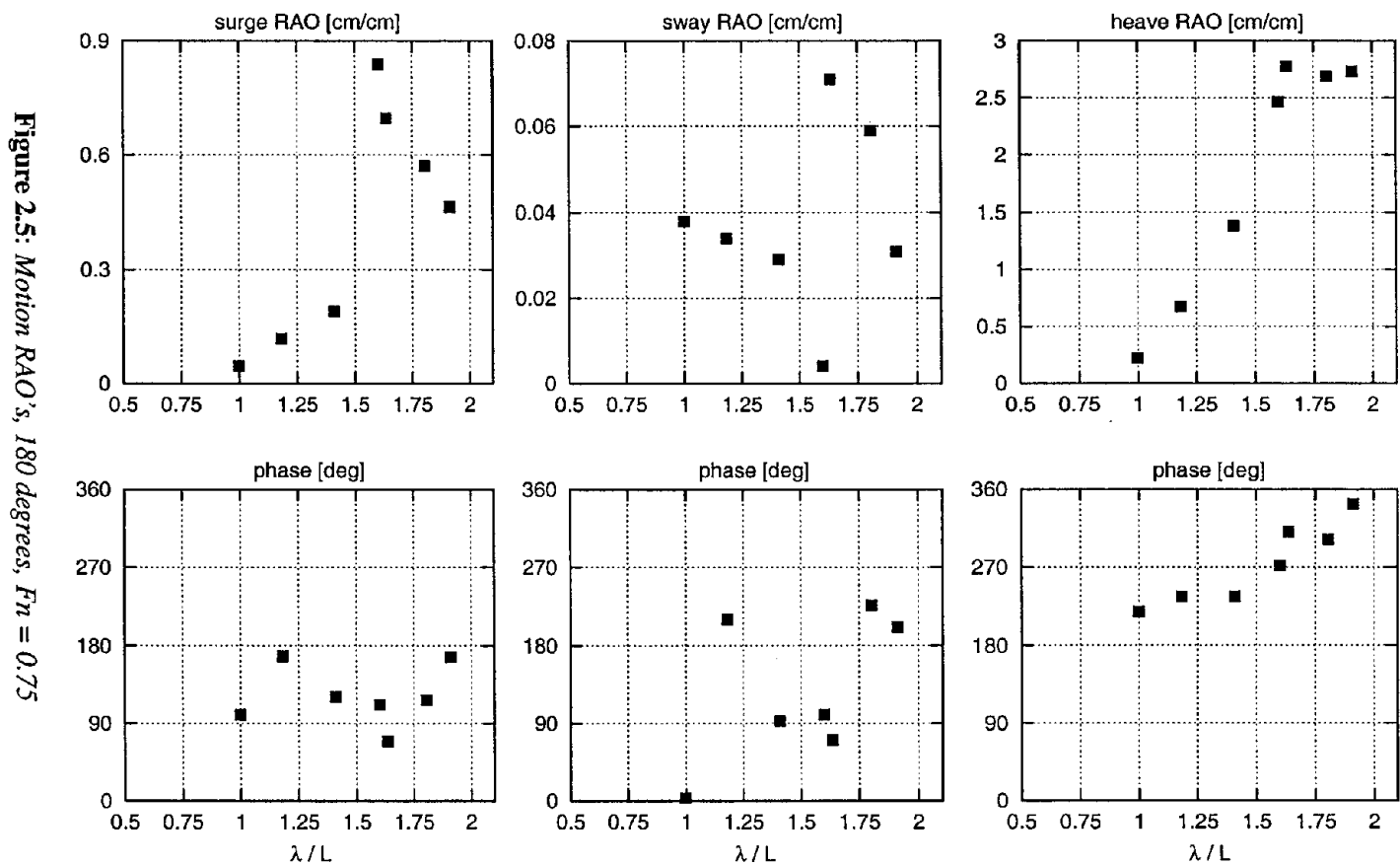
FORCES IN COG, 225 DEGREES															
Test no.	ω_0 [rad/s]	ω_e [rad/s]	λ/L	FX [N]	[deg]	FY [N]	[deg]	FZ [N]	[deg]	MX [Ncm]	[deg]	MY [Ncm]	[deg]	MZ [Ncm]	[deg]
Fn 0.75															
8753	3.206	6.258	1.999	2.813	9	2.353	310	2.285	189	0.512	242	3.090	3	0.826	340
8756	3.471	6.890	1.705	3.873	335	2.085	284	3.659	143	0.729	190	3.899	332	0.853	349
8772	3.653	7.540	1.540	2.907	327	1.150	343	3.401	129	1.011	144	1.892	349	0.940	65
8757	3.831	8.231	1.400	2.853	350	1.252	19	3.468	146	1.394	113	3.479	58	1.934	65
8786	4.054	8.850	1.250	3.705	352	1.910	9	5.623	135	1.534	82	7.423	44	2.891	74
8762	4.134	9.050	1.202	4.826	350	2.157	39	6.924	128	1.725	85	9.165	41	3.697	73
8787	4.217	9.456	1.155	5.240	332	3.760	16	6.642	103	1.764	52	10.498	18	3.529	46
8763	4.520	10.276	1.006	3.830	348	4.586	9	3.384	113	1.584	34	7.604	26	3.862	72
8768	5.068	12.442	0.800	3.499	342	4.993	353	2.299	91	1.200	17	9.494	6	5.534	82

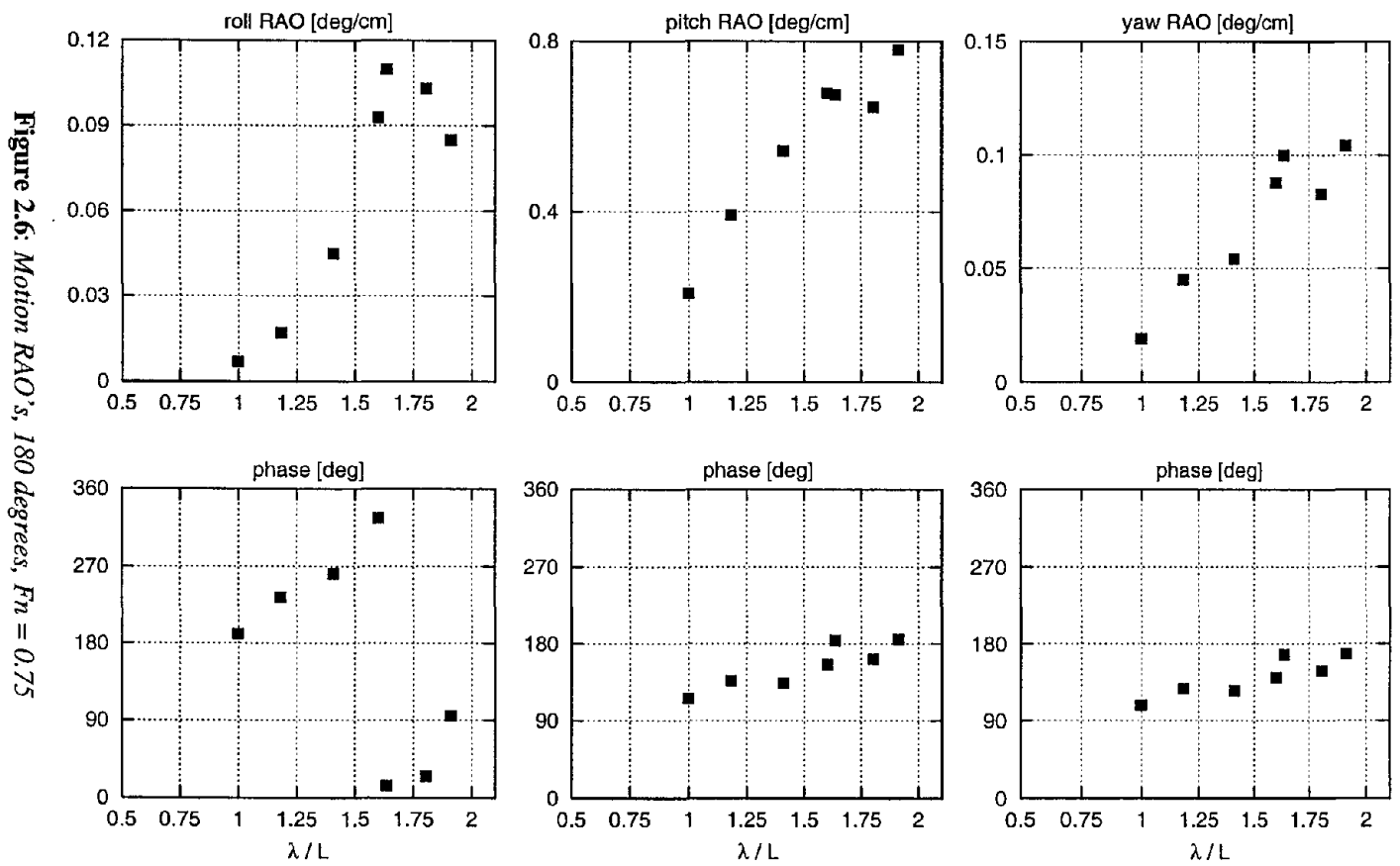


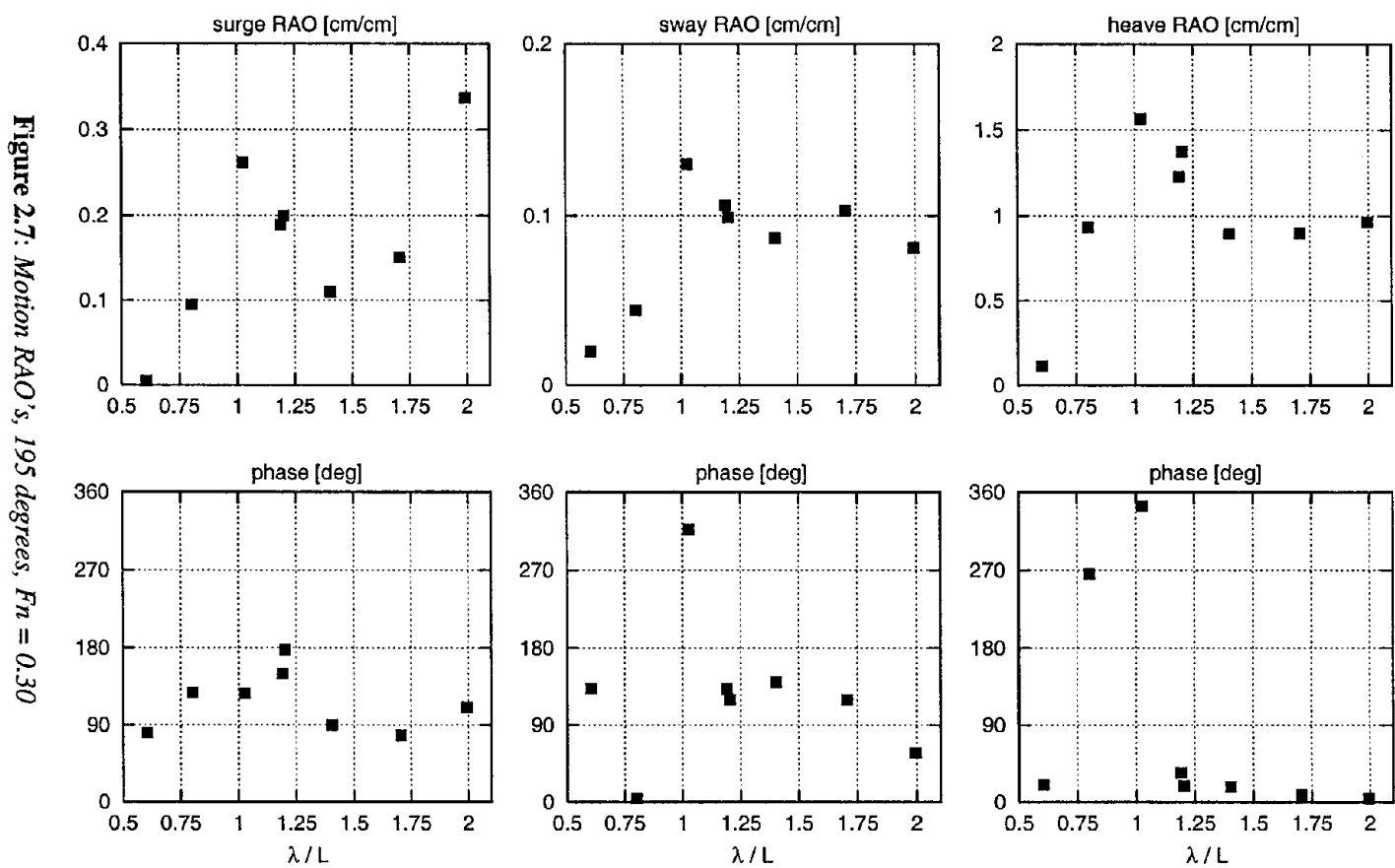


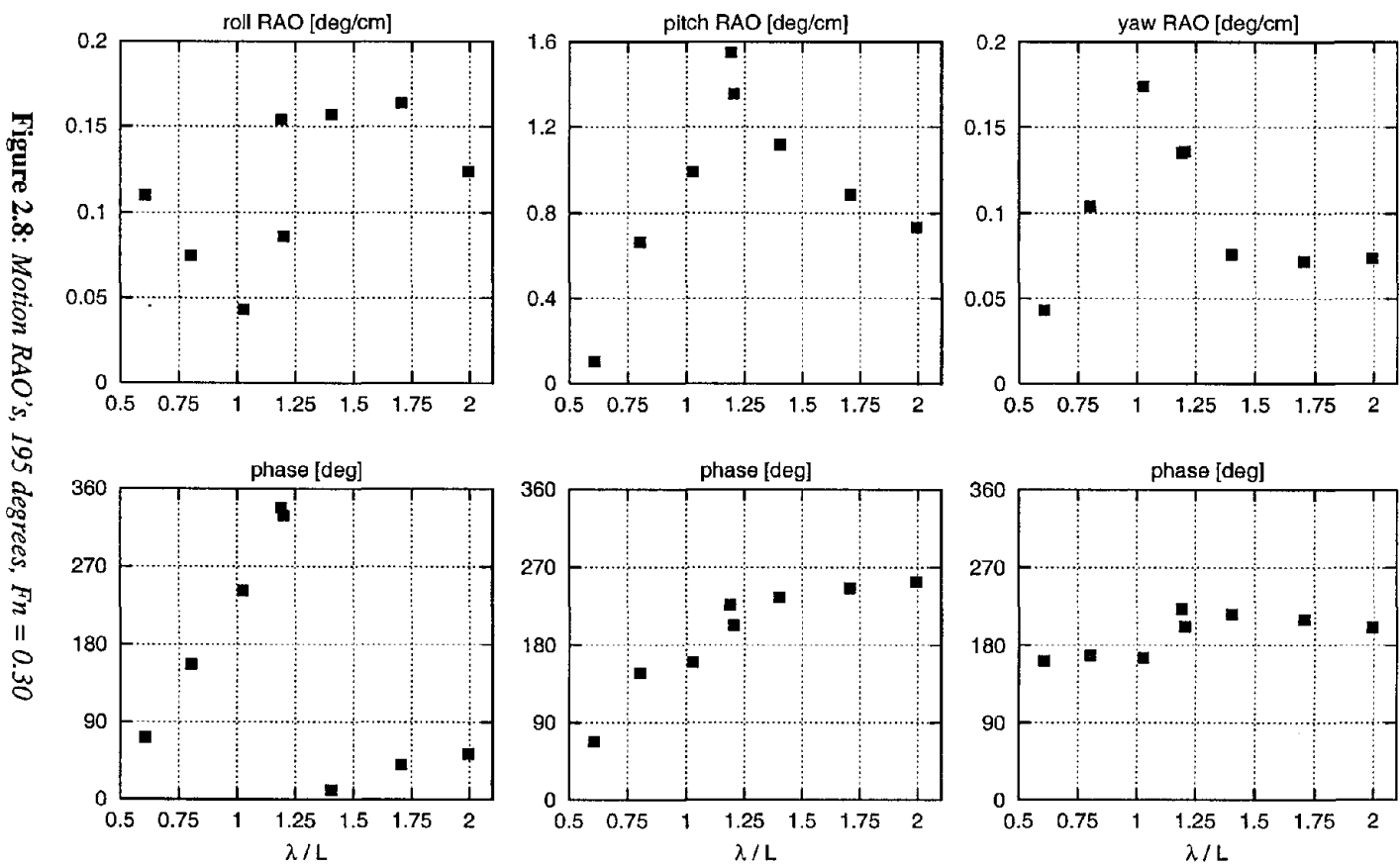


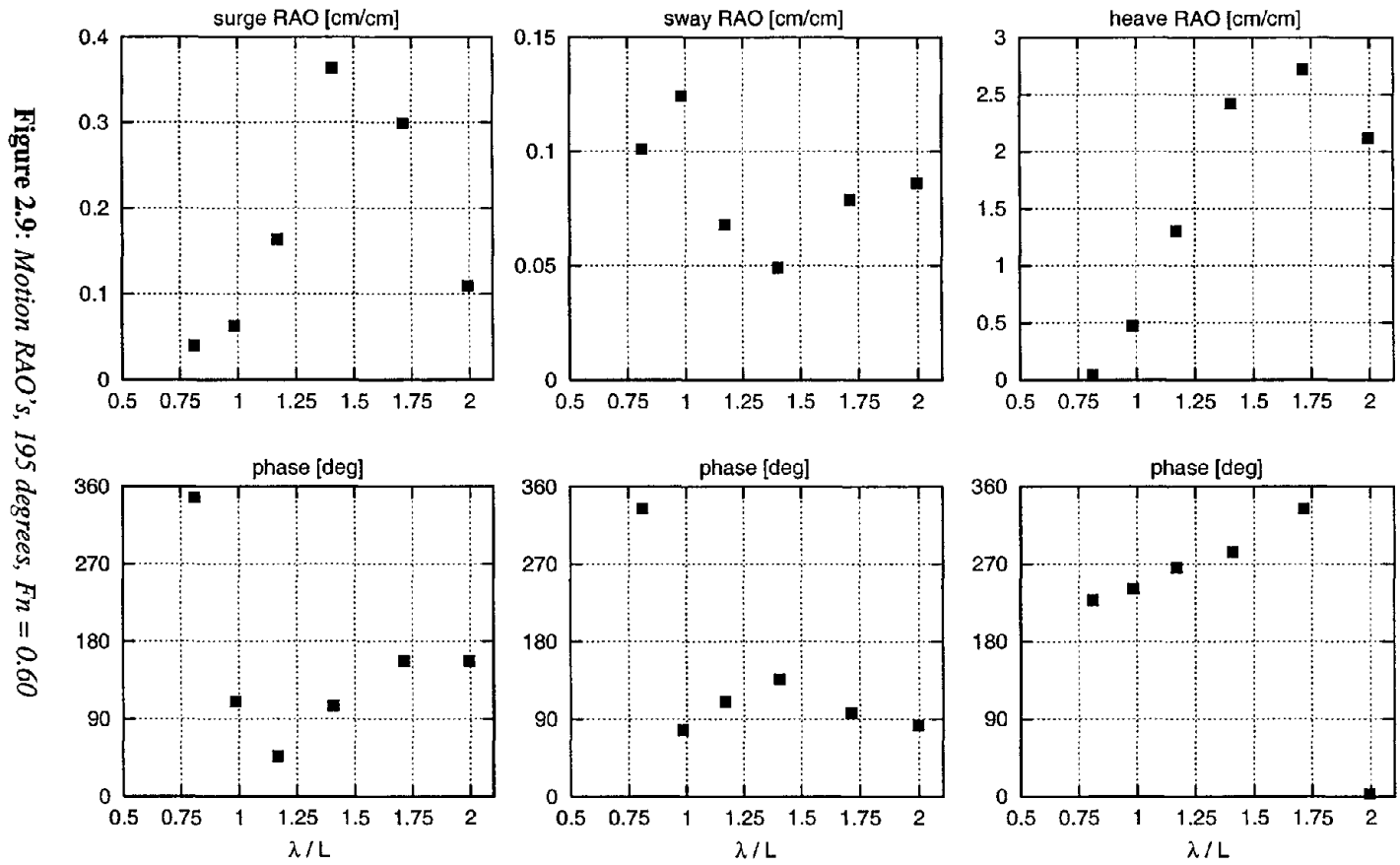


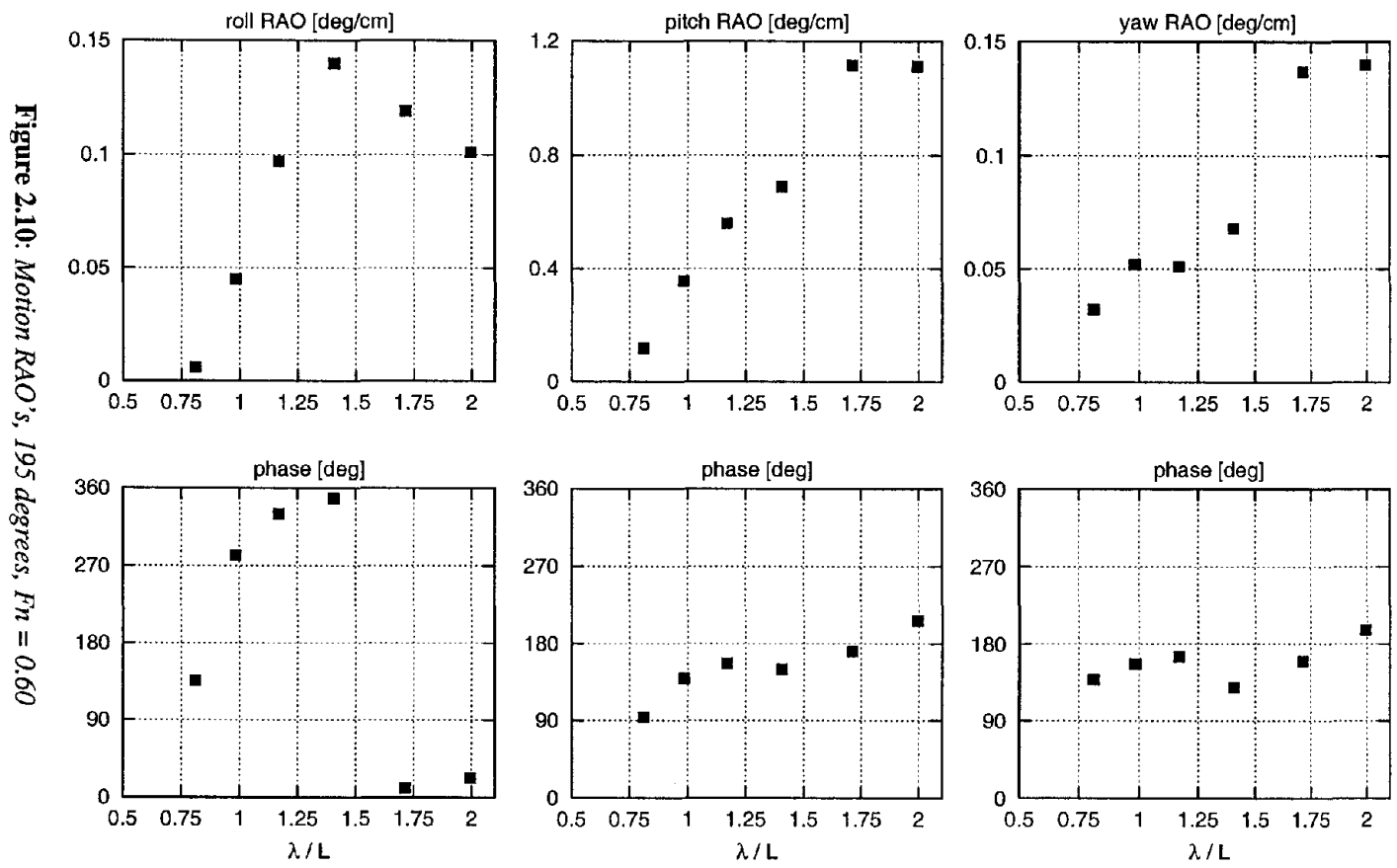


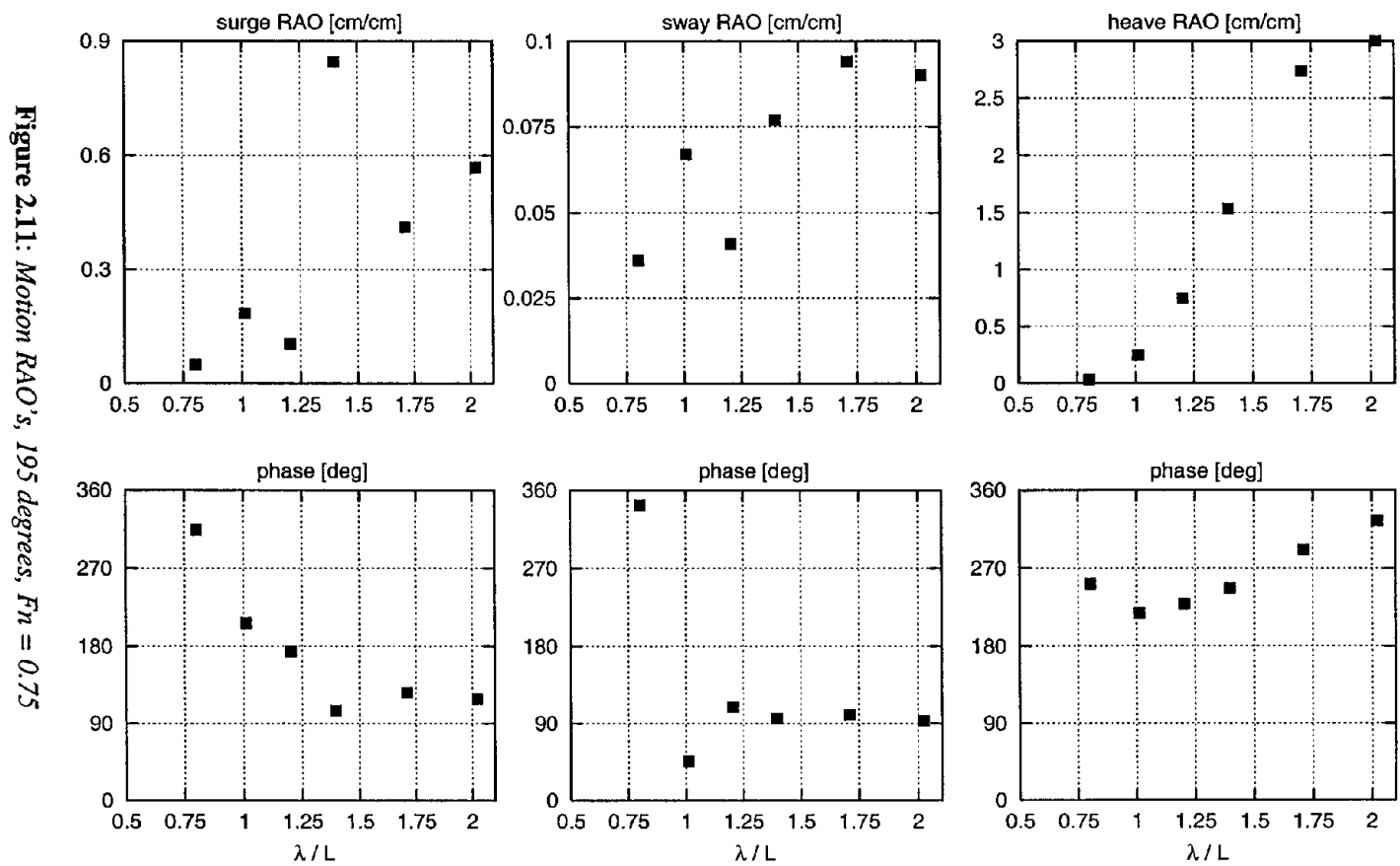


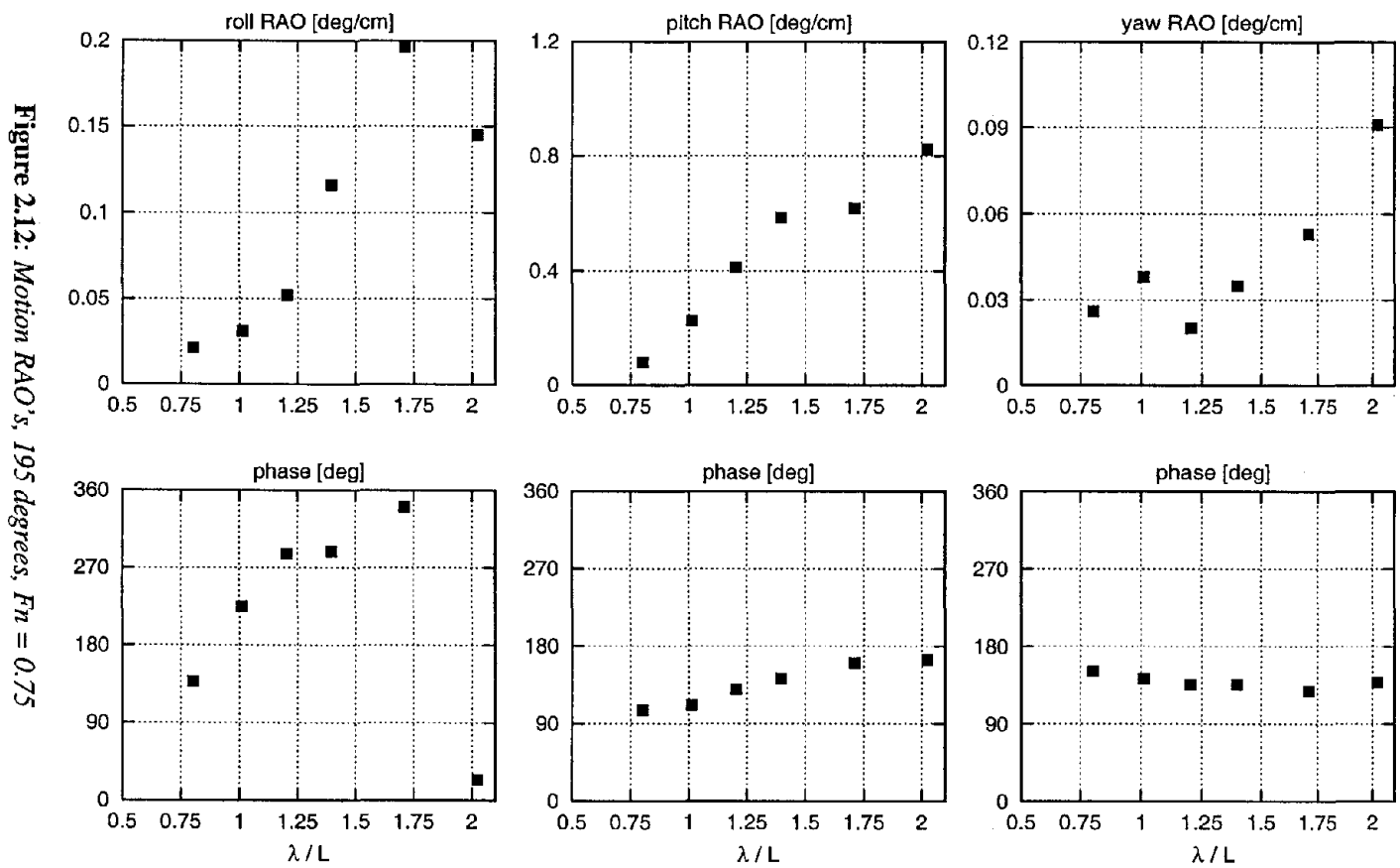


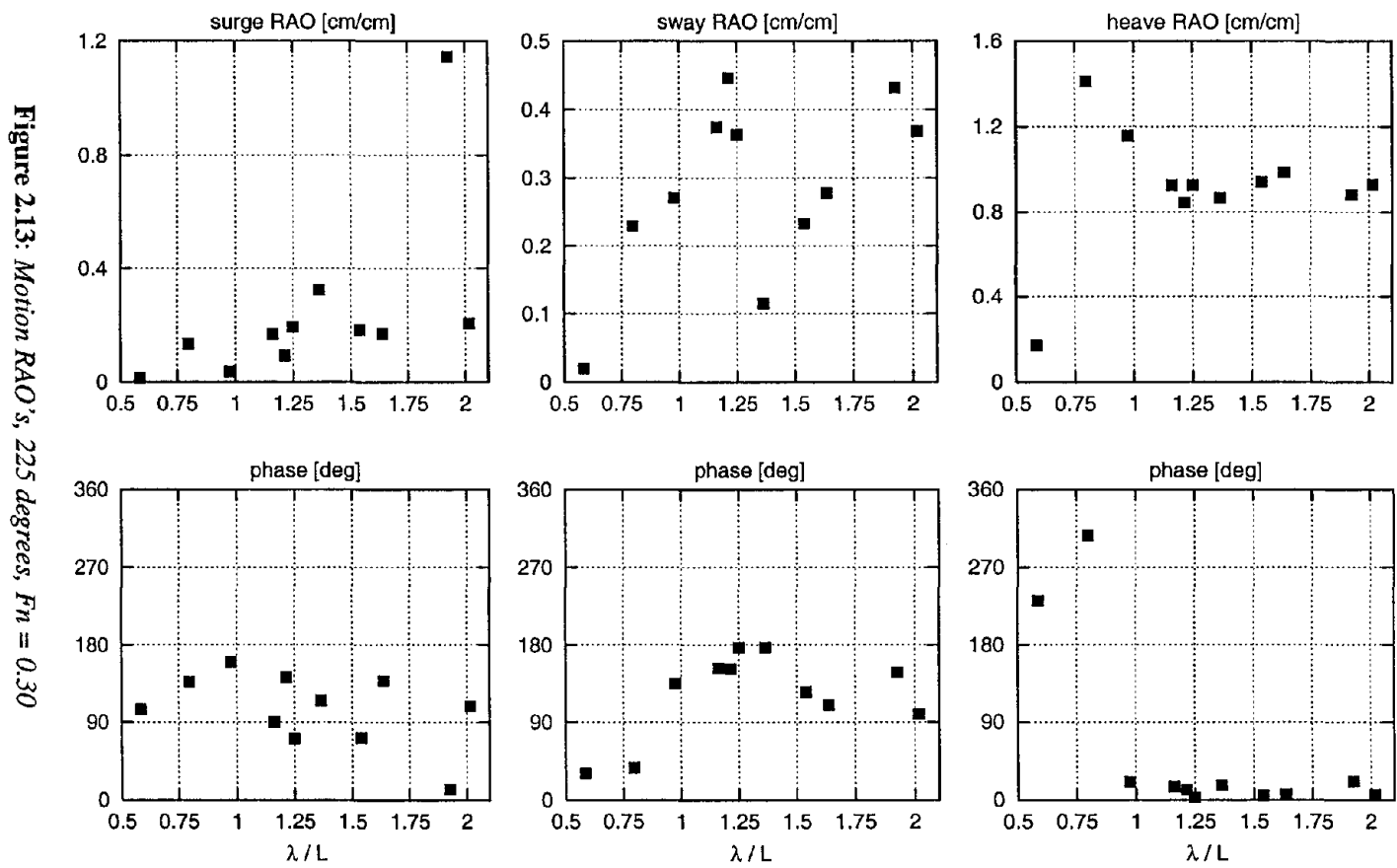


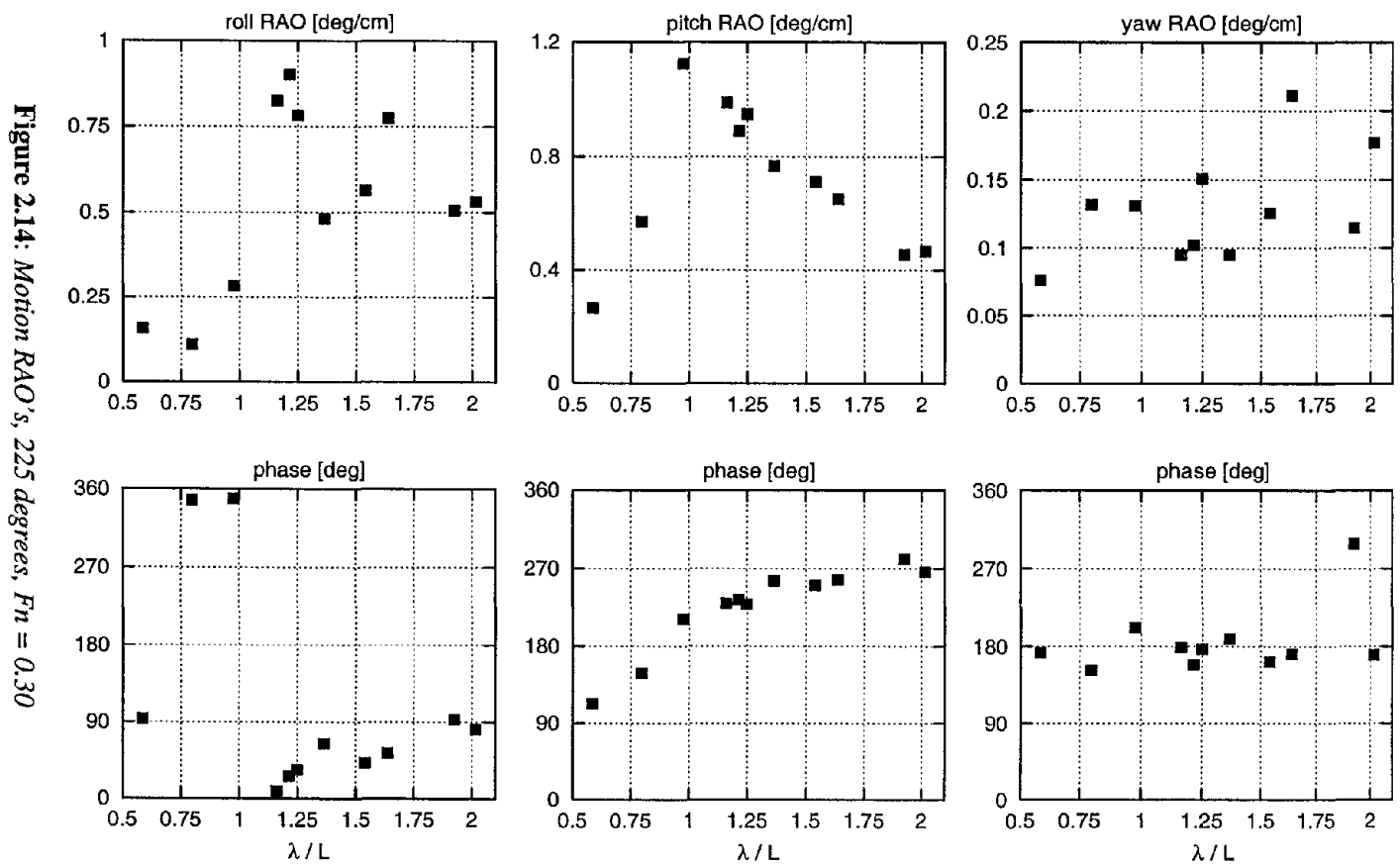


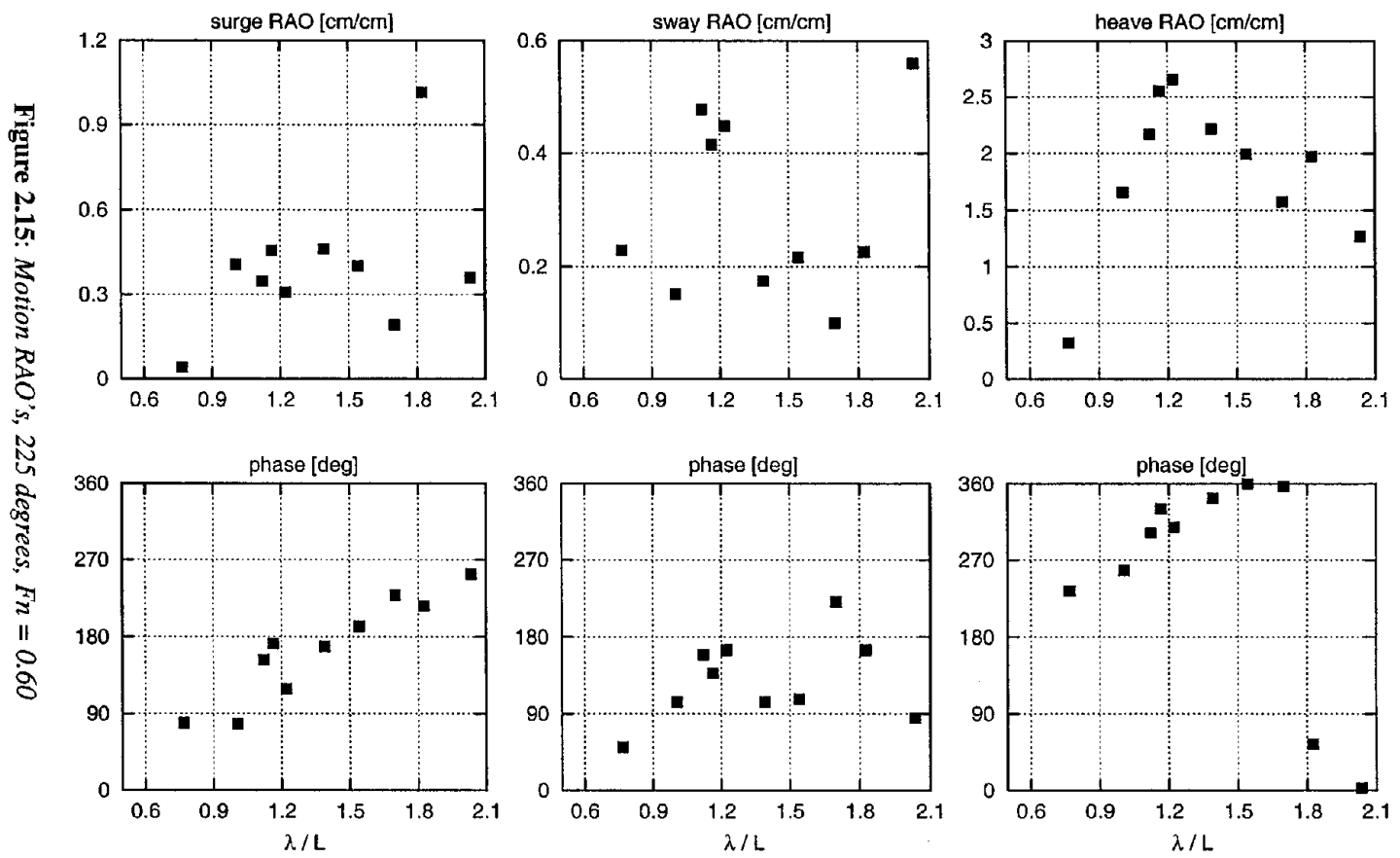


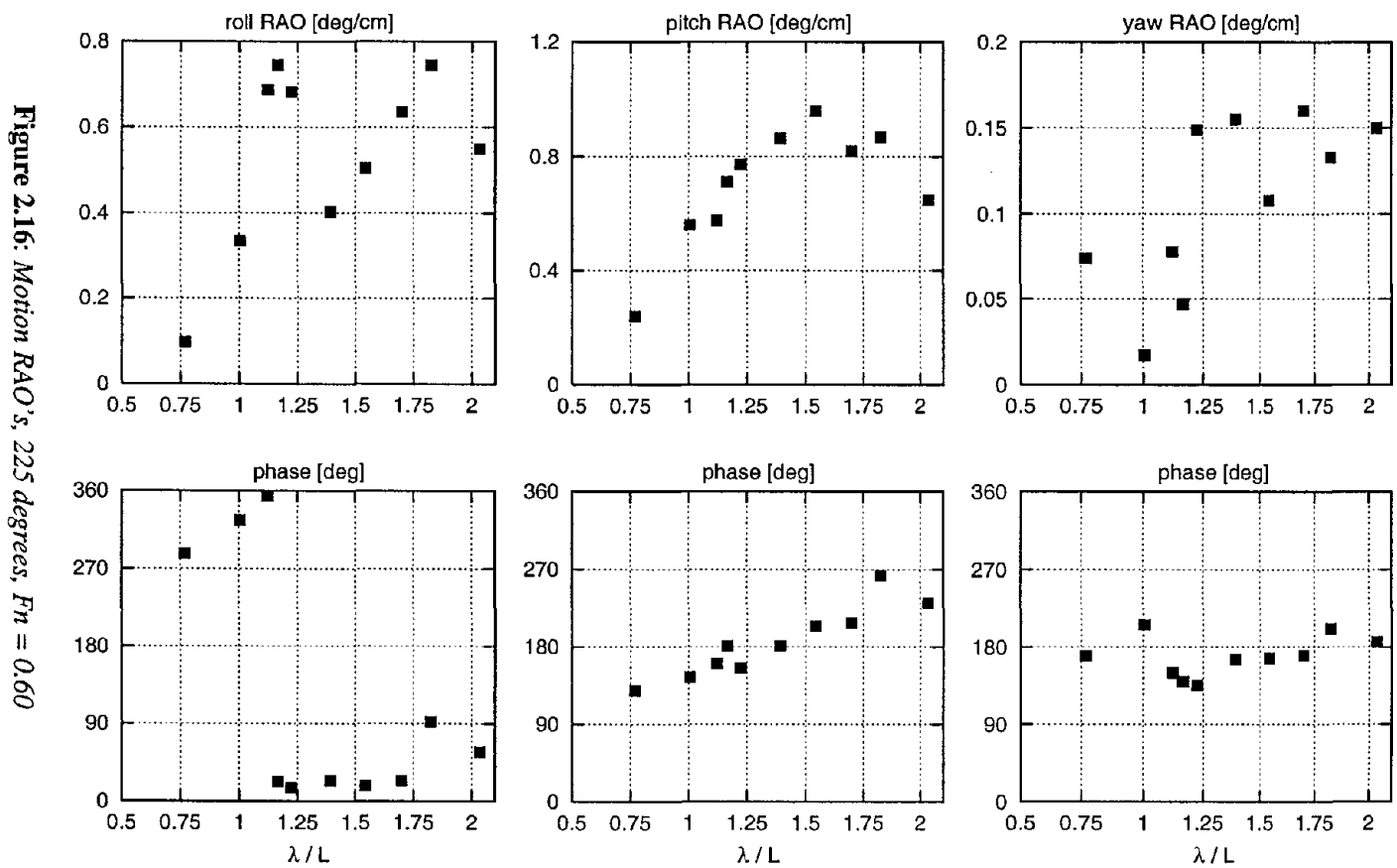


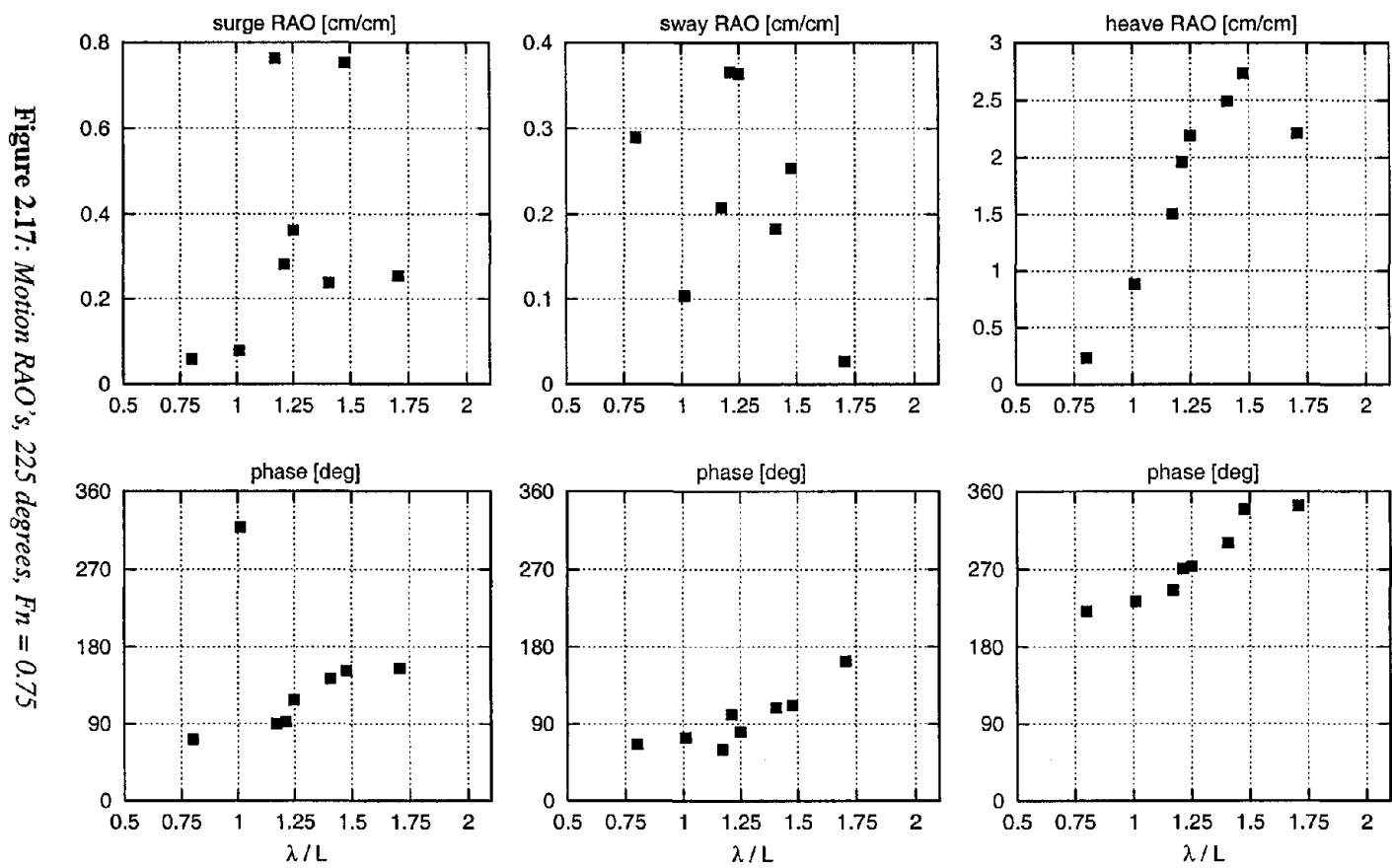


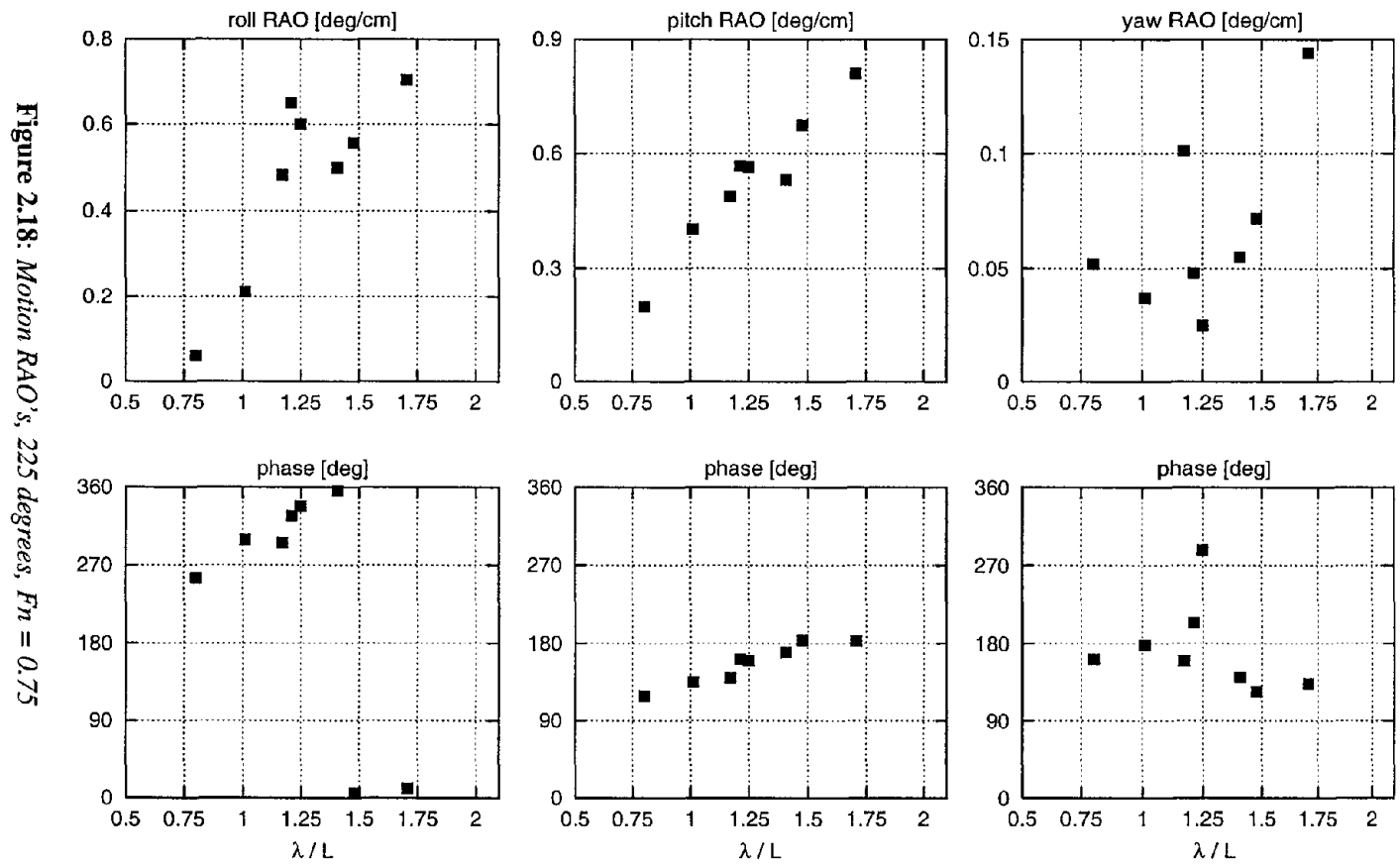












RESULTS CALCULATED FROM FILE: 8715

Wave frequency	3.589	rad/sec
Wave heading	180.000	deg
Encounter frequency	5.692	rad/sec
Number of oscillations	21	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.141	1.733	93	1.000
roll	deg	0.015	0.082	41	0.048
pitch	deg	-0.598	1.671	240	0.964
yaw	deg	-0.259	0.168	235	0.097
surge	cm	-31.368	0.374	63	0.216
sway	cm	-17.433	0.321	29	0.185
heave	cm	-0.564	1.499	10	0.865
ax cog	m/s2	0.059	0.171	49	
ay cog	m/s2	0.071	0.036	346	
az cog	m/s2	0.008	0.434	186	
ay ps fore	m/s2	0.003	0.043	261	
az ps fore	m/s2	0.001	1.779	229	
ay ps aft	m/s2	0.075	0.020	299	
az sb fore	m/s2	-0.018	1.731	229	
fx	N	0.455	0.549	41	
fy 1	N	1.346	5.511	221	
fy 2	N	0.379	4.731	48	
fy 3	N	-1.408	4.784	255	
fz 1	N	0.003	0.357	102	
fz 2	N	-0.085	0.591	240	
fy rod	N	0.102	0.581	275	
nozzle angle	deg	-0.350	2.688		
P nozzle sb	kPa	0.498	0.004		
P nozzle ps	kPa	0.444	0.010		
rpm	rpm	965.646	0.121		
speed	m/s	1.602	0.016		

RESULTS CALCULATED FROM FILE: 8714

Wave frequency 3.835 rad/sec
Wave heading 180.000 deg
Encounter frequency 6.412 rad/sec
Number of oscillations 20

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.087	1.237	141	1.000
roll	deg	0.086	0.053	336	0.043
pitch	deg	-0.689	1.512	225	1.222
yaw	deg	-0.172	0.245	219	0.198
surge	cm	-12.129	0.228	337	0.185
sway	cm	-38.179	0.404	324	0.326
heave	cm	-0.784	1.248	20	1.009
ax cog	m/s2	0.102	0.203	41	
ay cog	m/s2	0.061	0.039	315	
az cog	m/s2	0.004	0.429	200	
ay ps fore	m/s2	0.026	0.041	254	
az ps fore	m/s2	-0.038	2.093	220	
ay ps aft	m/s2	0.045	0.031	275	
az sb fore	m/s2	-0.000	2.057	221	
fx	N	0.605	0.594	22	
fy 1	N	1.475	6.095	195	
fy 2	N	0.199	6.220	22	
fy 3	N	-1.299	7.528	210	
fz 1	N	0.048	1.152	74	
fz 2	N	-0.054	0.958	234	
fy rod	N	0.001	0.947	253	
nozzle angle	deg	0.346	3.842		
P nozzle sb	kPa	0.574	0.006		
P nozzle ps	kPa	0.513	0.011		
rpm	rpm	1032.435	0.290		
speed	m/s	1.718	0.005		

RESULTS CALCULATED FROM FILE: 8713

Wave frequency	4.137	rad/sec
Wave heading	180.000	deg
Encounter frequency	6.970	rad/sec
Number of oscillations	25	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.079	1.447	232	1.000
roll	deg	0.027	0.043	305	0.030
pitch	deg	-0.787	1.971	203	1.362
yaw	deg	-0.184	0.267	200	0.185
surge	cm	-20.968	0.160	124	0.111
sway	cm	-26.789	0.348	102	0.240
heave	cm	-0.588	1.924	21	1.330
ax cog	m/s2	0.068	0.326	22	
ay cog	m/s2	0.087	0.052	281	
az cog	m/s2	0.077	0.779	204	
ay ps fore	m/s2	0.009	0.097	221	
az ps fore	m/s2	-0.035	3.385	203	
ay ps aft	m/s2	0.028	0.043	250	
az sb fore	m/s2	-0.022	3.357	204	
fx	N	0.054	0.805	2	
fy 1	N	1.292	9.878	158	
fy 2	N	0.547	10.551	349	
fy 3	N	-0.644	14.677	173	
fz 1	N	0.010	1.846	35	
fz 2	N	0.000	1.696	200	
fy rod	N	-0.700	1.155	232	
nozzle angle	deg	-0.528	4.113		
P nozzle sb	kPa	0.587	0.002		
P nozzle ps	kPa	0.750	0.023		
rpm	rpm	1032.181	0.152		
speed	m/s	1.623	0.004		

RESULTS CALCULATED FROM FILE: 8712

Wave frequency	4.519	rad/sec
Wave heading	180.000	deg
Encounter frequency	7.876	rad/sec
Number of oscillations	28	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.119	2.541	0	1.000
roll	deg	-0.084	0.054	262	0.021
pitch	deg	-0.686	2.156	170	0.848
yaw	deg	-0.221	0.258	167	0.101
surge	cm	-32.623	0.604	173	0.238
sway	cm	-18.496	0.199	179	0.078
heave	cm	-0.383	4.186	344	1.647
ax cog	m/s2	0.050	0.514	352	
ay cog	m/s2	0.102	0.095	290	
az cog	m/s2	0.385	2.389	166	
ay ps fore	m/s2	0.012	0.147	169	
az ps fore	m/s2	-0.001	5.985	169	
ay ps aft	m/s2	0.050	0.042	220	
az sb fore	m/s2	-0.025	5.925	169	
fx	N	-0.769	1.051	321	
fy 1	N	2.919	13.602	91	
fy 2	N	-0.950	18.378	279	
fy 3	N	0.630	35.080	88	
fz 1	N	0.059	2.368	356	
fz 2	N	-0.098	2.159	155	
fy rod	N	0.227	0.961	200	
nozzle angle	deg	-1.099	3.790		
P nozzle sb	kPa	0.428	0.004		
P nozzle ps	kPa	0.805	0.020		
rpm	rpm	1059.065	0.076		
speed	m/s	1.612	0.027		

RESULTS CALCULATED FROM FILE: 8711

Wave frequency	4.763	rad/sec
Wave heading	180.000	deg
Encounter frequency	8.710	rad/sec
Number of oscillations	27	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.186	2.611	59	1.000
roll	deg	0.059	0.030	251	0.012
pitch	deg	-0.805	1.864	163	0.714
yaw	deg	-0.144	0.290	141	0.111
surge	cm	-16.593	0.458	180	0.176
sway	cm	-37.029	0.303	226	0.116
heave	cm	-0.557	3.769	297	1.444
ax cog	m/s2	0.018	0.483	336	
ay cog	m/s2	0.154	0.086	303	
az cog	m/s2	0.171	2.702	117	
ay ps fore	m/s2	0.018	0.154	139	
az ps fore	m/s2	-0.012	6.009	145	
ay ps aft	m/s2	0.013	0.025	206	
az sb fore	m/s2	-0.006	5.926	145	
fx	N	0.065	0.589	292	
fy 1	N	2.845	8.251	46	
fy 2	N	-1.424	13.758	220	
fy 3	N	1.986	33.056	15	
fz 1	N	0.120	2.284	339	
fz 2	N	0.027	1.915	150	
fy rod	N	-0.234	0.701	166	
nozzle angle	deg	-0.787	2.902		
P nozzle sb	kPa	0.456	0.002		
P nozzle ps	kPa	0.571	0.017		
rpm	rpm	1090.223	0.248		
speed	m/s	1.706	0.029		

RESULTS CALCULATED FROM FILE: 8710

Wave frequency	5.061	rad/sec
Wave heading	180.000	deg
Encounter frequency	9.358	rad/sec
Number of oscillations	27	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.170	2.467	176	1.000
roll	deg	0.002	0.048	200	0.019
pitch	deg	-0.798	1.361	138	0.552
yaw	deg	-0.311	0.188	137	0.076
surge	cm	-27.767	0.339	110	0.138
sway	cm	-26.170	0.307	95	0.125
heave	cm	-0.721	1.620	254	0.657
ax cog	m/s2	0.084	0.375	314	
ay cog	m/s2	0.067	0.059	227	
az cog	m/s2	0.034	1.317	71	
ay ps fore	m/s2	0.027	0.083	104	
az ps fore	m/s2	-0.036	3.950	120	
ay ps aft	m/s2	0.020	0.061	209	
az sb fore	m/s2	-0.028	3.898	121	
fx	N	-0.112	0.337	349	
fy 1	N	2.131	4.371	339	
fy 2	N	-0.750	8.276	155	
fy 3	N	1.097	23.954	319	
fz 1	N	-0.025	1.717	303	
fz 2	N	0.034	1.558	113	
fy rod	N	0.120	0.995	163	
nozzle angle	deg	-0.762	2.784		
P nozzle sb	kPa	0.613	0.009		
P nozzle ps	kPa	0.810	0.019		
rpm	rpm	1059.672	0.184		
speed	m/s	1.645	0.016		

RESULTS CALCULATED FROM FILE: 8706

Wave frequency	5.890	rad/sec
Wave heading	180.000	deg
Encounter frequency	11.537	rad/sec
Number of oscillations	34	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.184	1.489	153	1.000
roll	deg	-0.050	0.026	47	0.018
pitch	deg	-0.613	0.140	62	0.094
yaw	deg	-0.503	0.012	61	0.008
surge	cm	-37.320	0.024	128	0.016
sway	cm	-28.277	0.269	307	0.181
heave	cm	-0.683	0.228	36	0.153
ax cog	m/s2	0.002	0.014	215	
ay cog	m/s2	0.090	0.014	85	
az cog	m/s2	-0.037	0.313	227	
ay ps fore	m/s2	0.039	0.043	31	
az ps fore	m/s2	-0.039	0.145	106	
ay ps aft	m/s2	0.041	0.027	150	
az sb fore	m/s2	-0.035	0.171	89	
fx	N	-0.130	0.277	11	
fy 1	N	1.018	1.819	259	
fy 2	N	0.196	2.967	59	
fy 3	N	-0.932	5.512	240	
fz 1	N	0.059	0.195	258	
fz 2	N	-0.018	0.140	131	
fy rod	N	0.208	0.040	119	
nozzle angle	deg	-0.288	0.096		
P nozzle sb	kPa	0.520	0.008		
P nozzle ps	kPa	0.675	0.009		
rpm	rpm	971.899	0.213		
speed	m/s	1.596	0.000		

RESULTS CALCULATED FROM FILE: 8722

Wave frequency	3.258	rad/sec
Wave heading	180.000	deg
Encounter frequency	6.714	rad/sec
Number of oscillations	10	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.224	1.376	25	1.000
roll	deg	-0.006	0.035	96	0.025
pitch	deg	-3.292	1.415	221	1.028
yaw	deg	-0.256	0.168	210	0.122
surge	cm	-50.224	0.278	164	0.202
sway	cm	-12.158	0.146	340	0.106
heave	cm	-0.826	2.504	15	1.820
ax cog	m/s2	0.309	0.090	20	
ay cog	m/s2	0.038	0.039	152	
az cog	m/s2	-0.004	0.000	21	
ay ps fore	m/s2	0.046	0.025	301	
az ps fore	m/s2	-0.167	2.595	212	
ay ps aft	m/s2	0.086	0.076	221	
az sb fore	m/s2	-0.112	2.190	215	
fx	N	-0.119	1.294	26	
fy 1	N	10.033	5.142	207	
fy 2	N	-4.990	5.255	20	
fy 3	N	0.630	4.071	189	
fz 1	N	0.133	1.376	40	
fz 2	N	-0.169	1.170	208	
fy rod	N	-0.458	1.163	232	
nozzle angle	deg	-0.944	2.767		
P nozzle sb	kPa	2.400	0.014		
P nozzle ps	kPa	3.151	0.015		
rpm	rpm	2034.761	0.737		
speed	m/s	3.196	0.010		

RESULTS CALCULATED FROM FILE: 8721

Wave frequency	3.434	rad/sec
Wave heading	180.000	deg
Encounter frequency	7.465	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.251	1.406	49	1.000
roll	deg	0.117	0.048	124	0.034
pitch	deg	-3.009	1.523	197	1.083
yaw	deg	-0.258	0.239	190	0.170
surge	cm	-23.266	0.074	243	0.052
sway	cm	-17.099	0.285	156	0.203
heave	cm	-0.355	3.754	358	2.670
ax cog	m/s2	0.315	0.149	17	
ay cog	m/s2	0.086	0.014	310	
az cog	m/s2	-0.004	0.001	34	
ay ps fore	m/s2	0.017	0.026	141	
az ps fore	m/s2	-0.140	3.904	190	
ay ps aft	m/s2	0.038	0.110	225	
az sb fore	m/s2	-0.093	3.910	189	
fx	N	-0.110	1.950	7	
fy 1	N	10.248	7.892	190	
fy 2	N	-5.812	8.317	2	
fy 3	N	3.308	4.801	156	
fz 1	N	0.728	1.922	16	
fz 2	N	-0.560	1.134	189	
fy rod	N	-0.942	1.605	208	
nozzle angle	deg	-0.622	3.846		
P nozzle sb	kPa	2.550	0.002		
P nozzle ps	kPa	3.362	0.025		
rpm	rpm	2100.128	0.098		
speed	m/s	3.354	0.024		

RESULTS CALCULATED FROM FILE: 8720

Wave frequency 3.583 rad/sec
Wave heading 180.000 deg
Encounter frequency 7.964 rad/sec
Number of oscillations 12

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
wave 1	cm	-0.207	1.794	92	1.000
roll	deg	-0.030	0.139	40	0.077
pitch	deg	-3.115	1.551	160	0.865
yaw	deg	-0.204	0.244	150	0.136
surge	cm	-32.231	0.505	153	0.281
sway	cm	-22.451	0.048	87	0.027
heave	cm	-0.230	5.026	312	2.802
ax cog	m/s2	0.272	0.206	330	
ay cog	m/s2	0.090	0.077	320	
az cog	m/s2	-0.005	0.000	84	
ay ps fore	m/s2	0.078	0.057	118	
az ps fore	m/s2	-0.096	5.455	145	
ay ps aft	m/s2	0.038	0.090	186	
az sb fore	m/s2	-0.057	5.385	145	
fx	N	-0.068	2.253	327	
fy 1	N	10.066	11.825	146	
fy 2	N	-5.509	12.338	315	
fy 3	N	3.316	6.697	106	
fz 1	N	0.522	2.315	322	
fz 2	N	-0.358	1.998	157	
fy rod	N	-1.002	1.638	168	
nozzle angle	deg	-0.979	3.958		
P nozzle sb	kPa	2.606	0.052		
P nozzle ps	kPa	3.368	0.027		
rpm	rpm	2109.502	0.298		
speed	m/s	3.347	0.044		

RESULTS CALCULATED FROM FILE: 8719

Wave frequency	3.849	rad/sec
Wave heading	180.000	deg
Encounter frequency	8.815	rad/sec
Number of oscillations	14	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.261	1.946	150	1.000
roll	deg	0.006	0.151	340	0.078
pitch	deg	-3.075	1.338	156	0.688
yaw	deg	-0.186	0.171	145	0.088
surge	cm	-17.746	0.542	115	0.279
sway	cm	-20.324	0.205	303	0.105
heave	cm	-0.536	4.496	285	2.310
ax cog	m/s2	0.267	0.237	316	
ay cog	m/s2	0.127	0.067	287	
az cog	m/s2	-0.005	0.000	19	
ay ps fore	m/s2	0.038	0.049	153	
az ps fore	m/s2	-0.082	5.426	129	
ay ps aft	m/s2	0.099	0.045	88	
az sb fore	m/s2	-0.048	5.203	127	
fx	N	0.019	1.875	304	
fy 1	N	11.078	12.509	124	
fy 2	N	-5.227	12.594	293	
fy 3	N	1.776	5.695	81	
fz 1	N	0.474	2.183	316	
fz 2	N	-0.392	2.600	145	
fy rod	N	-0.635	1.195	155	
nozzle angle	deg	-1.115	2.835		
P nozzle sb	kPa	2.606	0.025		
P nozzle ps	kPa	3.370	0.019		
rpm	rpm	2108.433	0.139		
speed	m/s	3.287	0.047		

RESULTS CALCULATED FROM FILE: 8718

Wave frequency	4.115	rad/sec
Wave heading	180.000	deg
Encounter frequency	9.689	rad/sec
Number of oscillations	14	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.244	2.347	237	1.000
roll	deg	0.010	0.096	266	0.041
pitch	deg	-3.088	1.331	132	0.567
yaw	deg	-0.195	0.149	127	0.064
surge	cm	-32.582	0.473	59	0.202
sway	cm	-18.001	0.077	82	0.033
heave	cm	-0.609	2.943	241	1.254
ax cog	m/s2	0.227	0.260	288	
ay cog	m/s2	0.172	0.100	233	
az cog	m/s2	-0.003	0.001	196	
ay ps fore	m/s2	0.013	0.068	60	
az ps fore	m/s2	-0.068	4.968	100	
ay ps aft	m/s2	0.081	0.033	138	
az sb fore	m/s2	-0.085	4.811	100	
fx	N	0.012	1.529	277	
fy 1	N	10.940	10.230	85	
fy 2	N	-5.195	10.174	252	
fy 3	N	0.324	4.106	43	
fz 1	N	0.546	2.009	288	
fz 2	N	-0.464	2.427	103	
fy rod	N	0.264	1.162	131	
nozzle angle	deg	-1.114	2.469		
P nozzle sb	kPa	2.517	0.026		
P nozzle ps	kPa	3.279	0.049		
rpm	rpm	2083.932	0.347		
speed	m/s	3.229	0.041		

RESULTS CALCULATED FROM FILE: 8717

Wave frequency	4.497	rad/sec
Wave heading	180.000	deg
Encounter frequency	11.177	rad/sec
Number of oscillations	18	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.161	2.555	5	1.000
roll	deg	0.027	0.028	213	0.011
pitch	deg	-3.166	0.833	109	0.326
yaw	deg	-0.223	0.072	105	0.028
surge	cm	-42.845	0.401	32	0.157
sway	cm	-10.569	0.036	105	0.014
heave	cm	-0.827	0.989	214	0.387
ax cog	m/s2	0.267	0.134	275	
ay cog	m/s2	0.032	0.081	200	
az cog	m/s2	-0.005	0.001	236	
ay ps fore	m/s2	0.090	0.046	134	
az ps fore	m/s2	-0.060	3.202	86	
ay ps aft	m/s2	0.054	0.016	235	
az sb fore	m/s2	-0.126	3.289	95	
fx	N	-0.224	0.853	270	
fy 1	N	11.211	4.397	65	
fy 2	N	-5.495	4.929	234	
fy 3	N	0.666	3.123	42	
fz 1	N	0.964	1.436	235	
fz 2	N	-0.863	1.883	61	
fy rod	N	0.084	0.558	113	
nozzle angle	deg	-1.208	1.087		
P nozzle sb	kPa	2.490	0.005		
P nozzle ps	kPa	3.227	0.014		
rpm	rpm	2066.907	0.896		
speed	m/s	3.240	0.019		

RESULTS CALCULATED FROM FILE: 8716

Wave frequency 4.738 rad/sec
Wave heading 180.000 deg
Encounter frequency 12.159 rad/sec
Number of oscillations 19

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.212	2.610	79	1.000
roll	deg	0.092	0.006	188	0.002
pitch	deg	-3.216	0.552	103	0.211
yaw	deg	-0.231	0.051	107	0.019
surge	cm	-38.400	0.240	145	0.092
sway	cm	-13.125	0.081	11	0.031
heave	cm	-0.821	0.401	209	0.154
ax cog	m/s2	0.253	0.081	305	
ay cog	m/s2	0.050	0.007	153	
az cog	m/s2	24.415	0.001	206	
ay ps fore	m/s2	0.009	0.064	103	
az ps fore	m/s2	-0.095	2.330	89	
ay ps aft	m/s2	0.003	0.031	237	
az sb fore	m/s2	-0.039	2.275	87	
fx	N	-0.256	0.700	279	
fy 1	N	10.579	3.006	76	
fy 2	N	-5.112	3.024	234	
fy 3	N	1.807	2.819	37	
fz 1	N	0.622	0.779	241	
fz 2	N	-0.512	1.094	61	
fy rod	N	-0.636	0.432	102	
nozzle angle	deg	-1.508	0.772		
P nozzle sb	kPa	2.463	0.022		
P nozzle ps	kPa	3.192	0.032		
rpm	rpm	2059.934	0.476		
speed	m/s	3.242	0.014		

RESULTS CALCULATED FROM FILE: 8729

Wave frequency	3.278	rad/sec
Wave heading	180.000	deg
Encounter frequency	7.758	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.141	1.375	22	1.000
roll	deg	0.145	0.117	95	0.085
pitch	deg	-3.040	1.073	185	0.781
yaw	deg	-0.216	0.143	168	0.104
surge	cm	-38.041	0.638	167	0.464
sway	cm	-15.532	0.042	201	0.031
heave	cm	-0.073	3.753	343	2.729
ax cog	m/s2	0.277	0.135	7	
ay cog	m/s2	0.053	0.039	77	
az cog	m/s2	-0.002	0.000	28	
ay ps fore	m/s2	0.018	0.137	193	
az ps fore	m/s2	-0.121	3.519	174	
ay ps aft	m/s2	0.038	0.062	138	
az sb fore	m/s2	-0.082	3.594	173	
fx	N	-0.186	1.475	355	
fy 1	N	11.611	8.360	178	
fy 2	N	-7.768	8.895	351	
fy 3	N	8.601	4.972	165	
fz 1	N	0.367	1.811	345	
fz 2	N	-0.138	1.167	188	
fy rod	N	-1.080	0.932	191	
nozzle angle	deg	-0.632	2.403		
P nozzle sb	kPa	3.244	0.067		
P nozzle ps	kPa	4.184	0.061		
rpm	rpm	2339.462	0.174		
speed	m/s	4.090	0.037		

RESULTS CALCULATED FROM FILE: 8728

Wave frequency 3.375 rad/sec
Wave heading 180.000 deg
Encounter frequency 8.135 rad/sec
Number of oscillations 8

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
wave 1	cm	-0.224	1.702	42	1.000
roll	deg	-0.043	0.176	25	0.103
pitch	deg	-2.986	1.098	162	0.645
yaw	deg	-0.204	0.142	148	0.083
surge	cm	-34.109	0.972	117	0.571
sway	cm	-18.068	0.101	226	0.059
heave	cm	0.004	4.581	302	2.691
ax cog	m/s2	0.215	0.245	305	
ay cog	m/s2	0.078	0.079	302	
az cog	m/s2	-0.004	0.000	122	
ay ps fore	m/s2	0.031	0.066	163	
az ps fore	m/s2	-0.065	4.560	139	
ay ps aft	m/s2	0.067	0.053	261	
az sb fore	m/s2	-0.041	4.519	138	
fx	N	-0.064	1.898	331	
fy 1	N	11.727	11.858	139	
fy 2	N	-7.622	12.412	312	
fy 3	N	8.311	6.838	125	
fz 1	N	0.404	1.872	307	
fz 2	N	0.041	2.086	154	
fy rod	N	-0.963	1.031	167	
nozzle angle	deg	-0.797	2.352		
P nozzle sb	kPa	3.309	0.050		
P nozzle ps	kPa	4.275	0.054		
rpm	rpm	2361.897	0.162		
speed	m/s	4.101	0.056		

RESULTS CALCULATED FROM FILE: 8727

Wave frequency	3.544	rad/sec
Wave heading	180.000	deg
Encounter frequency	8.789	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.134	1.453	84	1.000
roll	deg	-0.126	0.160	14	0.110
pitch	deg	-2.969	0.980	184	0.674
yaw	deg	-0.179	0.145	167	0.100
surge	cm	-33.764	1.012	69	0.696
sway	cm	-17.064	0.103	71	0.071
heave	cm	-0.110	4.035	311	2.776
ax cog	m/s2	0.287	0.207	341	
ay cog	m/s2	0.062	0.049	315	
az cog	m/s2	-0.005	0.000	325	
ay ps fore	m/s2	0.020	0.101	177	
az ps fore	m/s2	0.019	4.173	154	
ay ps aft	m/s2	0.050	0.050	257	
az sb fore	m/s2	0.097	4.083	152	
fx	N	-0.021	1.902	346	
fy 1	N	12.058	10.591	149	
fy 2	N	-8.306	11.539	321	
fy 3	N	7.874	6.625	137	
fz 1	N	0.078	1.821	320	
fz 2	N	0.313	2.176	159	
fy rod	N	-0.169	0.993	182	
nozzle angle	deg	-0.913	2.324		
P nozzle sb	kPa	3.350	0.013		
P nozzle ps	kPa	4.327	0.068		
rpm	rpm	2376.402	0.498		
speed	m/s	4.095	0.052		

RESULTS CALCULATED FROM FILE: 8726

Wave frequency	3.583	rad/sec
Wave heading	180.000	deg
Encounter frequency	8.949	rad/sec
Number of oscillations	9	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.188	1.778	86	1.000
roll	deg	0.031	0.165	326	0.093
pitch	deg	-3.043	1.206	156	0.678
yaw	deg	-0.175	0.157	140	0.088
surge	cm	-24.077	1.489	112	0.838
sway	cm	-19.994	0.008	100	0.004
heave	cm	0.238	4.385	272	2.466
ax cog	m/s2	0.228	0.162	297	
ay cog	m/s2	0.073	0.042	308	
az cog	m/s2	-0.002	0.000	197	
ay ps fore	m/s2	0.043	0.071	101	
az ps fore	m/s2	-0.098	4.959	119	
ay ps aft	m/s2	0.055	0.031	181	
az sb fore	m/s2	-0.015	4.866	119	
fx	N	0.032	2.224	312	
fy 1	N	11.792	12.942	111	
fy 2	N	-8.015	14.089	282	
fy 3	N	7.958	8.531	97	
fz 1	N	0.415	2.188	292	
fz 2	N	0.095	2.907	122	
fy rod	N	-0.303	1.164	157	
nozzle angle	deg	-0.746	2.689		
P nozzle sb	kPa	3.390	0.073		
P nozzle ps	kPa	4.415	0.067		
rpm	rpm	2389.240	0.115		
speed	m/s	4.102	0.054		

RESULTS CALCULATED FROM FILE: 8725

Wave frequency	3.818	rad/sec
Wave heading	180.000	deg
Encounter frequency	9.819	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.308	2.098	156	1.000
roll	deg	0.099	0.094	262	0.045
pitch	deg	-3.028	1.137	134	0.542
yaw	deg	-0.182	0.114	125	0.054
surge	cm	-35.273	0.401	121	0.191
sway	cm	-16.378	0.060	93	0.029
heave	cm	-0.083	2.895	236	1.380
ax cog	m/s2	0.237	0.194	264	
ay cog	m/s2	0.125	0.080	233	
az cog	m/s2	-0.002	0.001	17	
ay ps fore	m/s2	0.034	0.114	134	
az ps fore	m/s2	-0.014	4.414	96	
ay ps aft	m/s2	0.148	0.086	219	
az sb fore	m/s2	-0.003	4.353	96	
fx	N	0.045	1.862	279	
fy 1	N	11.770	11.161	82	
fy 2	N	-7.915	12.139	250	
fy 3	N	8.190	8.115	64	
fz 1	N	0.401	1.767	269	
fz 2	N	-0.094	2.153	94	
fy rod	N	-0.731	1.004	126	
nozzle angle	deg	-1.220	1.965		
P nozzle sb	kPa	3.262	0.078		
P nozzle ps	kPa	4.243	0.068		
rpm	rpm	2358.023	0.493		
speed	m/s	4.037	0.043		

RESULTS CALCULATED FROM FILE: 8724

Wave frequency	4.169	rad/sec
Wave heading	180.000	deg
Encounter frequency	11.315	rad/sec
Number of oscillations	13	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.205	2.067	262	1.000
roll	deg	0.000	0.036	234	0.017
pitch	deg	-3.147	0.810	137	0.392
yaw	deg	-0.172	0.093	127	0.045
surge	cm	-41.966	0.244	168	0.118
sway	cm	-18.941	0.071	210	0.034
heave	cm	0.058	1.390	236	0.673
ax cog	m/s2	0.270	0.163	271	
ay cog	m/s2	0.109	0.036	242	
az cog	m/s2	0.008	0.000	36	
ay ps fore	m/s2	0.004	0.039	169	
az ps fore	m/s2	-0.149	3.311	105	
ay ps aft	m/s2	0.120	0.088	207	
az sb fore	m/s2	-0.110	3.278	106	
fx	N	-0.088	1.599	279	
fy 1	N	11.837	7.183	85	
fy 2	N	-7.795	7.913	249	
fy 3	N	6.918	6.337	57	
fz 1	N	0.437	1.192	280	
fz 2	N	0.035	1.408	105	
fy rod	N	-0.082	0.816	125	
nozzle angle	deg	-0.904	1.438		
P nozzle sb	kPa	3.225	0.016		
P nozzle ps	kPa	4.202	0.071		
rpm	rpm	2341.461	0.530		
speed	m/s	4.033	0.024		

RESULTS CALCULATED FROM FILE: 8723

Wave frequency	4.538	rad/sec
Wave heading	180.000	deg
Encounter frequency	12.938	rad/sec
Number of oscillations	14	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.167	2.413	33	1.000
roll	deg	0.084	0.016	191	0.007
pitch	deg	-3.069	0.499	116	0.207
yaw	deg	-0.226	0.047	108	0.019
surge	cm	-54.996	0.109	100	0.045
sway	cm	-14.725	0.092	3	0.038
heave	cm	-0.036	0.526	219	0.218
ax cog	m/s2	0.256	0.136	277	
ay cog	m/s2	0.098	0.068	235	
az cog	m/s2	-0.002	0.001	347	
ay ps fore	m/s2	-0.056	0.055	128	
az ps fore	m/s2	-0.034	2.442	98	
ay ps aft	m/s2	0.136	0.129	187	
az sb fore	m/s2	-0.025	2.395	98	
fx	N	-0.372	1.155	269	
fy 1	N	12.112	4.295	75	
fy 2	N	-8.038	4.676	232	
fy 3	N	6.903	5.405	36	
fz 1	N	0.380	0.838	288	
fz 2	N	-0.158	0.942	89	
fy rod	N	0.012	0.351	111	
nozzle angle	deg	-1.056	0.629		
P nozzle sb	kPa	3.164	0.027		
P nozzle ps	kPa	4.100	0.087		
rpm	rpm	2310.617	0.548		
speed	m/s	4.002	0.019		

RESULTS CALCULATED FROM FILE: 8747

Wave frequency 3.210 rad/sec
Wave heading 195.000 deg
Encounter frequency 4.825 rad/sec
Number of oscillations 18

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.090	2.781	356	1.000
roll	deg	0.029	0.346	53	0.124
pitch	deg	-0.562	2.045	253	0.735
yaw	deg	0.010	0.206	201	0.074
surge	cm	-42.148	0.937	111	0.337
sway	cm	-10.051	0.225	58	0.081
heave	cm	-0.916	2.686	4	0.966
ax cog	m/s2	0.082	0.136	63	
ay cog	m/s2	0.055	0.028	27	
az cog	m/s2	-0.021	0.571	181	
ay ps fore	m/s2	-0.002	0.096	317	
az ps fore	m/s2	-0.006	1.631	232	
ay ps aft	m/s2	0.084	0.019	165	
az sb fore	m/s2	-0.022	1.559	231	
fx	N	-0.019	1.471	360	
fy 1	N	1.599	5.337	254	
fy 2	N	-0.057	3.833	79	
fy 3	N	-1.553	4.802	308	
fz 1	N	0.237	2.624	157	
fz 2	N	-0.251	1.911	323	
fy rod	N	0.240	0.335	234	
nozzle angle	deg	0.174	3.336		
P nozzle sb	kPa	0.518	0.011		
P nozzle ps	kPa	0.667	0.013		
rpm	rpm	1012.185	0.781		
speed	m/s	1.590	0.037		

RESULTS CALCULATED FROM FILE: 8744

Wave frequency	3.470	rad/sec
Wave heading	195.000	deg
Encounter frequency	5.386	rad/sec
Number of oscillations	19	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.078	2.813	52	1.000
roll	deg	0.218	0.461	41	0.164
pitch	deg	-0.667	2.493	246	0.886
yaw	deg	0.182	0.204	210	0.072
surge	cm	-34.833	0.425	78	0.151
sway	cm	-7.561	0.289	120	0.103
heave	cm	-0.730	2.531	9	0.900
ax cog	m/s2	0.085	0.223	57	
ay cog	m/s2	0.032	0.058	30	
az cog	m/s2	-0.038	0.686	184	
ay ps fore	m/s2	0.004	0.154	333	
az ps fore	m/s2	-0.029	2.457	231	
ay ps aft	m/s2	0.056	0.034	80	
az sb fore	m/s2	-0.026	2.256	232	
fx	N	-0.152	2.014	3	
fy 1	N	1.714	7.517	237	
fy 2	N	0.143	6.115	63	
fy 3	N	-0.819	7.079	281	
fz 1	N	0.318	3.891	164	
fz 2	N	-0.372	2.253	325	
fy rod	N	-0.361	0.529	255	
nozzle angle	deg	-1.064	2.801		
P nozzle sb	kPa	0.579	0.007		
P nozzle ps	kPa	0.718	0.014		
rpm	rpm	1052.358	0.590		
speed	m/s	1.615	0.030		

RESULTS CALCULATED FROM FILE: 8743

Wave frequency	3.826	rad/sec
Wave heading	195.000	deg
Encounter frequency	6.104	rad/sec
Number of oscillations	20	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.072	2.429	147	1.000
roll	deg	-0.043	0.380	11	0.157
pitch	deg	-0.680	2.715	235	1.118
yaw	deg	0.133	0.185	216	0.076
surge	cm	-43.189	0.267	90	0.110
sway	cm	-3.372	0.210	140	0.087
heave	cm	-0.581	2.178	18	0.897
ax cog	m/s2	0.072	0.329	49	
ay cog	m/s2	0.003	0.098	348	
az cog	m/s2	-0.013	0.684	193	
ay ps fore	m/s2	0.006	0.168	309	
az ps fore	m/s2	0.037	3.370	226	
ay ps aft	m/s2	0.061	0.039	17	
az sb fore	m/s2	-0.004	3.200	228	
fx	N	-0.072	1.932	8	
fy 1	N	2.086	9.426	206	
fy 2	N	-0.013	8.676	39	
fy 3	N	-2.216	10.638	246	
fz 1	N	0.289	4.181	159	
fz 2	N	-0.466	1.673	326	
fy rod	N	0.730	0.489	250	
nozzle angle	deg	-0.974	2.378		
P nozzle sb	kPa	0.551	0.007		
P nozzle ps	kPa	0.719	0.017		
rpm	rpm	1012.336	0.397		
speed	m/s	1.579	0.016		

RESULTS CALCULATED FROM FILE: 8737

Wave frequency 4.155 rad/sec
Wave heading 195.000 deg
Encounter frequency 6.740 rad/sec
Number of oscillations 26

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.085	2.541	225	1.000
roll	deg	0.071	0.392	338	0.154
pitch	deg	-0.927	3.938	227	1.550
yaw	deg	0.287	0.343	222	0.135
surge	cm	-30.138	0.480	150	0.189
sway	cm	-10.377	0.270	132	0.106
heave	cm	-0.403	3.126	34	1.230
ax cog	m/s2	0.098	0.577	38	
ay cog	m/s2	0.063	0.190	332	
az cog	m/s2	-0.042	1.133	213	
ay ps fore	m/s2	-0.247	0.230	290	
az ps fore	m/s2	-0.277	5.936	223	
ay ps aft	m/s2	0.062	0.133	307	
az sb fore	m/s2	0.004	5.759	226	
fx	N	-0.308	3.123	20	
fy 1	N	1.495	17.213	186	
fy 2	N	0.346	16.915	21	
fy 3	N	0.943	21.531	216	
fz 1	N	0.499	5.993	155	
fz 2	N	-0.929	1.895	5	
fy rod	N	-1.368	1.449	254	
nozzle angle	deg	-1.507	3.816		
P nozzle sb	kPa	0.645	0.015		
P nozzle ps	kPa	0.797	0.030		
rpm	rpm	1073.489	0.283		
speed	m/s	1.520	0.008		

RESULTS CALCULATED FROM FILE: 8738

Wave frequency 4.132 rad/sec
Wave heading 195.000 deg
Encounter frequency 6.938 rad/sec
Number of oscillations 21

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.084	2.509	216	1.000
roll	deg	0.088	0.216	329	0.086
pitch	deg	-1.025	3.401	203	1.355
yaw	deg	0.377	0.342	202	0.136
surge	cm	-26.960	0.501	178	0.200
sway	cm	-11.587	0.249	120	0.099
heave	cm	-0.532	3.448	19	1.374
ax cog	m/s2	0.062	0.555	19	
ay cog	m/s2	0.040	0.159	318	
az cog	m/s2	-0.035	1.407	201	
ay ps fore	m/s2	0.005	0.159	276	
az ps fore	m/s2	-0.009	5.882	202	
ay ps aft	m/s2	0.053	0.106	277	
az sb fore	m/s2	-0.045	5.720	203	
fx	N	-0.047	2.841	10	
fy 1	N	3.500	15.399	160	
fy 2	N	-0.497	17.219	348	
fy 3	N	-2.386	24.908	170	
fz 1	N	0.081	3.305	142	
fz 2	N	-0.462	0.647	42	
fy rod	N	0.696	1.551	229	
nozzle angle	deg	-2.231	3.721		
P nozzle sb	kPa	0.707	0.010		
P nozzle ps	kPa	0.900	0.026		
rpm	rpm	1128.465	0.284		
speed	m/s	1.668	0.009		

RESULTS CALCULATED FROM FILE: 8736

Wave frequency	4.473	rad/sec
Wave heading	195.000	deg
Encounter frequency	7.541	rad/sec
Number of oscillations	21	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.038	2.334	326	1.000
roll	deg	0.032	0.100	242	0.043
pitch	deg	-0.680	2.310	160	0.990
yaw	deg	-0.230	0.406	166	0.174
surge	cm	-37.049	0.612	127	0.262
sway	cm	-24.116	0.304	317	0.130
heave	cm	-0.258	3.653	344	1.565
ax cog	m/s2	0.074	0.488	340	
ay cog	m/s2	0.068	0.157	259	
az cog	m/s2	-0.040	1.876	168	
ay ps fore	m/s2	0.017	0.113	261	
az ps fore	m/s2	-0.035	5.506	162	
ay ps aft	m/s2	-0.017	0.210	192	
az sb fore	m/s2	-0.066	5.460	163	
fx	N	-0.319	3.685	341	
fy 1	N	3.022	14.825	96	
fy 2	N	-1.078	17.409	286	
fy 3	N	-0.595	30.086	99	
fz 1	N	0.354	4.824	117	
fz 2	N	-0.674	3.226	342	
fy rod	N	0.737	1.624	207	
nozzle angle	deg	-2.672	4.294		
P nozzle sb	kPa	0.592	0.007		
P nozzle ps	kPa	0.761	0.020		
rpm	rpm	1035.549	0.123		
speed	m/s	1.556	0.019		

RESULTS CALCULATED FROM FILE: 8731

Wave frequency 5.063 rad/sec
Wave heading 195.000 deg
Encounter frequency 9.149 rad/sec
Number of oscillations 35

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
wave 1	cm	-0.144	2.191	163	1.000
roll	deg	-0.009	0.164	157	0.075
pitch	deg	-0.583	1.456	148	0.665
yaw	deg	-0.182	0.227	168	0.104
surge	cm	-39.644	0.209	128	0.095
sway	cm	-13.924	0.097	4	0.044
heave	cm	-0.756	2.045	266	0.933
ax cog	m/s2	0.074	0.364	326	
ay cog	m/s2	0.043	0.061	205	
az cog	m/s2	-0.027	1.617	85	
ay ps fore	m/s2	0.015	0.175	53	
az ps fore	m/s2	-0.006	4.237	128	
ay ps aft	m/s2	0.111	0.189	193	
az sb fore	m/s2	-0.033	4.340	129	
fx	N	0.146	3.504	16	
fy 1	N	2.399	4.850	9	
fy 2	N	-1.185	9.806	173	
fy 3	N	0.813	27.959	336	
fz 1	N	-0.369	3.863	182	
fz 2	N	0.047	4.506	4	
fy rod	N	0.245	1.420	206	
nozzle angle	deg	-1.272	3.940		
P nozzle sb	kPa	0.608	0.009		
P nozzle ps	kPa	0.794	0.017		
rpm	rpm	1053.163	0.164		
speed	m/s	1.618	0.019		

RESULTS CALCULATED FROM FILE: 8730

Wave frequency	5.827	rad/sec
Wave heading	195.000	deg
Encounter frequency	11.172	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	0.006	2.194	66	1.000
roll	deg	-0.018	0.242	72	0.110
pitch	deg	-0.532	0.227	68	0.103
yaw	deg	0.253	0.093	162	0.043
surge	cm	-27.380	0.012	81	0.005
sway	cm	-33.252	0.044	133	0.020
heave	cm	-0.797	0.250	20	0.114
ax cog	m/s2	0.101	0.045	258	
ay cog	m/s2	0.028	0.065	91	
az cog	m/s2	-0.033	0.350	210	
ay ps fore	m/s2	-0.251	0.276	34	
az ps fore	m/s2	-0.431	0.290	110	
ay ps aft	m/s2	0.028	0.126	184	
az sb fore	m/s2	-0.014	0.623	88	
fx	N	-0.038	2.737	38	
fy 1	N	1.062	3.172	202	
fy 2	N	-0.071	4.476	35	
fy 3	N	0.547	8.228	222	
fz 1	N	-0.104	1.045	255	
fz 2	N	-0.072	0.794	232	
fy rod	N	-0.887	0.217	251	
nozzle angle	deg	-1.375	1.428		
P nozzle sb	kPa	0.548	0.009		
P nozzle ps	kPa	0.677	0.019		
rpm	rpm	991.829	0.408		
speed	m/s	1.598	0.002		

RESULTS CALCULATED FROM FILE: 8748

Wave frequency 3.209 rad/sec
Wave heading 195.000 deg
Encounter frequency 6.587 rad/sec
Number of oscillations 10

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.048	2.719	344	1.000
roll	deg	0.054	0.275	23	0.101
pitch	deg	-3.168	3.024	207	1.112
yaw	deg	0.280	0.381	197	0.140
surge	cm	-20.902	0.297	157	0.109
sway	cm	-15.569	0.234	83	0.086
heave	cm	-0.282	5.751	3	2.115
ax cog	m/s2	0.304	0.205	19	
ay cog	m/s2	0.015	0.038	317	
az cog	m/s2	-0.038	2.259	184	
ay ps fore	m/s2	-0.016	0.077	290	
az ps fore	m/s2	-0.027	5.693	198	
ay ps aft	m/s2	0.102	0.066	235	
az sb fore	m/s2	-0.025	5.489	198	
fx	N	-0.014	3.407	1	
fy 1	N	10.368	9.497	198	
fy 2	N	-5.132	10.925	8	
fy 3	N	2.959	7.835	178	
fz 1	N	0.758	2.606	86	
fz 2	N	-0.740	1.258	237	
fy rod	N	-1.213	1.932	212	
nozzle angle	deg	-1.231	4.406		
P nozzle sb	kPa	2.574	0.027		
P nozzle ps	kPa	3.363	0.040		
rpm	rpm	2100.575	0.439		
speed	m/s	3.327	0.021		

RESULTS CALCULATED FROM FILE: 8745

Wave frequency	3.462	rad/sec
Wave heading	195.000	deg
Encounter frequency	7.223	rad/sec
Number of oscillations	10	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.094	2.801	48	1.000
roll	deg	0.044	0.335	11	0.119
pitch	deg	-3.463	3.122	171	1.115
yaw	deg	0.252	0.384	160	0.137
surge	cm	-33.013	0.838	157	0.299
sway	cm	-13.442	0.221	97	0.079
heave	cm	-0.226	7.631	335	2.725
ax cog	m/s2	0.322	0.294	343	
ay cog	m/s2	0.023	0.095	308	
az cog	m/s2	-0.050	3.823	157	
ay ps fore	m/s2	-0.026	0.105	210	
az ps fore	m/s2	-0.121	8.180	165	
ay ps aft	m/s2	0.086	0.098	213	
az sb fore	m/s2	-0.109	7.905	164	
fx	N	-0.073	4.155	338	
fy 1	N	9.663	16.117	167	
fy 2	N	-3.843	16.964	337	
fy 3	N	0.358	11.563	136	
fz 1	N	1.258	1.485	35	
fz 2	N	-1.427	1.838	157	
fy rod	N	-0.937	1.760	176	
nozzle angle	deg	-1.422	4.033		
P nozzle sb	kPa	2.572	0.054		
P nozzle ps	kPa	3.327	0.024		
rpm	rpm	2101.607	0.329		
speed	m/s	3.185	0.062		

RESULTS CALCULATED FROM FILE: 8742

Wave frequency	3.823	rad/sec
Wave heading	195.000	deg
Encounter frequency	8.492	rad/sec
Number of oscillations	13	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.111	2.386	143	1.000
roll	deg	0.020	0.333	348	0.140
pitch	deg	-3.039	1.647	150	0.690
yaw	deg	0.338	0.162	129	0.068
surge	cm	-39.567	0.869	105	0.364
sway	cm	-11.433	0.117	136	0.049
heave	cm	-0.525	5.771	284	2.419
ax cog	m/s2	0.296	0.273	313	
ay cog	m/s2	-0.018	0.127	293	
az cog	m/s2	-0.044	4.065	105	
ay ps fore	m/s2	-0.005	0.099	119	
az ps fore	m/s2	-0.054	6.814	125	
ay ps aft	m/s2	0.024	0.052	129	
az sb fore	m/s2	-0.067	6.482	123	
fx	N	-0.176	2.605	323	
fy 1	N	10.155	15.282	119	
fy 2	N	-4.709	15.012	287	
fy 3	N	0.267	7.156	75	
fz 1	N	1.040	2.032	240	
fz 2	N	-1.119	2.679	106	
fy rod	N	0.176	0.657	139	
nozzle angle	deg	-2.063	1.482		
P nozzle sb	kPa	2.596	0.019		
P nozzle ps	kPa	3.320	0.060		
rpm	rpm	2108.376	0.202		
speed	m/s	3.246	0.058		

RESULTS CALCULATED FROM FILE: 8739

Wave frequency	4.193	rad/sec
Wave heading	195.000	deg
Encounter frequency	9.819	rad/sec
Number of oscillations	14	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.108	2.034	226	1.000
roll	deg	-0.107	0.197	330	0.097
pitch	deg	-3.117	1.141	157	0.561
yaw	deg	0.392	0.104	165	0.051
surge	cm	-20.186	0.334	47	0.164
sway	cm	-12.993	0.138	110	0.068
heave	cm	-0.331	2.658	266	1.307
ax cog	m/s2	0.309	0.194	312	
ay cog	m/s2	-0.010	0.128	284	
az cog	m/s2	-0.097	2.384	87	
ay ps fore	m/s2	-0.041	0.059	21	
az ps fore	m/s2	-0.159	4.319	124	
ay ps aft	m/s2	0.056	0.050	286	
az sb fore	m/s2	-0.119	4.000	123	
fx	N	0.007	1.984	360	
fy 1	N	10.626	9.187	105	
fy 2	N	-4.836	8.612	273	
fy 3	N	0.157	3.834	55	
fz 1	N	0.775	3.410	234	
fz 2	N	-0.922	3.897	93	
fy rod	N	0.328	0.612	188	
nozzle angle	deg	-2.231	0.871		
P nozzle sb	kPa	2.559	0.030		
P nozzle ps	kPa	3.275	0.018		
rpm	rpm	2091.674	0.215		
speed	m/s	3.251	0.035		

RESULTS CALCULATED FROM FILE: 8735

Wave frequency	4.569	rad/sec
Wave heading	195.000	deg
Encounter frequency	11.348	rad/sec
Number of oscillations	16	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.047	1.701	330	1.000
roll	deg	-0.051	0.076	282	0.045
pitch	deg	-3.114	0.609	140	0.358
yaw	deg	0.362	0.088	157	0.052
surge	cm	-15.139	0.108	110	0.063
sway	cm	-15.028	0.211	78	0.124
heave	cm	-0.509	0.808	241	0.475
ax cog	m/s2	0.307	0.131	300	
ay cog	m/s2	0.022	0.094	247	
az cog	m/s2	-0.011	0.877	65	
ay ps fore	m/s2	0.016	0.057	233	
az ps fore	m/s2	-0.192	2.751	100	
ay ps aft	m/s2	-0.061	0.060	249	
az sb fore	m/s2	-0.100	2.145	116	
fx	N	-0.040	2.678	5	
fy 1	N	9.955	3.593	85	
fy 2	N	-5.047	3.977	249	
fy 3	N	2.439	3.089	35	
fz 1	N	0.073	4.708	206	
fz 2	N	-0.280	4.765	49	
fy rod	N	-0.953	0.637	195	
nozzle angle	deg	-2.230	0.960		
P nozzle sb	kPa	2.541	0.023		
P nozzle ps	kPa	3.279	0.038		
rpm	rpm	2085.998	0.512		
speed	m/s	3.295	0.014		

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Wave frequency      5.033 rad/sec
Wave heading        195.000 deg
Encounter frequency 13.339 rad/sec
Number of oscillations 18

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First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.128	2.232	138	1.000
roll	deg	-0.049	0.013	136	0.006
pitch	deg	-3.104	0.266	94	0.119
yaw	deg	0.072	0.072	138	0.032
surge	cm	-29.283	0.090	347	0.040
sway	cm	-21.873	0.225	335	0.101
heave	cm	-0.405	0.100	228	0.045
ax cog	m/s2	0.307	0.030	273	
ay cog	m/s2	-0.001	0.111	204	
az cog	m/s2	-0.017	0.152	40	
ay ps fore	m/s2	0.062	0.236	339	
az ps fore	m/s2	-0.015	1.228	85	
ay ps aft	m/s2	0.033	0.183	187	
az sb fore	m/s2	-0.005	1.449	89	
fx	N	-0.139	4.144	349	
fy 1	N	11.128	1.440	46	
fy 2	N	-5.850	1.040	184	
fy 3	N	0.984	3.185	329	
fz 1	N	0.372	5.896	170	
fz 2	N	-0.588	5.866	4	
fy rod	N	0.546	0.684	178	
nozzle angle	deg	-1.103	1.111		
P nozzle sb	kPa	2.536	0.023		
P nozzle ps	kPa	3.283	0.011		
rpm	rpm	2088.113	0.535		
speed	m/s	3.327	0.007		

RESULTS CALCULATED FROM FILE: 8749

Wave frequency	3.187	rad/sec
Wave heading	195.000	deg
Encounter frequency	7.273	rad/sec
Number of oscillations	6	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.123	2.599	350	1.000
roll	deg	0.100	0.377	24	0.145
pitch	deg	-3.162	2.137	164	0.822
yaw	deg	0.312	0.238	138	0.091
surge	cm	-40.557	1.477	119	0.568
sway	cm	-11.810	0.234	93	0.090
heave	cm	0.379	7.794	325	2.999
ax cog	m/s2	0.324	0.318	342	
ay cog	m/s2	0.040	0.127	317	
az cog	m/s2	-0.080	4.031	147	
ay ps fore	m/s2	0.021	0.133	231	
az ps fore	m/s2	-0.219	6.885	155	
ay ps aft	m/s2	0.174	0.133	235	
az sb fore	m/s2	-0.154	6.843	154	
fx	N	-0.482	3.157	322	
fy 1	N	9.224	10.631	150	
fy 2	N	-7.364	14.731	327	
fy 3	N	11.268	8.838	150	
fz 1	N	1.085	0.955	352	
fz 2	N	-0.583	1.199	152	
fy rod	N	-1.268	1.228	144	
nozzle angle	deg	-2.256	2.863		
P nozzle sb	kPa	3.416	0.049		
P nozzle ps	kPa	4.411	0.122		
rpm	rpm	2401.351	0.482		
speed	m/s	4.081	0.075		

RESULTS CALCULATED FROM FILE: 8746

Wave frequency	3.467	rad/sec
Wave heading	195.000	deg
Encounter frequency	8.242	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.139	2.748	49	1.000
roll	deg	-0.031	0.539	341	0.196
pitch	deg	-3.103	1.701	161	0.619
yaw	deg	0.402	0.145	128	0.053
surge	cm	-32.537	1.134	126	0.412
sway	cm	-15.526	0.259	100	0.094
heave	cm	0.292	7.516	291	2.735
ax cog	m/s2	0.232	0.247	315	
ay cog	m/s2	0.003	0.190	293	
az cog	m/s2	-0.046	5.053	113	
ay ps fore	m/s2	-0.013	0.123	180	
az ps fore	m/s2	-0.116	7.380	131	
ay ps aft	m/s2	0.100	0.032	306	
az sb fore	m/s2	-0.072	7.157	130	
fx	N	-0.477	3.082	310	
fy 1	N	9.533	16.079	115	
fy 2	N	-7.220	19.682	294	
fy 3	N	9.589	11.649	116	
fz 1	N	1.288	1.847	268	
fz 2	N	-1.138	3.943	104	
fy rod	N	-0.212	0.823	129	
nozzle angle	deg	-2.267	1.906		
P nozzle sb	kPa	3.338	0.400		
P nozzle ps	kPa	4.148	0.642		
rpm	rpm	2478.888	2.979		
speed	m/s	4.031	0.104		

RESULTS CALCULATED FROM FILE: 8741

Wave frequency	3.834	rad/sec
Wave heading	195.000	deg
Encounter frequency	9.819	rad/sec
Number of oscillations	10	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.139	2.333	139	1.000
roll	deg	-0.008	0.270	289	0.116
pitch	deg	-2.879	1.368	143	0.586
yaw	deg	0.393	0.082	136	0.035
surge	cm	-30.946	1.971	105	0.845
sway	cm	-11.005	0.180	96	0.077
heave	cm	0.021	3.577	247	1.534
ax cog	m/s2	0.255	0.157	285	
ay cog	m/s2	0.015	0.188	253	
az cog	m/s2	-0.046	3.348	68	
ay ps fore	m/s2	0.032	0.069	121	
az ps fore	m/s2	-0.117	5.264	105	
ay ps aft	m/s2	0.055	0.081	240	
az sb fore	m/s2	-0.067	5.089	105	
fx	N	0.373	3.331	324	
fy 1	N	9.729	12.279	75	
fy 2	N	-7.624	13.605	252	
fy 3	N	10.688	8.135	74	
fz 1	N	0.676	3.153	228	
fz 2	N	-0.408	4.895	75	
fy rod	N	-1.011	0.419	143	
nozzle angle	deg	-2.548	0.925		
P nozzle sb	kPa	3.423	0.384		
P nozzle ps	kPa	4.499	0.237		
rpm	rpm	2468.935	2.885		
speed	m/s	4.129	0.064		

RESULTS CALCULATED FROM FILE: 8740

Wave frequency	4.131	rad/sec
Wave heading	195.000	deg
Encounter frequency	10.872	rad/sec
Number of oscillations	10	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.089	2.413	230	1.000
roll	deg	0.103	0.125	286	0.052
pitch	deg	-3.061	0.998	130	0.414
yaw	deg	0.301	0.048	136	0.020
surge	cm	-42.205	0.250	174	0.104
sway	cm	-8.085	0.098	109	0.041
heave	cm	-0.087	1.804	229	0.748
ax cog	m/s2	0.314	0.182	274	
ay cog	m/s2	-0.008	0.129	286	
az cog	m/s2	-0.031	2.065	50	
ay ps fore	m/s2	0.057	0.043	79	
az ps fore	m/s2	-0.081	4.059	98	
ay ps aft	m/s2	0.029	0.178	244	
az sb fore	m/s2	-0.064	3.817	98	
fx	N	0.059	2.910	340	
fy 1	N	10.487	8.003	58	
fy 2	N	-7.588	8.559	232	
fy 3	N	10.404	5.724	44	
fz 1	N	-0.011	4.764	217	
fz 2	N	0.260	5.413	65	
fy rod	N	-1.269	0.329	185	
nozzle angle	deg	-1.965	0.398		
P nozzle sb	kPa	3.150	0.110		
P nozzle ps	kPa	4.188	0.062		
rpm	rpm	2333.919	1.501		
speed	m/s	4.010	0.029		

RESULTS CALCULATED FROM FILE: 8734

Wave frequency	4.506	rad/sec
Wave heading	195.000	deg
Encounter frequency	12.568	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.184	2.293	338	1.000
roll	deg	0.131	0.071	225	0.031
pitch	deg	-3.012	0.521	112	0.227
yaw	deg	0.405	0.087	143	0.038
surge	cm	-37.869	0.424	207	0.185
sway	cm	-11.041	0.154	46	0.067
heave	cm	-0.092	0.564	218	0.246
ax cog	m/s2	0.301	0.120	297	
ay cog	m/s2	0.033	0.075	229	
az cog	m/s2	-0.021	0.853	36	
ay ps fore	m/s2	0.013	0.112	288	
az ps fore	m/s2	-0.044	2.471	91	
ay ps aft	m/s2	0.129	0.227	191	
az sb fore	m/s2	-0.057	2.346	94	
fx	N	-0.897	3.800	335	
fy 1	N	10.394	5.369	65	
fy 2	N	-8.491	4.605	229	
fy 3	N	10.495	3.914	22	
fz 1	N	-0.031	6.703	176	
fz 2	N	-0.189	7.462	17	
fy rod	N	-0.188	0.669	185	
nozzle angle	deg	-3.203	1.061		
P nozzle sb	kPa	3.156	0.007		
P nozzle ps	kPa	4.080	0.068		
rpm	rpm	2312.763	0.319		
speed	m/s	4.030	0.016		

RESULTS CALCULATED FROM FILE: 8733

Wave frequency	5.064	rad/sec
Wave heading	195.000	deg
Encounter frequency	15.226	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.150	2.096	148	1.000
roll	deg	0.164	0.043	138	0.021
pitch	deg	-3.033	0.168	106	0.080
yaw	deg	0.329	0.054	151	0.026
surge	cm	-28.969	0.103	314	0.049
sway	cm	-10.449	0.076	343	0.036
heave	cm	-0.059	0.070	252	0.033
ax cog	m/s2	0.259	0.037	355	
ay cog	m/s2	-0.006	0.054	342	
az cog	m/s2	-0.038	0.131	90	
ay ps fore	m/s2	-0.002	0.033	295	
az ps fore	m/s2	0.074	0.766	115	
ay ps aft	m/s2	0.084	0.234	193	
az sb fore	m/s2	-0.013	1.071	104	
fx	N	-0.718	3.678	356	
fy 1	N	10.792	2.525	103	
fy 2	N	-8.656	1.147	242	
fy 3	N	9.154	2.965	343	
fz 1	N	0.349	5.655	174	
fz 2	N	-0.387	5.767	2	
fy rod	N	0.731	0.590	197	
nozzle angle	deg	-2.292	0.941		
P nozzle sb	kPa	3.106	0.037		
P nozzle ps	kPa	4.078	0.062		
rpm	rpm	2304.979	0.260		
speed	m/s	4.022	0.005		

RESULTS CALCULATED FROM FILE: 8750

Wave frequency	3.267	rad/sec
Wave heading	225.000	deg
Encounter frequency	4.580	rad/sec
Number of oscillations	15	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.047	0.893	263	1.000
roll	deg	0.053	0.451	93	0.505
pitch	deg	-0.665	0.407	281	0.456
yaw	deg	0.056	0.102	299	0.115
surge	cm	-26.118	1.023	12	1.145
sway	cm	-15.997	0.385	148	0.432
heave	cm	-0.822	0.788	22	0.882
ax cog	m/s2	0.109	0.021	71	
ay cog	m/s2	0.016	0.023	87	
az cog	m/s2	0.010	0.148	204	
ay ps fore	m/s2	-0.006	0.091	20	
az ps fore	m/s2	-0.026	0.348	255	
ay ps aft	m/s2	0.007	0.050	226	
az sb fore	m/s2	-0.011	0.232	248	
fx	N	-0.219	1.259	33	
fy 1	N	1.425	2.319	348	
fy 2	N	0.043	1.672	165	
fy 3	N	-0.166	2.371	12	
fz 1	N	0.210	2.990	209	
fz 2	N	-0.128	2.004	18	
fy rod	N	-0.711	0.462	295	
nozzle angle	deg	-1.372	2.000		
P nozzle sb	kPa	0.505	0.009		
P nozzle ps	kPa	0.668	0.002		
rpm	rpm	1034.712	0.383		
speed	m/s	1.710	0.005		

RESULTS CALCULATED FROM FILE: 8751

Wave frequency	3.193	rad/sec
Wave heading	225.000	deg
Encounter frequency	4.379	rad/sec
Number of oscillations	16	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.084	1.533	258	1.000
roll	deg	0.079	0.814	82	0.531
pitch	deg	-0.588	0.714	266	0.466
yaw	deg	0.017	0.272	170	0.177
surge	cm	-41.601	0.316	109	0.206
sway	cm	-9.981	0.564	99	0.368
heave	cm	-0.980	1.425	6	0.929
ax cog	m/s2	0.114	0.054	76	
ay cog	m/s2	0.012	0.071	46	
az cog	m/s2	-0.002	0.278	184	
ay ps fore	m/s2	0.021	0.204	10	
az ps fore	m/s2	-0.012	0.586	238	
ay ps aft	m/s2	0.006	0.065	163	
az sb fore	m/s2	-0.012	0.404	225	
fx	N	-0.056	2.291	9	
fy 1	N	1.337	3.874	323	
fy 2	N	0.403	2.708	145	
fy 3	N	-0.537	5.002	357	
fz 1	N	0.135	4.729	193	
fz 2	N	-0.192	3.594	1	
fy rod	N	-0.650	1.000	223	
nozzle angle	deg	-0.658	4.501		
P nozzle sb	kPa	0.515	0.007		
P nozzle ps	kPa	0.679	0.002		
rpm	rpm	977.452	0.051		
speed	m/s	1.612	0.017		

RESULTS CALCULATED FROM FILE: 8754

Wave frequency 3.543 rad/sec
Wave heading 225.000 deg
Encounter frequency 5.022 rad/sec
Number of oscillations 16

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.074	1.136	313	1.000
roll	deg	-0.100	0.880	54	0.775
pitch	deg	-0.621	0.740	257	0.651
yaw	deg	0.201	0.240	171	0.211
surge	cm	-32.223	0.193	138	0.170
sway	cm	-26.113	0.316	110	0.278
heave	cm	-0.524	1.118	6	0.985
ax cog	m/s2	0.139	0.045	75	
ay cog	m/s2	0.027	0.087	33	
az cog	m/s2	-0.010	0.269	185	
ay ps fore	m/s2	0.033	0.227	356	
az ps fore	m/s2	-0.048	0.751	234	
ay ps aft	m/s2	-0.031	0.072	197	
az sb fore	m/s2	-0.023	0.478	233	
fx	N	0.043	2.849	356	
fy 1	N	1.870	3.469	318	
fy 2	N	-0.235	2.708	130	
fy 3	N	-2.300	5.012	344	
fz 1	N	0.028	6.180	178	
fz 2	N	-0.138	3.925	347	
fy rod	N	0.626	1.229	224	
nozzle angle	deg	-0.320	4.471		
P nozzle sb	kPa	0.550	0.007		
P nozzle ps	kPa	0.717	0.001		
rpm	rpm	1005.241	0.131		
speed	m/s	1.631	0.013		

RESULTS CALCULATED FROM FILE: 8770

Wave frequency	3.652	rad/sec
Wave heading	225.000	deg
Encounter frequency	5.301	rad/sec
Number of oscillations	18	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.084	2.414	327	1.000
roll	deg	-0.136	1.363	42	0.565
pitch	deg	-0.481	1.718	251	0.712
yaw	deg	0.000	0.304	162	0.126
surge	cm	-21.703	0.445	72	0.184
sway	cm	-13.359	0.563	125	0.233
heave	cm	-0.886	2.271	5	0.941
ax cog	m/s2	0.114	0.130	56	
ay cog	m/s2	0.016	0.175	21	
az cog	m/s2	-0.013	0.607	185	
ay ps fore	m/s2	-0.019	0.393	343	
az ps fore	m/s2	-0.040	1.881	229	
ay ps aft	m/s2	-0.026	0.089	102	
az sb fore	m/s2	-0.014	1.390	232	
fx	N	-0.663	4.647	351	
fy 1	N	1.161	7.778	314	
fy 2	N	0.243	6.596	121	
fy 3	N	0.095	11.339	320	
fz 1	N	0.414	10.774	174	
fz 2	N	-0.396	6.161	342	
fy rod	N	-0.995	1.575	225	
nozzle angle	deg	-0.213	6.345		
P nozzle sb	kPa	0.412	0.005		
P nozzle ps	kPa	0.763	0.003		
rpm	rpm	1029.949	0.088		
speed	m/s	1.710	0.023		

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Wave frequency      3.881 rad/sec
Wave heading        225.000 deg
Encounter frequency  5.667 rad/sec
Number of oscillations 18

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First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.058	2.021	22	1.000
roll	deg	0.039	0.974	65	0.482
pitch	deg	-0.709	1.551	256	0.767
yaw	deg	0.167	0.192	189	0.095
surge	cm	-42.835	0.655	115	0.324
sway	cm	-12.751	0.232	177	0.115
heave	cm	-0.650	1.753	17	0.867
ax cog	m/s2	0.132	0.140	67	
ay cog	m/s2	-0.001	0.160	49	
az cog	m/s2	-0.012	0.511	195	
ay ps fore	m/s2	-0.000	0.281	14	
az ps fore	m/s2	-0.052	1.805	240	
ay ps aft	m/s2	-0.006	0.053	76	
az sb fore	m/s2	-0.042	1.415	239	
fx	N	0.352	3.457	23	
fy 1	N	0.660	5.645	323	
fy 2	N	0.563	5.394	127	
fy 3	N	-0.060	9.263	322	
fz 1	N	0.242	8.255	206	
fz 2	N	-0.479	4.087	16	
fy rod	N	-1.117	1.080	258	
nozzle angle	deg	-1.069	3.956		
P nozzle sb	kPa	0.544	0.007		
P nozzle ps	kPa	0.542	0.011		
rpm	rpm	1005.918	0.188		
speed	m/s	1.643	0.018		

RESULTS CALCULATED FROM FILE: 8784

Wave frequency	4.055	rad/sec
Wave heading	225.000	deg
Encounter frequency	5.979	rad/sec
Number of oscillations	20	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.084	2.324	51	1.000
roll	deg	0.069	1.816	34	0.782
pitch	deg	-0.591	2.205	229	0.949
yaw	deg	0.070	0.351	177	0.151
surge	cm	-33.699	0.451	71	0.194
sway	cm	-21.049	0.844	177	0.363
heave	cm	-0.581	2.152	3	0.926
ax cog	m/s2	0.129	0.203	45	
ay cog	m/s2	0.030	0.330	16	
az cog	m/s2	-0.018	0.697	180	
ay ps fore	m/s2	-0.023	0.668	354	
az ps fore	m/s2	-0.018	3.057	217	
ay ps aft	m/s2	0.014	0.169	13	
az sb fore	m/s2	-0.022	2.202	218	
fx	N	-0.325	7.589	8	
fy 1	N	1.064	6.467	286	
fy 2	N	0.263	7.880	87	
fy 3	N	-2.044	13.235	283	
fz 1	N	0.952	20.007	190	
fz 2	N	-1.230	9.171	13	
fy rod	N	0.407	1.793	238	
nozzle angle	deg	-1.307	7.446		
P nozzle sb	kPa	0.576	0.009		
P nozzle ps	kPa	0.754	0.013		
rpm	rpm	1027.238	0.205		
speed	m/s	1.620	0.017		

RESULTS CALCULATED FROM FILE: 8760

Wave frequency	4.114	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.083	rad/sec
Number of oscillations	20	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.124	2.103	70	1.000
roll	deg	0.246	1.895	26	0.901
pitch	deg	-0.735	1.870	234	0.889
yaw	deg	0.261	0.214	159	0.102
surge	cm	-44.547	0.194	142	0.092
sway	cm	-12.446	0.938	152	0.446
heave	cm	-0.879	1.772	12	0.843
ax cog	m/s2	0.094	0.199	44	
ay cog	m/s2	0.038	0.443	9	
az cog	m/s2	-0.026	0.624	190	
ay ps fore	m/s2	0.000	0.771	350	
az ps fore	m/s2	-0.032	2.814	220	
ay ps aft	m/s2	0.008	0.179	344	
az sb fore	m/s2	-0.022	1.925	228	
fx	N	-0.311	8.055	5	
fy 1	N	-0.333	3.753	286	
fy 2	N	0.930	6.055	85	
fy 3	N	-0.713	11.308	286	
fz 1	N	0.876	22.959	185	
fz 2	N	-1.289	10.788	10	
fy rod	N	-0.724	1.602	234	
nozzle angle	deg	-1.477	4.975		
P nozzle sb	kPa	0.556	0.010		
P nozzle ps	kPa	0.720	0.014		
rpm	rpm	1005.522	0.265		
speed	m/s	1.611	0.013		

RESULTS CALCULATED FROM FILE: 8789

Wave frequency	4.204	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.278	rad/sec
Number of oscillations	20	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.104	2.196	83	1.000
roll	deg	0.030	1.812	9	0.825
pitch	deg	-0.524	2.172	230	0.989
yaw	deg	-0.021	0.209	179	0.095
surge	cm	-35.700	0.371	90	0.169
sway	cm	-12.371	0.821	153	0.374
heave	cm	-0.717	2.034	15	0.926
ax cog	m/s2	0.086	0.237	44	
ay cog	m/s2	0.039	0.418	352	
az cog	m/s2	-0.028	0.716	192	
ay ps fore	m/s2	-0.004	0.804	334	
az ps fore	m/s2	-0.017	3.311	217	
ay ps aft	m/s2	0.036	0.204	351	
az sb fore	m/s2	-0.031	2.527	227	
fx	N	-0.411	8.435	355	
fy 1	N	-0.009	2.115	281	
fy 2	N	0.701	6.106	74	
fy 3	N	-1.337	12.459	276	
fz 1	N	0.952	24.847	170	
fz 2	N	-1.273	11.888	358	
fy rod	N	-0.114	1.600	238	
nozzle angle	deg	-0.896	5.957		
P nozzle sb	kPa	0.595	0.010		
P nozzle ps	kPa	0.771	0.008		
rpm	rpm	1038.604	0.067		
speed	m/s	1.623	0.012		

RESULTS CALCULATED FROM FILE: 8765

Wave frequency	4.589	rad/sec
Wave heading	225.000	deg
Encounter frequency	7.034	rad/sec
Number of oscillations	24	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.058	1.990	174	1.000
roll	deg	0.033	0.560	349	0.282
pitch	deg	-0.706	2.238	212	1.125
yaw	deg	0.608	0.260	202	0.131
surge	cm	-43.508	0.072	160	0.036
sway	cm	-14.954	0.539	135	0.271
heave	cm	-0.512	2.302	21	1.157
ax cog	m/s2	0.106	0.320	25	
ay cog	m/s2	0.035	0.186	344	
az cog	m/s2	-0.074	0.897	200	
ay ps fore	m/s2	-0.022	0.362	315	
az ps fore	m/s2	0.062	3.930	206	
ay ps aft	m/s2	-0.038	0.169	302	
az sb fore	m/s2	0.014	3.619	211	
fx	N	-1.386	4.837	3	
fy 1	N	0.258	5.294	112	
fy 2	N	0.237	6.003	5	
fy 3	N	-2.681	11.650	213	
fz 1	N	1.214	12.682	163	
fz 2	N	-1.816	7.002	7	
fy rod	N	0.802	1.197	254	
nozzle angle	deg	-3.514	4.930		
P nozzle sb	kPa	0.426	0.006		
P nozzle ps	kPa	0.775	0.013		
rpm	rpm	1053.751	0.149		
speed	m/s	1.608	0.006		

RESULTS CALCULATED FROM FILE: 8766

Wave frequency	5.076	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.167	rad/sec
Number of oscillations	20	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.092	2.715	253	1.000
roll	deg	-0.019	0.299	347	0.110
pitch	deg	-0.643	1.553	149	0.572
yaw	deg	1.977	0.358	152	0.132
surge	cm	-4.732	0.360	137	0.133
sway	cm	-33.818	0.622	37	0.229
heave	cm	-0.594	3.833	307	1.412
ax cog	m/s2	0.069	0.335	326	
ay cog	m/s2	0.045	0.165	277	
az cog	m/s2	-0.034	2.353	129	
ay ps fore	m/s2	0.042	0.332	297	
az ps fore	m/s2	-0.043	5.275	140	
ay ps aft	m/s2	-0.000	0.245	200	
az sb fore	m/s2	-0.032	4.987	139	
fx	N	-0.684	7.984	336	
fy 1	N	0.388	19.617	37	
fy 2	N	-0.460	15.922	218	
fy 3	N	1.822	25.678	16	
fz 1	N	-0.078	15.889	137	
fz 2	N	-0.887	12.628	340	
fy rod	N	-0.608	1.157	188	
nozzle angle	deg	-5.173	3.309		
P nozzle sb	kPa	0.664	0.008		
P nozzle ps	kPa	0.847	0.015		
rpm	rpm	1124.324	0.223		
speed	m/s	1.661	0.026		

RESULTS CALCULATED FROM FILE: 8769

Wave frequency	5.927	rad/sec
Wave heading	225.000	deg
Encounter frequency	10.057	rad/sec
Number of oscillations	33	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.018	2.147	165	1.000
roll	deg	-0.071	0.337	94	0.157
pitch	deg	-0.693	0.572	113	0.267
yaw	deg	0.746	0.164	173	0.076
surge	cm	-39.126	0.028	105	0.013
sway	cm	-21.680	0.043	31	0.020
heave	cm	-0.568	0.370	231	0.172
ax cog	m/s2	0.088	0.144	293	
ay cog	m/s2	0.011	0.016	280	
az cog	m/s2	-0.044	0.324	43	
ay ps fore	m/s2	0.048	0.451	30	
az ps fore	m/s2	-0.016	1.441	104	
ay ps aft	m/s2	0.021	0.243	211	
az sb fore	m/s2	-0.040	1.826	102	
fx	N	-0.513	6.561	35	
fy 1	N	0.674	10.180	48	
fy 2	N	0.239	4.108	151	
fy 3	N	-0.126	16.524	269	
fz 1	N	-1.138	10.912	194	
fz 2	N	0.749	9.078	10	
fy rod	N	-0.328	0.900	255	
nozzle angle	deg	-7.594	2.054		
P nozzle sb	kPa	0.404	0.010		
P nozzle ps	kPa	0.734	0.011		
rpm	rpm	1018.162	0.083		
speed	m/s	1.627	0.007		

RESULTS CALCULATED FROM FILE: 8752

Wave frequency	3.177	rad/sec
Wave heading	225.000	deg
Encounter frequency	5.636	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.219	1.521	242	1.000
roll	deg	0.238	0.835	57	0.549
pitch	deg	-3.424	0.985	231	0.648
yaw	deg	0.207	0.228	186	0.150
surge	cm	-8.637	0.546	253	0.359
sway	cm	-20.710	0.852	85	0.560
heave	cm	-0.481	1.928	3	1.268
ax cog	m/s2	0.313	0.052	31	
ay cog	m/s2	0.002	0.063	1	
az cog	m/s2	-0.033	0.564	182	
ay ps fore	m/s2	-0.047	0.162	338	
az ps fore	m/s2	-0.016	1.480	213	
ay ps aft	m/s2	0.095	0.024	239	
az sb fore	m/s2	-0.029	1.077	207	
fx	N	0.040	3.104	1	
fy 1	N	11.112	2.363	303	
fy 2	N	-5.981	1.825	70	
fy 3	N	2.561	2.087	315	
fz 1	N	0.255	5.329	170	
fz 2	N	-0.264	3.142	341	
fy rod	N	-0.109	1.361	209	
nozzle angle	deg	-0.372	2.906		
P nozzle sb	kPa	2.552	0.016		
P nozzle ps	kPa	3.318	0.016		
rpm	rpm	2091.425	0.140		
speed	m/s	3.371	0.008		

RESULTS CALCULATED FROM FILE: 8755

Wave frequency	3.478	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.380	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.131	1.666	298	1.000
roll	deg	0.030	1.058	24	0.635
pitch	deg	-3.097	1.364	208	0.819
yaw	deg	-0.077	0.266	170	0.160
surge	cm	-25.345	0.322	228	0.193
sway	cm	-29.473	0.166	221	0.100
heave	cm	-0.307	2.627	357	1.577
ax cog	m/s2	0.318	0.070	7	
ay cog	m/s2	0.021	0.124	336	
az cog	m/s2	-0.048	1.012	179	
ay ps fore	m/s2	-0.045	0.187	319	
az ps fore	m/s2	-0.031	2.533	199	
ay ps aft	m/s2	-0.029	0.092	274	
az sb fore	m/s2	-0.008	2.078	196	
fx	N	-0.210	5.193	343	
fy 1	N	9.075	3.233	249	
fy 2	N	-5.244	3.912	37	
fy 3	N	3.721	2.489	275	
fz 1	N	0.877	8.794	144	
fz 2	N	-0.671	4.287	319	
fy rod	N	-1.238	1.883	200	
nozzle angle	deg	-0.942	3.642		
P nozzle sb	kPa	2.492	0.018		
P nozzle ps	kPa	3.249	0.030		
rpm	rpm	2073.356	0.271		
speed	m/s	3.320	0.009		

RESULTS CALCULATED FROM FILE: 8771

Wave frequency	3.650	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.772	rad/sec
Number of oscillations	10	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.067	2.348	332	1.000
roll	deg	0.126	1.186	18	0.505
pitch	deg	-3.296	2.246	204	0.957
yaw	deg	0.254	0.255	167	0.108
surge	cm	-32.962	0.943	192	0.402
sway	cm	-13.821	0.508	108	0.217
heave	cm	-0.327	4.680	360	1.993
ax cog	m/s2	0.289	0.157	16	
ay cog	m/s2	0.047	0.226	332	
az cog	m/s2	-0.070	1.957	183	
ay ps fore	m/s2	-0.007	0.316	312	
az ps fore	m/s2	-0.050	5.542	191	
ay ps aft	m/s2	0.056	0.016	164	
az sb fore	m/s2	-0.024	4.258	194	
fx	N	-0.476	5.854	346	
fy 1	N	9.305	3.533	256	
fy 2	N	-5.274	5.800	26	
fy 3	N	2.372	2.581	244	
fz 1	N	1.735	8.811	140	
fz 2	N	-1.646	3.063	304	
fy rod	N	-0.179	1.798	200	
nozzle angle	deg	-1.921	3.681		
P nozzle sb	kPa	2.476	0.018		
P nozzle ps	kPa	3.179	0.035		
rpm	rpm	2059.902	0.102		
speed	m/s	3.243	0.028		

RESULTS CALCULATED FROM FILE: 8758

Wave frequency	3.843	rad/sec
Wave heading	225.000	deg
Encounter frequency	7.307	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.120	2.563	6	1.000
roll	deg	0.158	1.034	24	0.403
pitch	deg	-3.186	2.209	181	0.862
yaw	deg	0.392	0.396	166	0.155
surge	cm	-29.090	1.181	169	0.461
sway	cm	-30.812	0.447	104	0.174
heave	cm	-0.162	5.680	343	2.216
ax cog	m/s2	0.292	0.184	349	
ay cog	m/s2	-0.007	0.137	335	
az cog	m/s2	-0.028	2.798	165	
ay ps fore	m/s2	-0.014	0.186	322	
az ps fore	m/s2	-0.057	6.086	175	
ay ps aft	m/s2	0.027	0.129	245	
az sb fore	m/s2	-0.055	5.488	171	
fx	N	-1.343	6.658	353	
fy 1	N	7.781	3.130	194	
fy 2	N	-4.189	6.945	350	
fy 3	N	3.193	1.444	120	
fz 1	N	2.110	6.624	153	
fz 2	N	-2.331	1.253	329	
fy rod	N	-1.541	1.737	189	
nozzle angle	deg	-0.949	3.990		
P nozzle sb	kPa	2.528	0.022		
P nozzle ps	kPa	3.259	0.024		
rpm	rpm	2092.127	0.322		
speed	m/s	3.246	0.042		

RESULTS CALCULATED FROM FILE: 8785

Wave frequency	4.099	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.057	rad/sec
Number of oscillations	12	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.175	2.100	55	1.000
roll	deg	-0.131	1.429	16	0.681
pitch	deg	-3.298	1.625	155	0.774
yaw	deg	0.290	0.314	136	0.149
surge	cm	-26.287	0.650	119	0.309
sway	cm	-33.815	0.940	165	0.448
heave	cm	-0.196	5.573	309	2.654
ax cog	m/s2	0.291	0.240	324	
ay cog	m/s2	-0.019	0.263	335	
az cog	m/s2	0.012	3.333	131	
ay ps fore	m/s2	-0.076	0.242	342	
az ps fore	m/s2	0.044	6.177	145	
ay ps aft	m/s2	0.025	0.058	273	
az sb fore	m/s2	0.043	5.557	137	
fx	N	-1.001	7.154	357	
fy 1	N	7.734	6.250	114	
fy 2	N	-4.503	8.620	303	
fy 3	N	3.205	3.519	34	
fz 1	N	1.967	9.310	191	
fz 2	N	-2.333	4.402	82	
fy rod	N	-0.653	1.489	170	
nozzle angle	deg	-1.231	3.085		
P nozzle sb	kPa	2.546	0.021		
P nozzle ps	kPa	3.320	0.038		
rpm	rpm	2101.780	0.282		
speed	m/s	3.259	0.057		

RESULTS CALCULATED FROM FILE: 8761

Wave frequency	4.199	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.308	rad/sec
Number of oscillations	11	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.090	1.721	85	1.000
roll	deg	-0.141	1.283	23	0.745
pitch	deg	-3.262	1.225	181	0.712
yaw	deg	0.701	0.081	140	0.047
surge	cm	-39.728	0.787	172	0.457
sway	cm	-5.888	0.715	138	0.415
heave	cm	-0.553	4.392	331	2.551
ax cog	m/s2	0.321	0.170	354	
ay cog	m/s2	-0.026	0.328	342	
az cog	m/s2	0.019	2.770	154	
ay ps fore	m/s2	-0.042	0.303	342	
az ps fore	m/s2	0.093	5.157	170	
ay ps aft	m/s2	0.079	0.094	295	
az sb fore	m/s2	0.025	4.313	161	
fx	N	-1.307	7.111	16	
fy 1	N	7.893	6.883	130	
fy 2	N	-4.873	7.699	319	
fy 3	N	1.942	4.361	57	
fz 1	N	1.969	11.069	193	
fz 2	N	-2.461	5.087	84	
fy rod	N	0.354	0.821	230	
nozzle angle	deg	-3.098	1.339		
P nozzle sb	kPa	2.516	0.027		
P nozzle ps	kPa	3.275	0.048		
rpm	rpm	2093.419	0.162		
speed	m/s	3.227	0.048		

Wave frequency	4.280	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.563	rad/sec
Number of oscillations	13	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
wave 1	cm	-0.092	1.787	95	1.000
roll	deg	-0.245	1.227	354	0.687
pitch	deg	-3.093	1.029	161	0.576
yaw	deg	0.268	0.139	150	0.078
surge	cm	-30.705	0.622	153	0.348
sway	cm	-25.618	0.852	159	0.477
heave	cm	-0.385	3.877	302	2.170
ax cog	m/s2	0.285	0.175	330	
ay cog	m/s2	0.010	0.278	317	
az cog	m/s2	0.038	2.580	124	
ay ps fore	m/s2	-0.062	0.253	333	
az ps fore	m/s2	0.029	4.688	143	
ay ps aft	m/s2	0.064	0.145	284	
az sb fore	m/s2	0.011	3.775	134	
fx	N	-1.434	7.282	352	
fy 1	N	7.765	7.305	97	
fy 2	N	-4.403	7.429	286	
fy 3	N	2.029	4.952	26	
fz 1	N	2.546	10.246	174	
fz 2	N	-2.995	6.053	67	
fy rod	N	-0.074	1.034	201	
nozzle angle	deg	-1.948	2.119		
P nozzle sb	kPa	2.543	0.024		
P nozzle ps	kPa	3.315	0.055		
rpm	rpm	2098.948	0.412		
speed	m/s	3.237	0.045		

RESULTS CALCULATED FROM FILE: 8764

Wave frequency	4.522	rad/sec
Wave heading	225.000	deg
Encounter frequency	9.196	rad/sec
Number of oscillations	12	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.155	2.729	165	1.000
roll	deg	-0.113	0.914	326	0.335
pitch	deg	-2.968	1.531	145	0.561
yaw	deg	0.868	0.047	206	0.017
surge	cm	-50.624	1.108	78	0.406
sway	cm	-3.551	0.411	104	0.151
heave	cm	-0.447	4.518	258	1.656
ax cog	m/s2	0.331	0.232	304	
ay cog	m/s2	0.008	0.333	278	
az cog	m/s2	-0.024	3.759	80	
ay ps fore	m/s2	-0.017	0.189	329	
az ps fore	m/s2	-0.039	6.473	112	
ay ps aft	m/s2	0.057	0.218	249	
az sb fore	m/s2	-0.045	5.583	108	
fx	N	-0.009	7.760	1	
fy 1	N	7.804	18.206	56	
fy 2	N	-3.570	14.067	231	
fy 3	N	-0.778	11.798	341	
fz 1	N	0.816	12.986	185	
fz 2	N	-1.536	12.143	51	
fy rod	N	0.697	1.226	244	
nozzle angle	deg	-4.248	1.898		
P nozzle sb	kPa	2.488	0.155		
P nozzle ps	kPa	3.302	0.056		
rpm	rpm	2100.875	1.157		
speed	m/s	3.168	0.056		

RESULTS CALCULATED FROM FILE: 8767

Wave frequency	5.173	rad/sec
Wave heading	225.000	deg
Encounter frequency	11.730	rad/sec
Number of oscillations	14	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.131	1.979	288	1.000
roll	deg	-0.048	0.193	287	0.097
pitch	deg	-3.167	0.475	129	0.240
yaw	deg	0.508	0.147	170	0.074
surge	cm	-15.457	0.080	79	0.040
sway	cm	-24.071	0.454	51	0.229
heave	cm	-0.422	0.646	234	0.326
ax cog	m/s2	0.277	0.087	284	
ay cog	m/s2	0.054	0.108	249	
az cog	m/s2	-0.056	0.803	51	
ay ps fore	m/s2	-0.033	0.268	322	
az ps fore	m/s2	-0.120	2.050	103	
ay ps aft	m/s2	0.016	0.220	218	
az sb fore	m/s2	-0.106	1.703	103	
fx	N	-0.160	7.398	356	
fy 1	N	9.183	10.775	39	
fy 2	N	-5.252	6.932	197	
fy 3	N	4.222	10.028	327	
fz 1	N	0.455	12.965	186	
fz 2	N	-0.986	13.484	31	
fy rod	N	-0.627	1.297	207	
nozzle angle	deg	-3.832	2.387		
P nozzle sb	kPa	2.619	0.030		
P nozzle ps	kPa	3.384	0.012		
rpm	rpm	2133.600	0.312		
speed	m/s	3.393	0.011		

RESULTS CALCULATED FROM FILE: 8753

Wave frequency	3.356	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.781	rad/sec
Number of oscillations	6	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.051	0.851	278	1.000
roll	deg	-0.051	0.634	92	0.745
pitch	deg	-2.977	0.740	263	0.869
yaw	deg	0.349	0.113	202	0.133
surge	cm	-26.269	0.864	216	1.015
sway	cm	-13.892	0.193	165	0.226
heave	cm	0.061	1.679	55	1.974
ax cog	m/s2	0.317	0.137	69	
ay cog	m/s2	0.084	0.090	76	
az cog	m/s2	-0.028	0.613	242	
ay ps fore	m/s2	0.037	0.090	323	
az ps fore	m/s2	-0.065	1.453	255	
ay ps aft	m/s2	0.081	0.045	124	
az sb fore	m/s2	-0.037	1.150	256	
fx	N	-0.693	2.700	49	
fy 1	N	9.222	2.247	318	
fy 2	N	-8.585	2.284	101	
fy 3	N	11.377	1.765	321	
fz 1	N	1.629	3.456	222	
fz 2	N	-0.935	1.014	22	
fy rod	N	0.459	0.548	228	
nozzle angle	deg	-2.641	1.421		
P nozzle sb	kPa	3.345	0.012		
P nozzle ps	kPa	4.349	0.019		
rpm	rpm	2379.368	0.174		
speed	m/s	4.211	0.005		

RESULTS CALCULATED FROM FILE: 8756

Wave frequency	3.470	rad/sec
Wave heading	225.000	deg
Encounter frequency	6.891	rad/sec
Number of oscillations	6	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.270	1.710	303	1.000
roll	deg	-0.090	1.204	11	0.704
pitch	deg	-3.118	1.385	183	0.810
yaw	deg	0.066	0.246	133	0.144
surge	cm	-37.834	0.434	155	0.254
sway	cm	-24.626	0.046	163	0.027
heave	cm	0.328	3.784	344	2.212
ax cog	m/s2	0.298	0.108	350	
ay cog	m/s2	0.008	0.173	347	
az cog	m/s2	-0.016	1.715	167	
ay ps fore	m/s2	0.065	0.206	304	
az ps fore	m/s2	-0.104	3.706	176	
ay ps aft	m/s2	0.023	0.063	288	
az sb fore	m/s2	-0.046	3.021	174	
fx	N	-0.979	6.624	335	
fy 1	N	8.979	2.810	231	
fy 2	N	-7.762	4.916	8	
fy 3	N	11.052	2.177	255	
fz 1	N	2.290	7.721	148	
fz 2	N	-1.692	1.624	351	
fy rod	N	-0.822	1.681	174	
nozzle angle	deg	-3.038	3.188		
P nozzle sb	kPa	3.045	0.021		
P nozzle ps	kPa	3.947	0.028		
rpm	rpm	2273.738	0.402		
speed	m/s	3.934	0.033		

RESULTS CALCULATED FROM FILE: 8772

Wave frequency	3.729	rad/sec
Wave heading	225.000	deg
Encounter frequency	7.772	rad/sec
Number of oscillations	7	

First harmonic component

channel	dimension	mean	amplitude	phase	RA0
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wave 1	cm	-0.148	2.099	355	1.000
roll	deg	0.041	1.170	6	0.558
pitch	deg	-3.035	1.413	184	0.673
yaw	deg	0.330	0.151	124	0.072
surge	cm	-47.286	1.581	152	0.753
sway	cm	-13.967	0.533	112	0.254
heave	cm	0.074	5.746	340	2.738
ax cog	m/s2	0.317	0.225	341	
ay cog	m/s2	-0.022	0.262	332	
az cog	m/s2	-0.065	3.152	163	
ay ps fore	m/s2	0.034	0.197	318	
az ps fore	m/s2	-0.185	5.467	172	
ay ps aft	m/s2	0.090	0.180	295	
az sb fore	m/s2	-0.118	4.720	170	
fx	N	-1.329	6.159	342	
fy 1	N	8.260	3.849	136	
fy 2	N	-7.854	7.716	339	
fy 3	N	10.933	1.091	196	
fz 1	N	1.496	5.878	158	
fz 2	N	-1.512	2.353	98	
fy rod	N	0.138	0.975	166	
nozzle angle	deg	-2.531	1.933		
P nozzle sb	kPa	3.199	0.034		
P nozzle ps	kPa	4.095	0.078		
rpm	rpm	2330.021	0.444		
speed	m/s	4.024	0.054		

RESULTS CALCULATED FROM FILE: 8757

Wave frequency	3.822	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.233	rad/sec
Number of oscillations	6	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.149	2.511	353	1.000
roll	deg	-0.062	1.254	356	0.500
pitch	deg	-2.937	1.338	170	0.533
yaw	deg	0.684	0.138	141	0.055
surge	cm	-14.021	0.596	143	0.237
sway	cm	-12.367	0.459	109	0.183
heave	cm	0.184	6.255	301	2.491
ax cog	m/s2	0.202	0.229	317	
ay cog	m/s2	0.024	0.354	312	
az cog	m/s2	-0.035	4.149	123	
ay ps fore	m/s2	0.057	0.079	276	
az ps fore	m/s2	-0.241	5.983	143	
ay ps aft	m/s2	0.083	0.215	275	
az sb fore	m/s2	-0.054	5.542	136	
fx	N	-1.505	7.162	348	
fy 1	N	6.313	8.975	89	
fy 2	N	-7.381	11.891	293	
fy 3	N	10.991	2.909	108	
fz 1	N	1.813	7.319	200	
fz 2	N	-2.003	7.581	91	
fy rod	N	0.874	1.011	174	
nozzle angle	deg	-3.034	1.776		
P nozzle sb	kPa	3.473	0.033		
P nozzle ps	kPa	4.463	0.082		
rpm	rpm	2434.634	0.449		
speed	m/s	4.184	0.075		

RESULTS CALCULATED FROM FILE: 8786

Wave frequency	4.056	rad/sec
Wave heading	225.000	deg
Encounter frequency	8.837	rad/sec
Number of oscillations	9	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.217	2.264	66	1.000
roll	deg	0.005	1.358	338	0.600
pitch	deg	-3.213	1.279	160	0.565
yaw	deg	0.265	0.057	288	0.025
surge	cm	-55.668	0.818	118	0.361
sway	cm	-13.504	0.824	81	0.364
heave	cm	0.182	4.956	273	2.190
ax cog	m/s2	0.225	0.289	291	
ay cog	m/s2	0.023	0.428	298	
az cog	m/s2	-0.030	3.776	95	
ay ps fore	m/s2	0.036	0.227	288	
az ps fore	m/s2	-0.206	5.671	125	
ay ps aft	m/s2	0.063	0.292	285	
az sb fore	m/s2	-0.066	4.836	115	
fx	N	-0.939	8.389	353	
fy 1	N	8.440	10.644	68	
fy 2	N	-7.120	11.160	262	
fy 3	N	8.944	3.441	57	
fz 1	N	0.212	13.353	196	
fz 2	N	-0.730	13.163	73	
fy rod	N	0.471	0.943	260	
nozzle angle	deg	-2.778	1.582		
P nozzle sb	kPa	3.241	0.130		
P nozzle ps	kPa	4.173	0.055		
rpm	rpm	2372.089	0.979		
speed	m/s	4.028	0.063		

RESULTS CALCULATED FROM FILE: 8762

Wave frequency	4.117	rad/sec
Wave heading	225.000	deg
Encounter frequency	9.089	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.164	2.043	64	1.000
roll	deg	0.054	1.327	327	0.650
pitch	deg	-3.004	1.158	162	0.567
yaw	deg	0.651	0.097	204	0.048
surge	cm	-34.260	0.573	92	0.281
sway	cm	-7.058	0.748	101	0.366
heave	cm	-0.149	4.006	271	1.961
ax cog	m/s2	0.285	0.212	294	
ay cog	m/s2	0.029	0.445	292	
az cog	m/s2	-0.054	3.344	94	
ay ps fore	m/s2	0.046	0.220	314	
az ps fore	m/s2	-0.096	5.331	124	
ay ps aft	m/s2	0.028	0.301	251	
az sb fore	m/s2	-0.076	4.087	118	
fx	N	-1.138	9.928	347	
fy 1	N	7.983	11.785	71	
fy 2	N	-7.549	11.124	263	
fy 3	N	9.499	4.119	61	
fz 1	N	-0.021	14.483	188	
fz 2	N	-0.539	15.191	65	
fy rod	N	0.481	1.501	232	
nozzle angle	deg	-3.504	2.341		
P nozzle sb	kPa	3.291	0.104		
P nozzle ps	kPa	4.275	0.067		
rpm	rpm	2371.472	0.918		
speed	m/s	4.061	0.053		

Wave frequency	4.188	rad/sec
Wave heading	225.000	deg
Encounter frequency	9.426	rad/sec
Number of oscillations	9	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.165	2.097	65	1.000
roll	deg	-0.208	1.013	296	0.483
pitch	deg	-2.901	1.025	140	0.489
yaw	deg	0.460	0.211	160	0.101
surge	cm	-16.167	1.601	90	0.764
sway	cm	-22.276	0.437	60	0.208
heave	cm	0.067	3.162	246	1.508
ax cog	m/s2	0.231	0.197	273	
ay cog	m/s2	0.038	0.359	266	
az cog	m/s2	-0.028	2.792	67	
ay ps fore	m/s2	0.060	0.086	288	
az ps fore	m/s2	-0.166	4.448	102	
ay ps aft	m/s2	0.005	0.235	223	
az sb fore	m/s2	-0.031	3.565	97	
fx	N	-1.083	10.940	326	
fy 1	N	6.900	13.559	37	
fy 2	N	-7.110	11.596	228	
fy 3	N	11.507	6.768	21	
fz 1	N	1.366	16.110	167	
fz 2	N	-1.471	17.200	36	
fy rod	N	-0.279	1.792	195	
nozzle angle	deg	-4.082	3.185		
P nozzle sb	kPa	3.385	0.028		
P nozzle ps	kPa	4.418	0.065		
rpm	rpm	2399.130	0.112		
speed	m/s	4.133	0.046		

RESULTS CALCULATED FROM FILE: 8763

Wave frequency	4.509	rad/sec
Wave heading	225.000	deg
Encounter frequency	10.344	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.168	2.702	150	1.000
roll	deg	0.109	0.568	299	0.210
pitch	deg	-2.912	1.090	135	0.403
yaw	deg	0.684	0.099	178	0.037
surge	cm	-36.549	0.212	319	0.078
sway	cm	2.283	0.280	74	0.104
heave	cm	0.131	2.386	233	0.883
ax cog	m/s2	0.219	0.176	266	
ay cog	m/s2	0.028	0.247	265	
az cog	m/s2	-0.057	2.540	55	
ay ps fore	m/s2	0.040	0.109	348	
az ps fore	m/s2	-0.046	4.644	100	
ay ps aft	m/s2	0.066	0.323	230	
az sb fore	m/s2	-0.031	3.899	97	
fx	N	0.990	10.438	345	
fy 1	N	11.327	17.727	38	
fy 2	N	-7.286	13.758	212	
fy 3	N	6.638	11.141	353	
fz 1	N	-0.886	14.869	185	
fz 2	N	0.252	15.311	41	
fy rod	N	0.183	1.333	219	
nozzle angle	deg	-2.998	2.099		
P nozzle sb	kPa	3.083	0.350		
P nozzle ps	kPa	4.202	0.086		
rpm	rpm	2361.649	2.952		
speed	m/s	3.975	0.041		

RESULTS CALCULATED FROM FILE: 8768

Wave frequency	5.064	rad/sec
Wave heading	225.000	deg
Encounter frequency	12.444	rad/sec
Number of oscillations	8	

First harmonic component

channel	dimension	mean	amplitude	phase	RAO
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wave 1	cm	-0.186	2.688	289	1.000
roll	deg	-0.091	0.163	255	0.061
pitch	deg	-3.005	0.532	119	0.198
yaw	deg	0.973	0.141	162	0.052
surge	cm	-42.004	0.157	72	0.058
sway	cm	-2.344	0.779	66	0.290
heave	cm	0.081	0.614	222	0.228
ax cog	m/s2	0.284	0.075	262	
ay cog	m/s2	0.091	0.155	232	
az cog	m/s2	-0.049	0.966	40	
ay ps fore	m/s2	-0.032	0.327	320	
az ps fore	m/s2	-0.082	2.564	94	
ay ps aft	m/s2	-0.030	0.230	177	
az sb fore	m/s2	-0.048	2.218	98	
fx	N	-0.372	9.409	341	
fy 1	N	10.512	16.086	40	
fy 2	N	-7.966	10.001	195	
fy 3	N	10.000	15.100	324	
fz 1	N	-0.767	17.641	175	
fz 2	N	0.503	18.248	15	
fy rod	N	-0.802	1.619	200	
nozzle angle	deg	-3.412	2.586		
P nozzle sb	kPa	3.123	0.034		
P nozzle ps	kPa	4.084	0.108		
rpm	rpm	2311.588	0.313		
speed	m/s	3.986	0.012		

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